



University of Haripur

Khyber Pakhtoonkhwa

Lab Reports

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Lab Number #01

What is UNIX?

- The UNIX is operating system capable of handling activities from multiple users
- The UNIX operating system is a set of programs that acts as a link between the computer and user.
- The computer programs that allocate system resources and coordinate all the details of the computer internals is called the operating system or kernel
- Users communicate with kernel through a program known as shell

UNIX Artitecture?

Kernel: The kernel is the heart of the operating system. It interacts with hardware and most of the tasks like memory management, task scheduling and file management

Shell: The shell is the utility that processes your requests. When you type in a command at your terminal, the shell interprets the command and calls the program that you want. The shell uses standard syntax for all commands. C Shell, Bourne Shell and Korn Shell are most famous shells which are available with most of the Unix variants.

Login

- When you first connect to a UNIX system, you usually see a prompt such as the following:
login:
- Have your userid (user identification) and password ready.
- Type your userid at the login prompt, then press ENTER. Your userid is case-sensitive
- Type your password at the password prompt, then press ENTER. Your password is also case-sensitive.
- If you provided correct userid and password then you would be allowed to enter into the system.

To view Calender

You would be provided with a command prompt (sometime called \$ prompt) where you would type your all the commands. For example to check calendar you need to type cal command as follows:


```
hammad@ubutu:~$ cal
October 2025
Su Mo Tu We Th Fr Sa
          1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

Change Password

All Unix systems require passwords to help ensure that your files and data remain your own and that the system itself is secure from hackers and crackers. Here are the steps to change your password:

To start, type `passwd` at command prompt as shown below.

Enter your old password the one you're currently using.

Type in your new password. Always keep your password complex enough so that no body can guess it. But make sure, you remember it.

You would need to verify the password by typing it again.

```
hammad@ubutu:~$ passwd
Changing password for hammad.
Current password:
New password:
```

Listing Directories and Files

All data in UNIX is organized into files. All files are organized into directories. These directories are organized into a tree-like structure called the filesystem.

You can use `ls` command to list out all the files or directories available in a directory. Following is the example of using `ls` command with `-l` option

```
hammad@ubutu:~$ ls
Desktop  Downloads  Pictures  snap      Templates
Documents Music      Public   'sudo reboot now'  Videos

hammad@ubutu:~$ ls -l
total 48
drwxr-xr-x 2 hammad hammad 4096 Oct  3 16:02 Desktop
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Documents
drwxr-xr-x 2 hammad hammad 4096 Oct  3 11:30 Downloads
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Music
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Pictures
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Public
drwx----- 7 hammad hammad 4096 Oct  3 12:00 snap
-rw-r--r-- 1 hammad hammad 8669 Oct  3 15:41 'sudo reboot now'
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Templates
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Videos

hammad@ubutu:~$
```

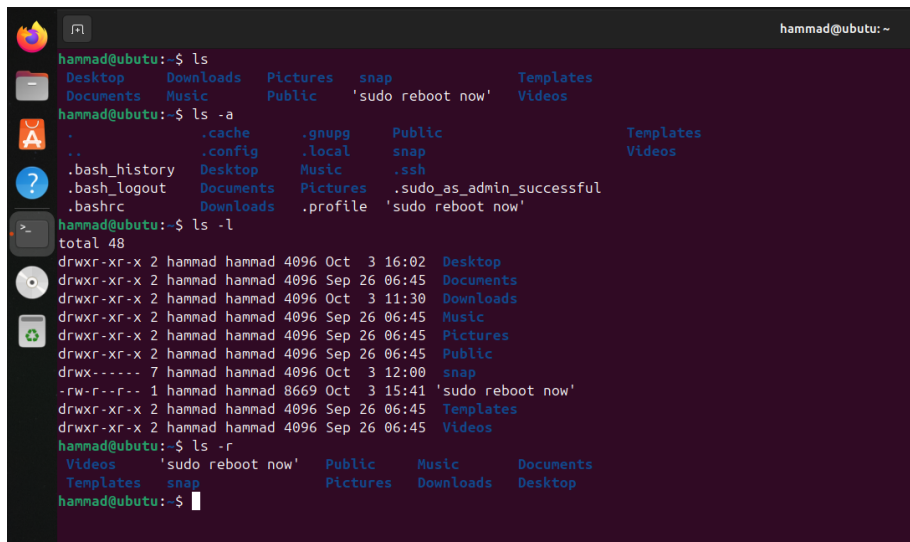
Directory Commands

Command Description

ls List the file in the directory, just like **dir** command in DOS.

Option:

- a** Display all the files, and subdirectories, including hidden files.
- l** Display detailed information about each file, and directory.
- r** Display files in the reverse order.



```
hammad@ubuntu:~$ ls
Desktop  Downloads  Pictures  snap      Templates
Documents Music      Public   'sudo reboot now'  Videos

hammad@ubuntu:~$ ls -a
.         .cache      .gnupg    Public      Templates
..        .config     .local    snap        Videos
.bash_history Desktop     Music     .ssh
.bash_logout Documents   Pictures  .sudo_as_admin_successful
.bashrc    Downloads  .profile  'sudo reboot now'

hammad@ubuntu:~$ ls -l
total 48
drwxr-xr-x 2 hammad hammad 4096 Oct  3 16:02 Desktop
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Documents
drwxr-xr-x 2 hammad hammad 4096 Oct  3 11:30 Downloads
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Music
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Pictures
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Public
drwx----- 7 hammad hammad 4096 Oct  3 12:00 snap
-rw-r--r-- 1 hammad hammad 8669 Oct  3 15:41 'sudo reboot now'
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Templates
drwxr-xr-x 2 hammad hammad 4096 Sep 26 06:45 Videos

hammad@ubuntu:~$ ls -r
Videos  'sudo reboot now' Public  Music  Documents
Templates snap    Pictures Downloads Desktop
hammad@ubuntu:~$
```

Who are you?

While you're logged in to the system, you might be willing to know : Who am I?

The easiest way to find out "who you are" is to enter the whoami command:

```
$ whoami
```



```
hammad@ubuntu:~$ whoiam
whoiam: command not found
hammad@ubuntu:~$ whoami
hammad
hammad@ubuntu:~$
```

Who is Logged in?

- Sometime you might be interested to know who is logged in to the computer at the same time
- There are three commands are available to get you this information, based on how much you'd like to learn about the other users: `users`, `who`, and `w`.
`$ users`
`$ who`
- Try `w` command on your system to check the output. This would list down few more information associated with the users logged in the system.

```
hammad@ubuntu:~$ who
hammad  seat0      2025-10-11 01:10
hammad  tty2       2025-10-11 01:10
hammad@ubuntu:~$ user
Command 'user' not found, did you mean:
  command 'fuser' from deb psmisc (23.7-2)
  command 'users' from deb coreutils (9.5-1ubuntu1.25.04.2)
Try: sudo apt install <deb name>
hammad@ubuntu:~$ users
hammad hammad
hammad@ubuntu:~$ w
01:27:07 up 17 min,  2 users,  load average: 0.02, 0.06, 0.11
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
hammad    tty2     -               01:10    17:30  0.06s  0.06s  /usr/libexec/gnome-session-binary --session=ubuntu
hammad    -       -               01:10    17:15  0.00s  1.66s  /usr/lib/systemd/systemd --user
hammad@ubuntu:~$
```

How to Shutdown the system?

- The most consistent way to shut down a Unix system properly via the command line is to use one of the following commands:
`$poweroff`
`$reboot`
`$shutdown`
- You typically need to be the superuser or root (the most privileged account on a Unix system) to shut down the system

Create, Display and Merge Command

Command: `cat` —`cat` is short for concatenate. This command is used to create, view and concatenate files.

Example:

- `cat>file1`

This command creates the file, enter the contents into it and to save the content, press `Ctrl+D`.

- `cat file1`

This command shows the content of the file.

- `cat file1 file2 > file3`

This command combines the contents of the first two files into the third file.

Command: `pwd` "`pwd`" stands for print working directory. It displays your current position in the UNIX/LINUX file system.

Example:

- `pwd`

It is simply used to report your current working directory.

```
hammad@ubuntu:~$ cat-hammad
My my name is Hannad Younis Abbasi
hammad@ubuntu:~$ cat-hammad1
I am student of computer science
hammad@ubuntu:~$ cat hammad hammad1 > hammad2
hammad@ubuntu:~$ cat hammad2
My my name is Hannad Younis Abbasi
I am student of computer science
hammad@ubuntu:~$ pwd
/home/hammad
hammad@ubuntu:~$
```

Activities

Activity #01: Verify that you are in your home directory. Make the directory LABS using the following command

```
hammad@ubuntu:~$ pwd
/home/hammad
hammad@ubuntu:~$ mkdir LABS
```

Activity #02: List the files in the current directory to verify that the directory LABS has been made correctly Change directories to LABS

```
hammad@ubuntu:~$ ls
Desktop  hammad  I        LABS      Public      Templates
Documents hammad1 IBAS     Music     snap        Videos
Downloads hammad2 IBAS     Pictures  'sudo reboot now'
hammad@ubuntu:~$ cd LABS
hammad@ubuntu:~/LABS$ cat>file1
```

Activity #03: Create the file named file? List the contents of the file file1 to the screen. Make a copy of the file file1 under the name file2. Verify that the files file 1 and file2 both exist.

```
hammad@ubuntu:~/LABS$ cat>file1
My name is hammadhammad@ubuntu:~/LABS$ cat>file2
I am a Coder
hammad@ubuntu:~/LABS$ cat file1
My name is hammadhammad@ubuntu:~/LABS$ cp file1 file2
hammad@ubuntu:~/LABS$ ls
file1 file2
hammad@ubuntu:~/LABS$
```

Activity #04: List the contents of both file1 and file2 to the monitor screen. Then delete the file file1. Clear the window. Rename file2 to thefile

```

My name is hammadhammad@ubutu:~/LABS$ cp file1 file2
hammad@ubutu:~/LABS$ ls
file1 file2
hammad@ubutu:~/LABS$ cat file1 file2
My name is hammadMy name is hammadhammad@ubutu:~/LABS$ rm file1
hammad@ubutu:~/LABS$ clear

hammad@ubutu:~/LABS$ mv file2 thefile
hammad@ubutu:~/LABS$ cat thefile
My name is hammadhammad@ubutu:~/LABS$

```

Activity #05: Copy thefile to your home directory.

```

hammad@ubutu:~/LABS$ cp thefile ~
hammad@ubutu:~/LABS$

```

Activity #06: Remove thefile from the current directory

```

hammad@ubutu:~/LABS$ rm thefile
hammad@ubutu:~/LABS$

```

Activity #07: Copy thefile from your home directory to the current directory Change directories to your home directory Remove thefile from your home directory and from directory LABS. Verify thefile is removed from the directory LABS Remove the directory LABS from your home directory with the following command. Verify that thefile and LABS are gone from your home directory

```

Oct 11 17:20
hammad@ubutu:~
hammad@ubutu:~/LABS$ cp ~/thefile
cp: missing destination file operand after '/home/hammad/thefile'
Try 'cp --help' for more information.
hammad@ubutu:~/LABS$ cd ~
hammad@ubutu:~$ rm thefile
hammad@ubutu:~$ cd ~/LABS
hammad@ubutu:~/LABS$ rm thefile
rm: cannot remove 'thefile': No such file or directory
hammad@ubutu:~/LABS$ rm thefile
rm: cannot remove 'thefile': No such file or directory
hammad@ubutu:~/LABS$ cd ~
hammad@ubutu:~$ rmdir LABS
hammad@ubutu:~$ ls
Desktop  Documents  Downloads  hammad  hammad1  hammad2  I  IBAs  IBAS  Music  Pictures  Public  snap  'sudo reboot now'  Templates  Videos
hammad@ubutu:~$

```

Lab number #02

Files: All data in UNIX is organized into files. All files are organized into directories. These directories are organized into a tree-like structure called the filesystem.

When you work with UNIX, one way or another you spend most of your time working with files. This lecture would teach you how to create and remove files, copy and rename them, create links to them etc.

Types of Files:

In UNIX there are three basic types of files:

1. Ordinary Files: An ordinary file is a file on the system that contains data, text, or program instructions. In this tutorial, you look at working with ordinary files.

2. Directories: Directories store both special and ordinary files. For users familiar with Windows or Mac OS, UNIX directories are equivalent to folders.

3. Special Files: Some special files provide access to hardware such as hard drives, CD-ROM drives, modems, and Ethernet adapters. Other special files are similar to aliases or shortcuts and enable you to access a single file using different names

Ls -l Command: The command ls supports the -l option which would help you to get more information about the listed files:

```
syed@syed-VirtualBox:~$ ls -l
total 44
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Desktop
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Documents
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Downloads
-rw-r--r-- 1 syed syed 8445 Feb  9 14:31 examples.desktop
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Music
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Pictures
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Public
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Templates
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Videos
```

Here is the information about all the listed columns:

1. First Column: represents file type and permission given on the file. Below is the description of all type of files.
2. Second Column: represents the number of memory blocks taken by the file or directory.
3. Third Column: represents owner of the file. This is the Unix user who created this file.
4. Fourth Column: represents group of the owner. Every Unix user would have an associated group.
5. Fifth Column: represents file size in bytes.
6. Sixth Column: represents date and time when this file was created or modified last time.
7. Seventh Column: represents file or directory name.

Meta Character: Meta characters have special meaning in Unix. For example * and ? are metacharacters. We use * to match 0 or more characters, a question mark ? matches with single character.

```
$ls ch*.doc
```

Displays all the files whose name start with ch and ends with .doc:

```
ch01-1.doc ch010.doc ch02.doc ch03-2.doc
ch04-1.doc ch040.doc ch05.doc ch06-2.doc
ch01-2.doc ch02-1.doc
```

Here * works as meta character which matches with any character. If you want to display all the files ending with just .doc then you can use following command:

```
$ls *.doc
```

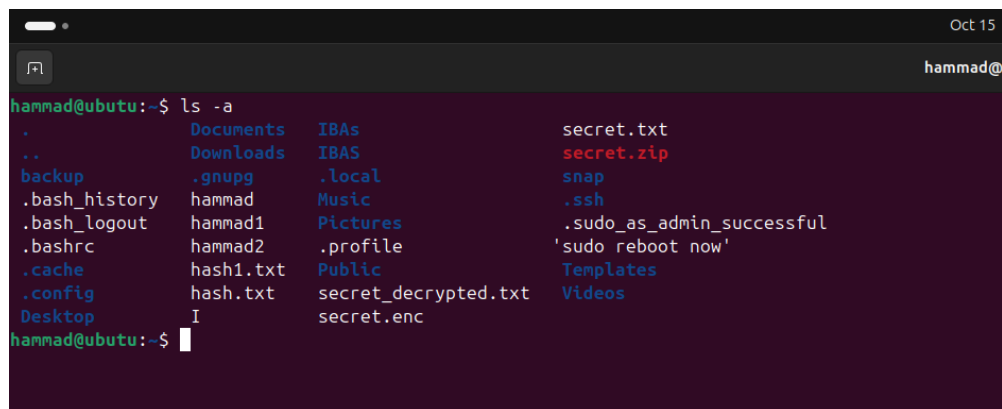
Hidden File:

An invisible file is one whose first character is the dot (.). UNIX programs (including the shell) use most of these files to store configuration information.

Some common examples of hidden files include the files:

- .profile: the Bourne shell (sh) initialization script
- .kshrc: the Korn shell (ksh) initialization script
- .cshrc: the C shell (csh) initialization script
- .rhosts: the remote shell configuration file

To list invisible files, specify the -a option to ls:



```
hammad@ubutu:~$ ls -a
.      Documents  IBAs      secret.txt
..     Downloads  IBAs      secret.zip
backup .gnupg      .local    snap
.bash_history hammad      Music     .ssh
.bash_logout  hammad1     Pictures  .sudo_as_admin_successful
.bashrc       hammad2     .profile  'sudo reboot now'
.cache        hash1.txt   Public    Templates
.config       hash.txt   secret_decrypted.txt  Videos
.Desktop      I          secret.enc
```

Vim Editor

```
sudo apt-get install vim
```

Command mode: letters or sequence of letters interactively command vim. Commands are case sensitive. The ESC key can end a command.

Edit mode: Text is inserted. The ESC key ends Edit mode and returns you to command mode. One can enter edit mode with the "i"

```
:q
```

quitting vim editor

```
:wq filename  
saving and quitting
```

Creating File: You can use vi editor to create ordinary files on any Unix system. You simply need to give following command:

```
$ vi filename
```

Above command would open a file with the given filename. You would need to press key i to come into edit mode.

Once you are done, do the following steps:

Press key esc to come out of edit mode.

Press two keys Shift + ZZ together to come out of the file completely.

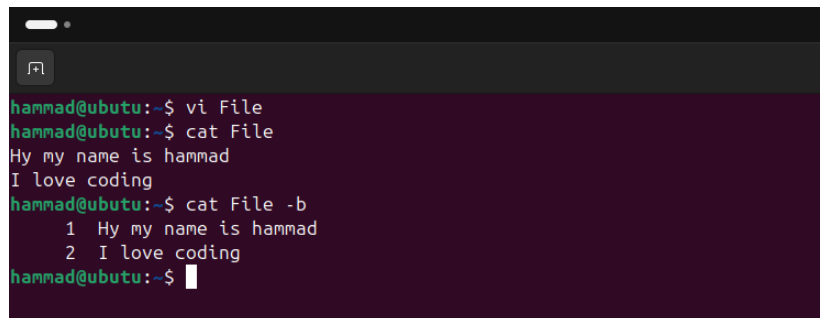
Display The Content of a File:

You can use cat command to see the content of a file. Following is the simple example to see the content of above created file:

```
$ cat filename
```

You can display line numbers by using -b option along with cat command as follows:

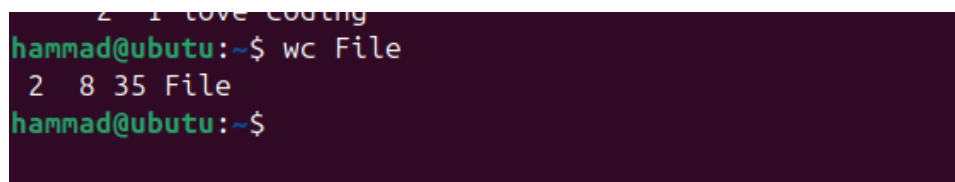
```
$ cat filename -b
```

A terminal window with a dark purple background. The prompt is hammad@ubutu:~\$. The user enters 'vi File', then 'cat File', which outputs 'Hy my name is hammad' and 'I love coding'. Then the user enters 'cat File -b', which outputs '1 Hy my name is hammad' and '2 I love coding'.

```
hammad@ubutu:~$ vi File  
hammad@ubutu:~$ cat File  
Hy my name is hammad  
I love coding  
hammad@ubutu:~$ cat File -b  
1 Hy my name is hammad  
2 I love coding  
hammad@ubutu:~$
```

Counting Words in a File: You can use the wc command to get a count of the total number of lines, words, and characters contained in a file. Following is the simple example to see the information about above created file:

```
$ wc filename
```

A terminal window with a dark purple background. The prompt is hammad@ubutu:~\$. The user enters 'wc File', which outputs '2 8 35 File'.

```
hammad@ubutu:~$ wc File  
2 8 35 File  
hammad@ubutu:~$
```


Here is the detail of all the four columns:

First Column: represents total number of lines in the file.

Second Column: represents total number of words in the file.

Third Column: represents total number of bytes in the file. This is actual size of the file.

Fourth Column: represents file name.

Copying File: To make a copy of a file use the cp command. The basic syntax of the command is:

\$ cp source_file destination_file

```
hammad@ubuntu:~$ vi abbasi
hammad@ubuntu:~$ cat hammad
Hy my name is Hammad Younis Abbasi
hammad@ubuntu:~$ cat abbasi
Inside abbasi file
hammad@ubuntu:~$ cp abbasi hammad
hammad@ubuntu:~$ cp abbasi hammad
hammad@ubuntu:~$ cat hammad
Inside abbasi file
```

Renaming File: To change the name of a file use the mv command. Its basic syntax is:

\$ mv old_file new_file

```
hammad@ubuntu:~$ mv hammad Hammad
hammad@ubuntu:~$ rm abbasi
```

Removing File: To delete an existing file use the rm command. Its basic syntax is:

\$ rm filename

Deleting a file can delete useful information. So be careful while using this command. It is recommended to use -i option along with rm command.

```
hammad@ubuntu:~$ rm abbasi
hammad@ubuntu:~$ ls -la
.          Documents I          secret.enc
..         Downloads IBAs   secret.txt
.Abba1i.swp File    IBAs   secret.zip
.backup   .gnupg .local snap
.bash_history Hammad Music  .ssh
.bash_logout hammad1 .myfile.swp .sudo_as_admin_successful
.bashrc     hammad2 Pictures 'sudo reboot now'
.cache      hammad .profile  Templates
.config     hash1.txt Public    Videos
.Desktop    hash.txt Public    vininfo
hammad@ubuntu:~$ rm hammad
hammad@ubuntu:~$ ls -la
.          .bashrc Desktop File hammad1 hash.txt IBAs .myfile.swp Public secret_decrypted.txt secret.txt .ssh Templates
..         .bash_history .cache Documents .gnupg hammad2 I .local Pictures secret_decrypted.txt secret.zip .sudo_as_admin_successful Videos
.Abba1i.swp .bash_logout .config Downloads Hammad hash1.txt IBAs Music .profile secret.enc snap 'sudo reboot now' vininfo
hammad@ubuntu:~$
```

Standard UNIX Streams:

Under normal circumstances every Unix program has three streams (files) opened for it when it starts up:

stdin : This is referred to as standard input and associated file descriptor is 0. This is also represented as STDIN. Unix program would read default input from STDIN.

stdout : This is referred to as standard output and associated file descriptor is 1. This is also represented as STDOUT. Unix program would write default output at STDOUT

stderr : This is referred to as standard error and associated file descriptor is 2. This is also represented as STDERR. Unix program would write all the error message at STDERR.

Lab Number #03

UNIX Directories: A directory is a file whose sole job is to store file names and related information. All files whether ordinary, special, or directory, are contained in directories.

UNIX uses a hierarchical structure for organizing files and directories. This structure is often referred to as a directory tree . The tree has a single root node, the slash character (/), and all other directories are contained below it.

Home Directories: The directory in which you find yourself when you first login is called your home directory.

You will be doing much of your work in your home directory and subdirectories that you'll be creating to organize your files.

You can go in your home directory anytime using the following command:

```
$cd ~
```

Absolute/Relative Pathnames: Directories are arranged in a hierarchy with root (/) at the top.

The position of any file within the hierarchy is described by its pathname.

Elements of a pathname are separated by a / . A pathname is absolute if it is described in relation to root, so absolute pathnames always begin with a / .

These are some example of absolute filenames

```
/etc/passwd
```

Listing Directories: To list the files in a directory you can use the following syntax:

```
$ls dirname
```

```
hammad@ubutu:~$ ls abbasi
hammad@ubutu:~$ cat > abbasi
bash: abbasi: Is a directory
```

Creating Directories: Directories are created by the following command:

```
$mkdir dirname
```

Here, directory is the absolute or relative pathname of the directory you want to create. For example, the command:

```
$mkdir mydir
```

Creates the directory mydir in the current directory.

Here is another example:

```
$mkdir /tmp/test-dir
```

This command creates the directory test-dir in the /tmp directory. The mkdir command produces no output if it successfully creates the requested directory.

```
hammad@ubutu:~$ mkdir Abbasi
mkdir: cannot create directory 'Abbasi': File exists
hammad@ubutu:~$ mkdir abbasi
hammad@ubutu:~$ ls abbasi
```

Removing Directories: Directories can be deleted using the rmdir command as follows:

`$rmdir dirname`

Note: To remove a directory make sure it is empty which means there should not be any file or sub-directory inside this directory.

You can remove multiple directories at a time as follows:

`$rmdir dirname1 dirname2 dirname3`

Above command removes the directories dirname1, dirname2, and dirname2 if they are empty. The rmdir command produces no output if it is successful.

```
hammad@ubutu:~$ rmdir abbasi
hammad@ubutu:~$ ls abbasi
ls: cannot access 'abbasi': No such file or directory
hammad@ubutu:~$ mkdir file1
```

Changing Directories: You can use the cd command to do more than change to a home directory:

You can use it to change to any directory by specifying a valid absolute or relative path. The syntax is as follows:

`$cd dirname`

Renaming Directories: The mv (move) command can also be used to rename a directory. The syntax is as follows:

`$mv olddir newdir`

You can rename a directory dir1 to dir2 as follows:

`$mv dir1 dir2`

```
hammad@ubutu:~$ cd file1
hammad@ubutu:~/file1$ mv file1 file2
mv: cannot stat 'file1': No such file or directory
hammad@ubutu:~/file1$ cd ~
hammad@ubutu:~$ mv file1 file2
hammad@ubutu:~$ ls file2
```

The directories . (dot) and .. (dot dot): The filename . (dot) represents the current working directory; and the filename .. (dot dot) represent the directory one level above the current working directory, often referred to as the parent directory.

If we enter the command to show a listing of the current working directories files and use the -a option to list all the files and the -l option provides the long listing, this is the result.

\$ls -la

```
hammad@ubutu:~$ ls file2
hammad@ubutu:~$ ls -la
total 188
drwxr-x--- 21 hammad hammad 4096 Oct 24 04:38 .
drwxr-xr-x  3 root   root   4096 Sep 26 18:45 ..
drwxrwxr-x  2 hammad hammad 4096 Oct 17 16:56 Abbasi
-rw-----  1 hammad hammad 12288 Oct 15 16:35 .Abbasi.swp
drwxrwxr-x  2 hammad hammad 4096 Oct 15 14:18 backup
-rw-----  1 hammad hammad 3805 Oct 17 17:00 .bash_history
-rw-r--r--  1 hammad hammad 220 Mar  5 2025 .bash_logout
-rw-r--r--  1 hammad hammad 3771 Mar  5 2025 .bashrc
drwx----- 13 hammad hammad 4096 Oct 17 15:54 .cache
drwxr-xr-x 14 hammad hammad 4096 Oct 17 15:19 .config
-rw-rw-r--  1 hammad hammad  26 Oct 17 07:27 decrypted.txt
drwxr-xr-x  2 hammad hammad 4096 Oct  3 16:02 Desktop
drwxr-xr-x  2 hammad hammad 4096 Sep 26 06:45 Documents
drwxr-xr-x  2 hammad hammad 4096 Oct 17 14:52 Downloads
-rw-rw-r--  1 hammad hammad  35 Oct 15 16:50 File
drwxrwxr-x  2 hammad hammad 4096 Oct 24 04:37 file2
drwx-----  2 hammad hammad 4096 Oct 17 15:54 .gnupg
-rw-rw-r--  1 hammad hammad  40 Oct 24 04:29 hammad
```

Lab Number #04:

UNIX File Permissions Setup:

File permissions:

File ownership is an important component of UNIX that provides a secure method for storing files. Every file in UNIX has the following attributes:

- Owner permissions: The owner's permissions determine what actions the owner of the file can perform on the file.
- Group permissions: The group's permissions determine what actions a user, who is a member of the group that a file belongs to, can perform on the file.
- Other (world) permissions: The permissions for others indicate what action all other users can perform on the file.

The Permission Indicators:

ls -l command displays various information related to file permission as follows:

```
ubuntu@ubuntu:~$ ls -l
total 0
drwxr-xr-x 2 ubuntu ubuntu 60 Oct 24 11:46 Desktop
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 11:48 Documents
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 11:48 Downloads
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 11:48 Music
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 11:48 Pictures
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 11:48 Public
drwx----- 4 ubuntu ubuntu 80 Oct 24 11:50 snap
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 11:48 Templates
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 11:48 Videos
ubuntu@ubuntu:~$
```

The Permission Indicators:

Here first column represents different access mode ie. permission associated with a file or directory. The permissions are broken into groups of threes, and each position in the group denotes a specific permission, in this order: read (r), write (w), execute (x): The first three characters (2-4) represent the permissions for the file's owner. For example -drwxr-xr-x- represents that owner has read (r), write (w) and execute (x) permission. (User) The second group of three characters (5-7) consists of the permissions for the group to which the file belongs. For example -rwxr-xr-- represents that group has read (r) and execute (x) permission but no write permission. (Group) The last group of three characters (8-10) represents the permissions for everyone else. For example -rwxr-xr-- represents that other world has read (r) only permission. (Others)

File Access Mods:

The basic building blocks of Unix permissions are the read, write, and execute permissions, which are described below: Read: Grants the capability to read ie. view the contents of the file. Write: Grants the capability to modify, or remove the content of the file. Execute: User with execute permissions can run a file as a program.

Directory Access Mods:

Directory access modes are listed and organized in the same manner as any other file. There are a few differences that need to be mentioned: Read: Access to a directory means that the user can read the contents. The user can look at the filenames inside the directory. Write: Access means that the user can add or delete files to the contents of the directory. Execute: Executing a directory doesn't really make a

lot of sense so think of this as a traverse permission. A user must have execute access to the bin directory in order to execute ls or cd command.

Changing Permissions:

To change file or directory permissions, use the chmod (change mode) command. There are two ways to use chmod: 1) symbolic mode
2) absolute mode

Using Ch Mode in Symbolic Mode:

With symbolic permissions you can add, delete, or specify the permissions using the operators in the following table.

Chmod operator	Description
+	Adds the designated permission(s) to a file or directory.
-	Removes the designated permission(s) from a file or directory.
=	Sets the designated permission(s).

Here's an example using testfile. Running ls -l on testfile shows that the file's permissions are as follows:

```
ubuntu@ubuntu:~$ ls -l testfile
-rw-rw-r-- 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Adding write and execute permissions for others permissions to test file\$chmod o+wx testfile

```
ubuntu@ubuntu:~$ chmod o+wx testfile
ubuntu@ubuntu:~$ chmod o +wx testfile
chmod: invalid mode: 'o'
Try 'chmod --help' for more information.
ubuntu@ubuntu:~$ ls -l testfile
-rw-rw-rwx 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Removing write permissions for group on testfile \$chmod g-w testfile

```
ubuntu@ubuntu:~$ chmod g-w testfile
ubuntu@ubuntu:~$ ls -l testfile
-rw-r--rwx 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Removing read permissions for owner (user) on testfile \$chmod u-r testfile

```
ubuntu@ubuntu:~$ chmod u-r testfile
ubuntu@ubuntu:~$ ls -l testfile
--w-r--rwx 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Using chmod with Absolute Permissions:

The second way to modify permissions with the chmod command is to use a number to specify each set of permissions for the file. Each permission is assigned a value, as the following table shows, and the total of each set of permissions provides a number for that set.

Number	Octal Permission Representation	Ref
0	No permission	---
1	Execute permission	--x
2	Write permission	-w-
3	Execute and write permission: 1 (execute) + 2 (write) = 3	-wx
4	Read permission	r--
5	Read and execute permission: 4 (read) + 1 (execute) = 5	r-x
6	Read and write permission: 4 (read) + 2 (write) = 6	rw-
7	All permissions: 4 (read) + 2 (write) + 1 (execute) = 7	rwx

Here's an example using testfile. Running ls -l on testfile shows that the file's permissions are as follows:

```
ubuntu@ubuntu:~$ chmod 754 testfile
ubuntu@ubuntu:~$ ls -l testfile
-rwxr-xr-- 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```


Then each example chmod command from the preceding table is run on testfile, followed by ls -l so you can see the permission changes:

```
ubuntu@ubuntu:~$ chmod 444 testfile
ubuntu@ubuntu:~$ ls -l testfile
-r--r--r-- 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Changing Owners and Groups: While creating an account on Unix, it assigns a owner ID and a group ID to each user. All the permissions mentioned above are also assigned based on Owner and Groups. Two commands are available to change the owner and the group of files: chown: The chown command stands for "change owner" and is used to change the owner of a file.

chgrp: The chgrp command stands for "change group" and is used to change the group of a file.

Changing Ownership: Syntax:\$chown username filenameExmaple:\$chown user2 testfileChanges the owner of the testfile to the user user2 The super user, root, has the unrestricted capability to change the ownership of a any file but normal users can change only the owner of files they own.

```
ubuntu@ubuntu:~$ sudo chown root testfile
ubuntu@ubuntu:~$ ls -l testfile
-r--r--r-- 1 root ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Changing Group Ownership:

Syntax:chgrp groupname filename Example:chgrp group2 testfileChanges the group of the testfile to group2

```
ubuntu@ubuntu:~$ sudo chgrp sudo testfile
ubuntu@ubuntu:~$ ls -l testfile
-r--r--r-- 1 root sudo 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Create Directory and Practise Directory Permissions:

1)mkdir testdir

2)chmod 755 testdir

3)ls -ld testdir

```
ubuntu@ubuntu:~$ mkdir testdir
ubuntu@ubuntu:~$ chmod 755 testdir
ubuntu@ubuntu:~$ ls -ld testdir
drwxr-xr-x 2 ubuntu ubuntu 40 Oct 24 12:06 testdir
ubuntu@ubuntu:~$
```

Lab Number #06:

Task 1: Using the tee Command

The tee command reads from standard input and writes both to standard output and to a file.

Command:

echo "This line will be displayed and saved." | tee output.txt

```
Endsohaib@ubuntu:~$ echo "This line will be displayed and saved." | tee output.txt
This line will be displayed and saved.
sohaib@ubuntu:~$ cat output.txt
This line will be displayed and saved.

sohaib@ubuntu:~$ grep "text" sohaib.txt
this file contain only text and is only for testing purpose.
sohaib@ubuntu:~$ grep "only" sohaib.txt
this file contain only text and is only for testing purpose.

sohaib@ubuntu:~$ grep -i "for" sohaib.txt
this file contain only text and is only for testing purpose.
```

Task 2: Using the grep Command

Search for lines containing a specific string in a file.

Command 2.1 : Basic Search

Command 2.2: Case-Insensitive

Search (-i) `grep -i "hello" file1.txt`

Command 2.3: Invert Match (-v) - Show lines that do NOT contain the

pattern `grep -v "test" file1.txt`

Command 2.4: Show Lines Around

Match (-A, -B, -C) *Show 1 line after the*

match `grep -A1 "multiple" file1.txt`

Show 1 line before the

match `grep -B1`

`"world" file1.txt`

```
sohaib@ubuntu:~$ grep -v "text" sohaib.txt
this is my personal file, name is sohaib hassan.
End
sohaib@ubuntu:~$ cat sohaib.tx
cat: sohaib.tx: No such file or directory
sohaib@ubuntu:~$ cat sohaib.txt
this is my personal file, name is sohaib hassan.
this file contain only text and is only for testing purpose.

sohaib@ubuntu:~$ grep -A1 "hassan" sohaib.txt
this is my personal file, name is sohaib hassan.
this file contain only text and is only for testing purpose.

sohaib@ubuntu:~$ grep -B1 "hassan" sohaib.txt
this is my personal file, name is sohaib hassan.

sohaib@ubuntu:~$ grep -C1 "hassan" sohaib.txt
this is my personal file, name is sohaib hassan.
this file contain only text and is only for testing purpose.
```

Show 1 line before and 1 line after the match

(Context) `grep -C1 "hello" file1.txt`

Task 3: Using the tr Command

Translate or delete characters.

Command 3.1: Translate one character to another

`echo "hello" | tr 'e' 'E'`

```
sohaib@ubuntu:~$ echo "sohaib" | tr 'o' 'O'  
sOhaib
```

Command 3.2: Convert lowercase to

uppercase `echo "Hello World" | tr`

`'[:lower:]' '[:upper:]'`

```
sohaib@ubuntu:~$ echo "Sohaib Hassan" | tr '[:lower:]' '[:upper:]'  
tr: misaligned [:upper:] and/or [:lower:] construct  
sohaib@ubuntu:~$ echo "Sohaib Hassan" | tr '[:lower:]' '[:upper:]'  
SOHAIB HASSAN
```

Command 3.3: Translate newlines to spaces (combine lines into one line)

`cat file1.txt | tr '\n' ' '`

```
sohaib@ubuntu:~$ cat sohaib.txt | tr '\n' ' '  
this is my personal file, name is sohaib hassan. this file contain only text and is only for testing purpose. Endssohaib@
```

Task 4: Using the wc Command

Count lines, words, and characters in a file.

Command: `wc file1.txt`

```
ubuntu:~$ wc sohaib.txt
2  21 113 sohaib.txt
```

Task 5: Using the sort Command

Sort the lines of a text file alphabetically.

Command: `sort unsorted.txt`

```
sohaib@ubuntu:~$ sort sohaib.txt
End
this file contain only text and is only for testing purpose.
this is my personal file, name is sohaib hassan.
```

Task 6: Using the uniq Command

Remove duplicate lines from a **sorted** file.

`uniq` only removes *adjacent* duplicates, so the input must be sorted

first. Command: `sort unsorted.txt | uniq`

```
sohaib@ubuntu:~$ sort sohaib.txt | uniq
End
this file contain only text and is only for testing purpose.
this is my personal file, name is sohaib hassan.
```

Task 7: Combining Commands (Pipeline)

This is a powerful concept. Use the output of one command as the input for the next using the pipe `|` operator.

Command 7.1: Count how many times the word "test" (case-insensitive) appears in `file1.txt`. `grep -i "test" file1.txt | wc -l`

```
sohaib@ubuntu:~$ grep -i "hassan" sohaib.txt | wc -l
wc: invalid option -- '1'
Try 'wc --help' for more information.
```

Command 7.2: List all unique words from a file.

```
cat file1.txt | tr ' '\n' | sort | uniq
```



```
sohaib@ubuntu:~$ cat sohaib.txt | tr ' '\n' | sort | uniq
End
and
contain
file
file,
for
hassan.
is
my
name
only
personal
purpose.
sohaib
testing
text
this
sohaib@ubuntu:~$
```

Lab Number #07:

Commands & Arguments (REDUCED & COMBINED FORMAT)

TOPIC 1: Command Line Scan, Arguments & White Space Handling

Description:

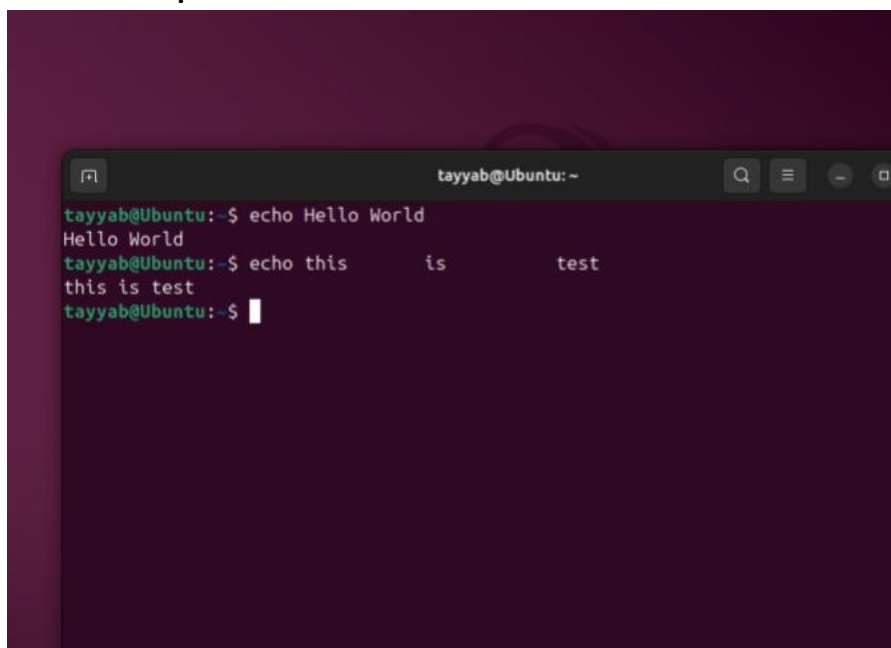
The shell reads a command line, splits it into arguments, and removes extra white spaces. Multiple spaces are treated as one unless quotes are used

Commands:

```
echo Hello World
```

```
echo this    is    test
```

Screenshot Space:



```
tayyab@Ubuntu: ~$ echo Hello World
Hello World
tayyab@Ubuntu: ~$ echo this    is    test
this is test
tayyab@Ubuntu: ~$
```

TOPIC 2: Quoting Mechanisms (Single & Double Quotes)

Description:

Quotes prevent the shell from removing spaces or modifying text.

- Single quotes preserve everything exactly.
- Double quotes preserve spaces but allow variable expansion.

Commands:

```
echo 'multiple spaces kept'
```

```
echo "double quotes also keep spaces"
```

Screenshot Space:

A terminal window with a dark purple background. The prompt is 'tayyab@Ubuntu:~\$'. The first command is 'echo 'multiple spaces kept'', and the output is 'multiple spaces kept'. The second command is 'echo "double quotes also keep spaces"', and the output is 'double quotes also keep spaces'. The prompt is now 'tayyab@Ubuntu:~\$' with a cursor.

```
tayyab@Ubuntu:~$ echo 'multiple spaces kept'
multiple spaces kept
tayyab@Ubuntu:~$ echo "double quotes also keep spaces"
double quotes also keep spaces
tayyab@Ubuntu:~$
```

TOPIC 3: echo with Escape Sequences (echo -e)

Description:

The -e option enables special characters like newline (\n) and tab (\t).

Commands:

```
echo -e "line1\nline2"
```

```
echo -e "col1\tcol2\tcol3"
```

Screenshot Space:

```
tayyab@Ubuntu: ~  
tayyab@Ubuntu:~$ echo Hello World  
Hello World  
tayyab@Ubuntu:~$ echo this      is      test  
this is test  
tayyab@Ubuntu:~$ echo 'multiple  spaces  kept'  
multiple  spaces  kept  
tayyab@Ubuntu:~$ echo "double quotes also keep      spaces"  
double quotes also keep      spaces  
tayyab@Ubuntu:~$ echo -e "line1\nline2"  
line1  
line2  
tayyab@Ubuntu:~$ echo -e "col1\tcol2\tcol3"  
col1    col2    col3  
tayyab@Ubuntu:~$
```

TOPIC 4: Built-in vs External Commands (type & full path)

Description:

- Some commands are built into the shell (e.g., cd).
- Others are external programs in /bin, /usr/bin.
- type tells which one it is.
- Full path runs the external version explicitly.

Commands:

type echo

type pwd

/bin/echo "This is external echo"

Screenshot Space:

```
tayyab@Ubuntu:~$ echo -e "col1\tcol2\tcol3"  
col1    col2    col3  
tayyab@Ubuntu:~$ type echo  
echo is a shell builtin  
tayyab@Ubuntu:~$ type pwd  
pwd is a shell builtin  
tayyab@Ubuntu:~$ /bin/echo"This is external echo"  
bash: /bin/echoThis is external echo: No such file or directory  
tayyab@Ubuntu:~$ type echo type pwd /bin/echo "This is external echo"  
echo is a shell builtin  
type is a shell builtin  
pwd is a shell builtin  
/bin/echo is /bin/echo
```


TOPIC 5: Finding Command Locations (which)

Description:

The which command searches the system PATH and shows the exact file location of external commands.

Commands:

which mv

which cp

Screenshot Space:

```
tayyab@Ubuntu:~$ which mv
/usr/bin/mv
tayyab@Ubuntu:~$ which cp
/usr/bin/cp
tayyab@Ubuntu:~$
```

TOPIC 6: Alias Management (create, view, remove)

Description:

Aliases allow you to create shortcuts or default options for commands.

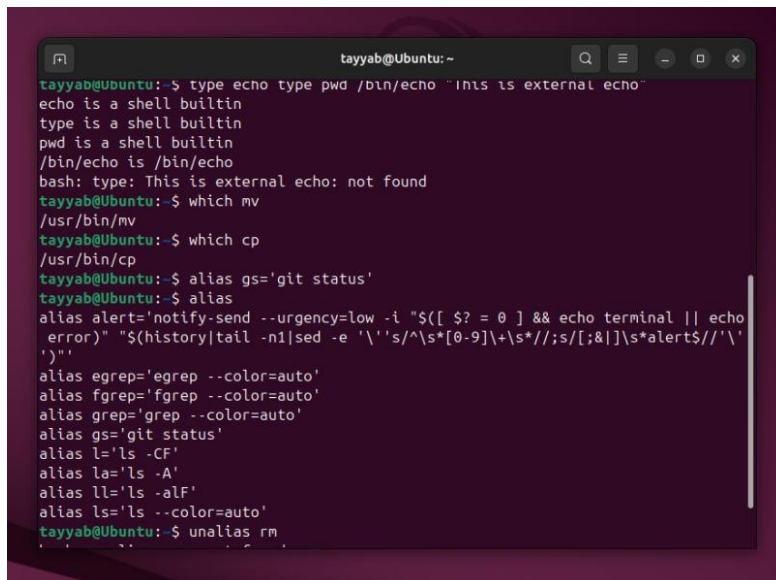
Commands:

alias gs='git status'

alias

unalias rm

Screenshot Space:



```
tayyab@Ubuntu:~$ type echo type pwd /bin/echo "this is external echo"
echo is a shell builtin
type is a shell builtin
pwd is a shell builtin
/bin/echo is /bin/echo
bash: type: This is external echo: not found
tayyab@Ubuntu:~$ which mv
/usr/bin/mv
tayyab@Ubuntu:~$ which cp
/usr/bin/cp
tayyab@Ubuntu:~$ alias gs='git status'
tayyab@Ubuntu:~$ alias
alias alert='notify-send --urgency=low -i "${[ $? = 0 ]} && echo terminal || echo error" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[;&]\s*alert$//'\`)'
tayyab@Ubuntu:~$ alias egrep='egrep --color=auto'
tayyab@Ubuntu:~$ alias fgrep='fgrep --color=auto'
tayyab@Ubuntu:~$ alias grep='grep --color=auto'
tayyab@Ubuntu:~$ alias gs='git status'
tayyab@Ubuntu:~$ alias l='ls -CF'
tayyab@Ubuntu:~$ alias la='ls -A'
tayyab@Ubuntu:~$ alias ll='ls -alF'
tayyab@Ubuntu:~$ alias ls='ls --color=auto'
tayyab@Ubuntu:~$ unalias rm
```

All Main Topic Names (Only the 6 Topics)

1. Command Line Scan, Arguments & White Space Handling
2. Quoting Mechanisms (Single & Double Quotes)
3. echo with Escape Sequences (echo -e)
4. Built-in vs External Commands (type & full path)
5. Finding Command Locations (which)
6. Alias Management (create, view, remove)

Lab Number #08:

UNIX PROCESS MANAGEMENT

Objective:

To understand and practice process management in UNIX/Linux systems using Ubuntu terminal commands.

Introduction

A process is a program in a running state. Each process is uniquely identified by a Process ID (PID). Processes can run in the foreground or background, and users can manage them using various commands.

Commands to Use in Ubuntu Linux Terminal

`ps` - Display Running Processes

Description: Shows the currently running processes.

Example:

```
```bash
```

ps

``

## **`ps -f` - Detailed Process List**

Description: Displays detailed information about running processes, including PPID (Parent Process ID).

Example:

```bash

ps -f

``

Run a Process in the Background

Description: Appending `&` to a command runs it in the background.

Example:

```bash

ls -l &

...

## **`jobs` - List Background and Suspended Jobs**

Description: Shows all background and suspended processes.

Example:

```
```bash
```

```
jobs
```

```
```
```

``fg`` - Bring a Background Process to the Foreground

Description: Brings a background job to the foreground using its job number.

Example:

```
```bash
```

```
fg %1
```

``bg`` - Run a Suspended Process in the Background

Description: Resumes a suspended job in the background.

Example:

```
```bash
```

```
bg %1
```

```
```
```

7. ``kill`` - Terminate a Process

Description: Stops a process using its PID or job number.

Example:

```
```bash
```

```
kill %1
```

```
...
```

or

```
```bash
```

```
kill 1234
```

```
...
```

`sleep` - Create a Timed Process

Description: Pauses execution for a specified number of seconds. Useful for testing foreground/background behavior.

Example:

```
```bash
```

```
sleep 30
```

```
...
```

## Lab Tasks

### Task 1: Foreground Process

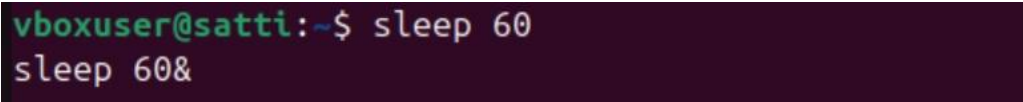
1. Open the terminal.

2. Run the following command:

```
``bash
```

```
sleep 60
```

```
...
```



```
vboxuser@satti:~$ sleep 60
sleep 60&
```

3. Observe that the terminal is unresponsive until the process completes.

## Task 2: Background Process

1. Run the same command in the background:

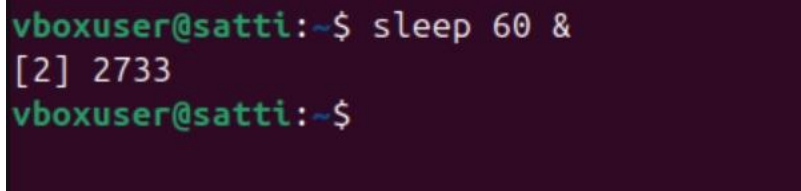
```
``bash
```

```
sleep 60 &
```

```
...
```

2. Note the job number and PID displayed.

3. Use `jobs` to see the background process.



```
vboxuser@satti:~$ sleep 60 &
[2] 2733
vboxuser@satti:~$
```

```
vboxuser@satti:~$ jobs
[1]- Done sleep 60
[2]+ Done sleep 60
vboxuser@satti:~$
```

### Task 3: Process Listing

1. Use `ps` to list all running processes.
2. Use `ps -f` to see detailed information.

```
vboxuser@satti:~$ ps
 PID TTY TIME CMD
 2720 pts/0 00:00:00 bash
 2747 pts/0 00:00:00 ps
vboxuser@satti:~$ ps -f
UID PID PPID C STIME TTY TIME CMD
vboxuser 2720 2713 0 10:33 pts/0 00:00:00 bash
vboxuser 2748 2720 99 10:38 pts/0 00:00:00 ps -f
```

### Task 4: Process Management

1. Start a process in the background:

```
``bash
```

```
sleep 120 &
```

```
...
```

2. Bring it to the foreground using `fg %job\_number`.
3. Suspend it with `Ctrl + Z`.
4. Resume it in the background with `bg %job\_number`.

```
vboxuser@satti:~$ sleep 120 &
[1] 2750
vboxuser@satti:~$ fg %job_number
bash: fg: %job_number: no such job
vboxuser@satti:~$ bg %job_number
bash: bg: %job_number: no such job
```

## Conclusion

In this lab, we learned how to:

- Differentiate between foreground and background processes.
- Use commands like `ps`, `jobs`, `fg`, `bg`, and `kill` to manage processes.
- Understand the relationship between parent and child processes.

These skills are essential for effective system administration and process control in UNIX-like operating systems.

## A-Z Index of the **Bash** command line for Linux

**alias** Create an alias

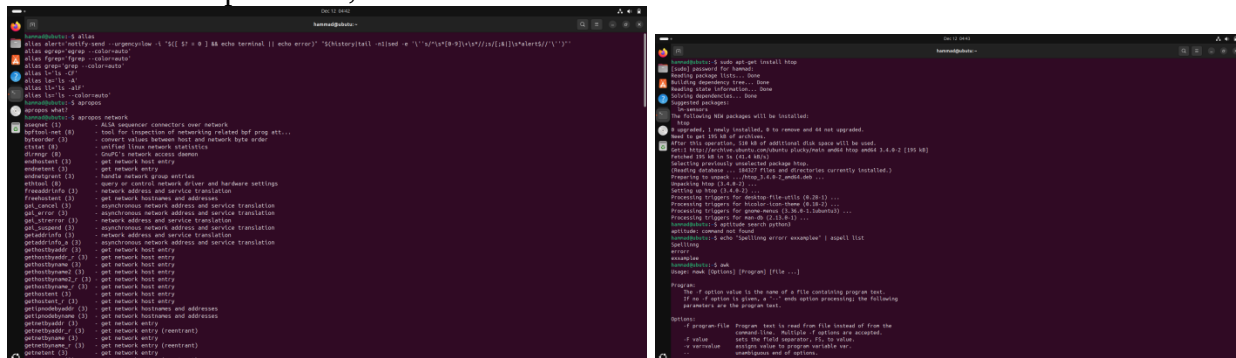
**apropos** Search Help manual pages (man -k)

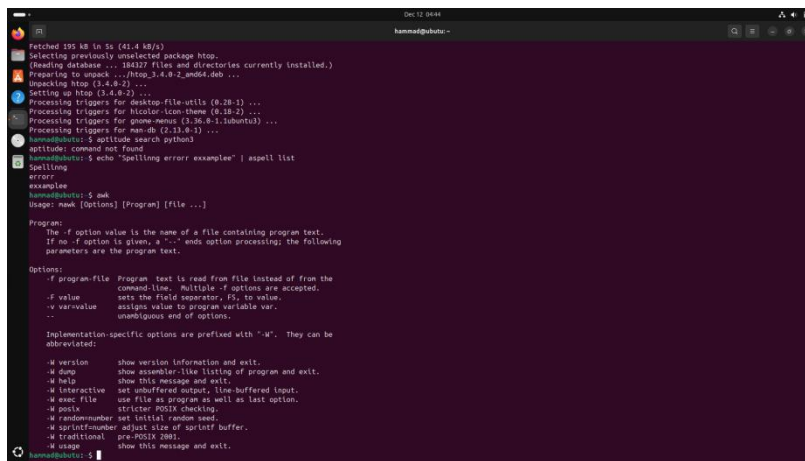
**apt-get** Search for and install software packages (Debian/Ubuntu)

**aptitude** Search for and install software packages (Debian/Ubuntu)

**aspell** Spell Checker

**awk** Find and Replace text, database sort/validate/index





**basename** Strip directory and suffix fromfilenames

## bash GNU Bourne-Again SHell

## bc Arbitrary precision calculator language

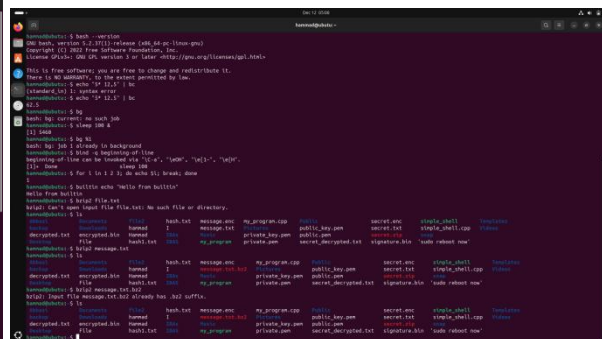
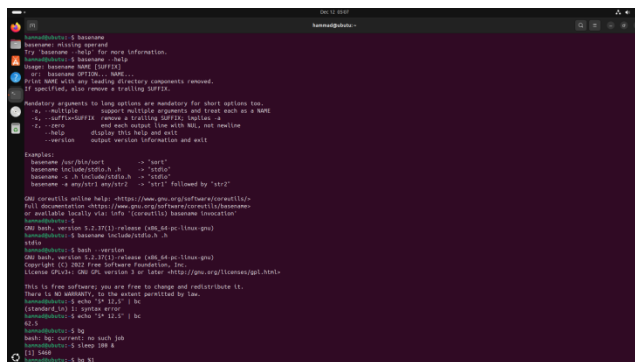
**bg** Send to background

**bind** Set or display readline key and function bindings

**break** Exit from a loop

**builtin** Run a shell builtin

**bzip2** Compress or decompress named file(s)



## cal Display a calendar

**case** Conditionally perform a command

**cat** Concatenate and print (display) the content of files

cd Change Directory

## cfdisk Partition table manipulator for Linux

## chattr Change file attributes on a Linux filesystem

chgrp Change group ownership

**chmod** Change access permissions

**chown** Change file owner and group

**chroot** Run a command with a different rootdirectory

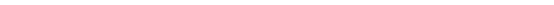
## chkconfig System services(runlevel)

**cksum** Print CRC checksum and bytecounts

**clear** Clear terminal screen

**cmp** Compare two files



[illegible]

**date** Display or change the date & time  
**dc** Desk Calculator  
**dd** Convert and copy a file, write disk headers, bootrecords  
**ddrescue** Data recovery tool  
**declare** Declare variables and give them attributes•  
**df** Display free disk space  
**diff** Display the differences between two files  
**diff3** Show differences among three files  
**dig** DNS lookup  
**dir** Briefly list directory contents  
**dircolors** Colour setup for 'ls'  
**dirname** Convert a full pathname to just a path  
**dirs** Display list of remembered directories  
**dmesg** Print kernel & driver messages  
**du** Estimate file space usage

```

mntazir@ubuntu: ~
d=00;90:*.*.pkg=tmp=00;90:*.*.old=00;90:*.*.orig=00;90:*.*.part=00;90:*.*.ra=00;90:*.*.rpmnew=00;90:*.*.rpmorig=00;90:*.*.rpmsave=00;90:*.*.sup=00;90:*.*.tmp=00;90:*.*.ucf-dist=00;90:*.*.ucf-new=00;90:*.*.ucf-old=00;90:*.*;
export L5_COLORS
mntazir@ubuntu: ~$ dirname /home/user/file.txt
/home/user
mntazir@ubuntu: ~$ df -h
Filesystem Size Used Avail Use% Mounted on
tmpfs 446M 1.0M 444M 1% /run
/dev/sda2 25G 9.2G 15G 40% /
tmpfs 2.0G 0 2.0G 0% /dev/shm
tmpfs 5.0M 8.0K 5.0M 1% /run/lock
tmpfs 2.0G 9.0M 2.0G 1% /tmp
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-resolved.service
tmpfs 446M 136K 445M 1% /run/user/1000
/dev/sr0 5.0G 5.0G 0 100% /media/mntazir/Ubuntu 25.04 amd64
mntazir@ubuntu: ~$ du -h
4.0K ./Music
12K ./local/state/wireplumber
20K ./local/state
76K ./local/share/gvfs-metadata
12K ./local/share/keyrings
4.0K ./local/share/backgrounds
4.0K ./local/share/applications
20K ./local/share/gnome-shell
4.0K ./local/share/ibus-table
4.0K ./local/share/Trash/files/DigitalSignatureLab
4.0K ./local/share/Trash/files/Project
2.0M ./local/share/Trash/files
34K ./local/share/Trash/info
mntazir@ubuntu: ~$ date
Fri Dec 5 04:29:31 AM UTC 2025
mntazir@ubuntu: ~$ df -h
Filesystem Size Used Avail Use% Mounted on
tmpfs 446M 1.0M 444M 1% /run
/dev/sda2 25G 7.6G 16G 33% /
tmpfs 2.0G 0 2.0G 0% /dev/shm
tmpfs 5.0M 8.0K 5.0M 1% /run/lock
tmpfs 2.0G 16K 2.0G 1% /tmp
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-resolved.service
tmpfs 446M 128K 445M 1% /run/user/1000
/dev/sr0 5.0G 5.0G 0 100% /media/mntazir/Ubuntu 25.04 amd64
mntazir@ubuntu: ~$ du -h
4.0K ./Music
12K ./local/state/wireplumber
16K ./local/state
76K ./local/share/gvfs-metadata
12K ./local/share/keyrings
4.0K ./local/share/backgrounds
4.0K ./local/share/applications
20K ./local/share/gnome-shell
4.0K ./local/share/ibus-table
4.0K ./local/share/Trash/files/DigitalSignatureLab
12K ./local/share/Trash/files
12K ./local/share/Trash/info
20K ./local/share/Trash
332K ./local/share/nautilus/tags
4.0K ./local/share/nautilus/scripts
140K ./local/share/nautilus
4.0K ./local/share/flatpak/db
0.0K ./local/share/flatpak
4.0K ./local/share/icc
4.0K ./local/share/org.gnome.TextEditor/drafts
14K ./local/share/org.gnome.TextEditor

```



5. **eval** — Evaluate several commands or arguments.

cmd="ls"; eval \$cmd

6. **exec** — Execute a command by replacing the shell.

exec ls

7. **exit** — Exit the shell.

exit

8. **expect** — Automate interactive terminal applications.

expect -c 'spawn ftp localhost; expect "Name"; send "user\r"'

9. **expand** — Convert tabs to spaces.

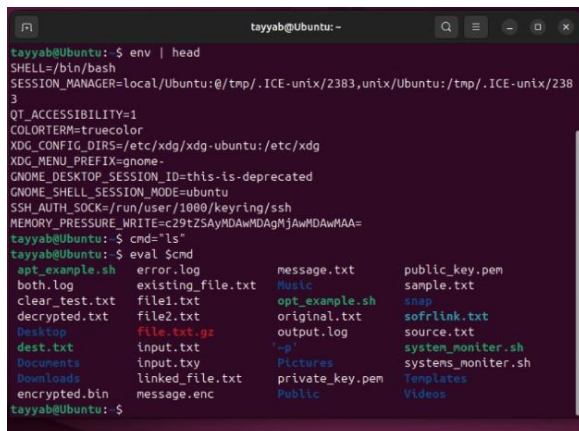
expand file.txt

10. **export** — Set an environment variable.

export MYVAR=10

11. **expr** — Evaluate expressions.

expr 10 + 5

A terminal window titled 'tayyab@Ubuntu: ~' showing the output of 'env | head' and 'eval \$cmd'. The 'env | head' command lists various system and user environment variables. The 'eval \$cmd' command runs 'ls', displaying a directory listing of files and folders in the current directory, including 'apt\_example.sh', 'both.log', 'clear\_test.txt', 'decrypted.txt', 'Desktop', 'dest.txt', 'Documents', 'Downloads', 'encrypted.bin', 'error.log', 'existing\_file.txt', 'file1.txt', 'file2.txt', 'file.txt.gz', 'input.txt', 'input.txy', 'linked\_file.txt', 'message.enc', 'message.txt', 'Music', 'opt\_example.sh', 'original.txt', 'output.log', 'private\_key.pem', 'Public', 'public\_key.pem', 'sample.txt', 'snap', 'sofmlink.txt', 'source.txt', 'system\_monitor.sh', 'systems\_monitor.sh', 'Templates', and 'Videos'.

12. **false** — Do nothing and return failure status.

false

13. **fdformat** — Low-level format a floppy disk.

sudo fdformat /dev/fd0

14. **fdisk** — Partition table manipulator for Linux.

sudo fdisk -l

15. **fg** — Send background job to foreground.

fg %1

16. **fgrep** — Search file(s) for a fixed string.

echo "hello world" | fgrep world

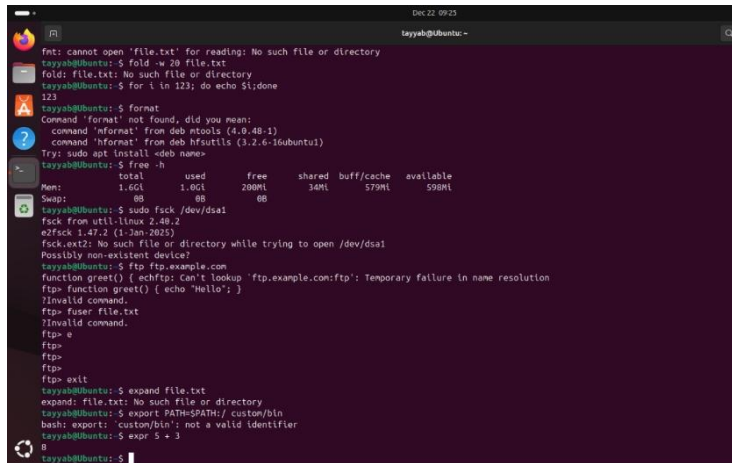
17. **file** — Determine file type.

file test.txt

18. **find** — Search for files matching criteria.

find . -name "\*.txt"

Screenshot :



```
Dec 22, 09:23
tayyab@Ubuntu:~$
fmt: cannot open 'file.txt' for reading: No such file or directory
tayyab@Ubuntu:~$ fold -w 20 file.txt
fold: file.txt: No such file or directory
tayyab@Ubuntu:~$ for i in 123; do echo $i;done
123
tayyab@Ubuntu:~$ format
Command 'format' not found, did you mean:
 command 'mformat' from deb nttools (4.0.4b-1)
 command 'hformat' from deb hfsutils (3.2.6-16ubuntu1)
Try: sudo apt install <deb name>
tayyab@Ubuntu:~$ free -h
 total used free shared buff/cache available
Mem: 1.6Gi 1.0Gi 200Mi 34Mi 579Mi 598Mi
Swap: 0B 0B 0B
tayyab@Ubuntu:~$ sudo fsck /dev/sda1
fsck from util-linux 2.40.2
e2fsck 1.47.2 (1-Jan-2025)
fsck.ext2: No such file or directory while trying to open /dev/sda1
Possibly non-existent device?
tayyab@Ubuntu:~$ ftp ftp.example.com
function greet() { echftp: Can't lookup 'ftp.example.com:ftp': Temporary failure in name resolution
ftp> function greet() { echo "Hello"; }
?Invalid command.
ftp> fuser file.txt
?Invalid command.
ftp> e
ftp>
ftp>
ftp>
ftp> exit
tayyab@Ubuntu:~$ expand file.txt
expand: file.txt: No such file or directory
tayyab@Ubuntu:~$ export PATH=$PATH:/ custom/bin
bash: export: 'custom/bin': not a valid identifier
tayyab@Ubuntu:~$ expr 5 + 3
8
tayyab@Ubuntu:~$
```

19. **fmt** — Reformat paragraph text.

fmt file.txt

20. **fold** — Wrap text to a specified width.

fold -w 20 file.txt

21. **for** — Loop through values and execute commands.

for i in 1 2 3; do echo \$i; done

22. **format** — Format disks or tapes.

format

23. **free** — Display memory usage.

free -h

24. **fsck** — Check and repair file system.

sudo fsck /dev/sda1

25. **ftp** — Transfer files using FTP protocol.

ftp ftp.example.com

26. **function** — Define a shell function.

myfunc(){ echo "Hello"; }

myfunc

27. **fuser** — Identify process using a file.

fuser file.txt

Screenshot :

```
Dec 22 09:22
tayyab@Ubuntu: ~
tayyab@Ubuntu:~$ expect -c 'spawn ftp localhost; expect "Name"; send "user\r"'
Command 'expect' not found, but can be installed with:
sudo snap install expect
tayyab@Ubuntu:~$ expand file.txt
expand: file.txt: No such file or directory
tayyab@Ubuntu:~$ false
tayyab@Ubuntu:~$ sudo fdformat /dev/fd0
[sudo] password for tayyab:
sudo: fdformat: command not found
tayyab@Ubuntu:~$ sudo fdisk -l
Disk /dev/loop0: 73.95 MiB, 77545472 bytes, 151456 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop1: 4 KiB, 4096 bytes, 8 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop2: 73.91 MiB, 77500416 bytes, 151368 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop3: 11.75 MiB, 12320768 bytes, 24064 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop4: 22.57 MiB, 23666688 bytes, 46224 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

1. **gawk** — Pattern scanning and text processing.  
echo "Hello World" | gawk '{print \$1}'
2. **getopts** — Parse command-line options in shell scripts.  
while getopts ":a" opt; do case \$opt in a) echo "Option -a";; esac; done <<(echo "-a")
3. **grep** — Search text using patterns.  
echo -e "apple\nbanana" | grep banana
4. **groupadd** — Create a new user group.  
sudo groupadd testgroup\_lab

**Screenshot :**



```
tayyab@Ubuntu:~$ sudo apt install gawk net-tools plocate lsof
gawk is already the newest version (1:5.2.1-2build3).
net-tools is already the newest version (2.10-1.1ubuntu1.25.04.4).
plocate is already the newest version (1.1.19-2ubuntu2).
lsof is already the newest version (4.99.4dfsg-2).
Summary:
 Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
tayyab@Ubuntu:~$ sudo updatedb
tayyab@Ubuntu:~$ echo "Hello World" | gawk '{print $1}'
Hello
tayyab@Ubuntu:~$ while getopts ":a" opt; do
> case $opt in
> a) echo "Option -a was triggered!";;
> esac
> done < <{(echo "-a")}
getopts:a: command not found
tayyab@Ubuntu:~$ while getopts ":a" opt; do case $opt in a) echo "Option -a was triggered!";; esac; done < <{(echo "-a")}
tayyab@Ubuntu:~$ echo -e "apple\nbanana\nberry" | grep
Usage: grep [OPTION]... PATTERNS [FILE]...
Try 'grep --help' for more information.
tayyab@Ubuntu:~$ echo -e "apple\nbanana\nberry" | grep "banana"
banana
tayyab@Ubuntu:~$ sudo groupadd testgroup_lab
tayyab@Ubuntu:~$ tail /etc/group | grep testgroup_lab
testgroup_lab:x:1001:
tayyab@Ubuntu:~$ sudo groupmod -n new_testgroup
Usage: groupmod [options] GROUP

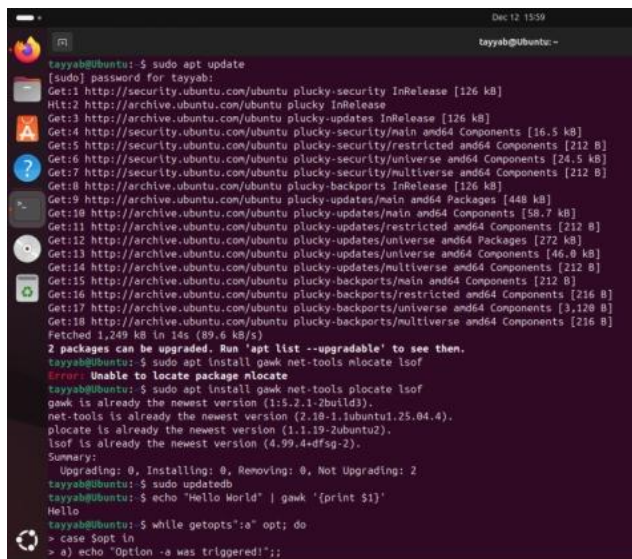
Options:
 -a, --append append the users mentioned by -U option to the group
 without removing existing user members
 -g, --gid GID change the group ID to GID
 -h, --help display this help message and exit
 -n, --new-name NEW_GROUP change the name to NEW_GROUP
 -o, --non-unique allow to use a duplicate (non-unique) GID
 -p, --password PASSWORD change the password to this (encrypted)
 PASSWORD
```

## Installation / Update Space:

sudo apt update

sudo apt install gawk

## Screenshot :



## 5. **groupmod** — Modify an existing group.

sudo groupmod -n new\_testgroup testgroup\_lab

## 6. **groups** — Show groups of current user.

groups

## 7. **groupdel** — Delete a group.

sudo groupdel new\_testgroup

## 8. **gzip** — Compress files.

touch file.txt && gzip file.txt

## 9. **hash** — Display command hash table.

hash

Screenshot):

```
Options:
-a, --append append the users mentioned by -U option to the group
 without removing existing user members
-g, --gid GID change the group ID to GID
-h, --help display this help message and exit
-n, --new-name NEW_GROUP change the name to NEW_GROUP
-o, --non-unique allow to use a duplicate (non-unique) GID
-p, --password PASSWORD change the password to this (encrypted)
 PASSWORD
-R, --root CHROOT_DIR directory to chroot into
-P, --prefix PREFIX_DIR prefix directory where are located the /etc/* files
-U, --users USERS list of user members of this group

tayyab@Ubuntu:~$ testgroup_lab
testgroup_lab: command not found
tayyab@Ubuntu:~$ sudo groupmod -n new_testgroup testgroup_lab
tayyab@Ubuntu:~$ tail /etc/group | grep new_testgroup
new_testgroup:x:1001:
tayyab@Ubuntu:~$ groups
tayyab sudo vboxsf
tayyab@Ubuntu:~$ sudo groupdel new_testgroup
tayyab@Ubuntu:~$ touch file.txt
tayyab@Ubuntu:~$ gzip file.txt
tayyab@Ubuntu:~$ ls file.txt.gz
file.txt.gz
tayyab@Ubuntu:~$ hash
hits command
1 /usr/bin/groups
1 /usr/bin/ls
1 /usr/bin/gzip
8 /usr/bin/sudo
1 /usr/bin/touch
tayyab@Ubuntu:~$ head -n 3 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
```

10. **head** — Show first lines of a file.

head -n 3 /etc/passwd

11. **help** — Get help for shell built-ins.

help cd

Screenshot :

```
1 /usr/bin/ls
1 /usr/bin/gzip
8 /usr/bin/sudo
1 /usr/bin/touch
tayyab@Ubuntu:~$ head -n 3 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
tayyab@Ubuntu:~$ help cd
cd: cd [-L|[-P [-e]] [-@]] [dir]
Change the shell working directory.

Change the current directory to DIR. The default DIR is the value of the
HOME shell variable. If DIR is ".", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing
DIR. Alternative directory names in CDPATH are separated by a colon (:).
A null directory name is the same as the current directory. If DIR begins
with a slash (/), then CDPATH is not used.

If the directory is not found, and the shell option 'cdable_vars' is set,
the word is assumed to be a variable name. If that variable has a value,
its value is used for DIR.

Options:
-L force symbolic links to be followed: resolve symbolic
links in DIR after processing instances of '..'
-P use the physical directory structure without following
symbolic links: resolve symbolic links in DIR before
processing instances of '..'
-e if the -P option is supplied, and the current working
directory cannot be determined successfully, exit with
a non-zero status
-@ on systems that support it, present a file with extended
attributes as a directory containing the file attributes

The default is to follow symbolic links, as if '-L' were specified.
```



---

12. **history** — View command history.

history | tail -n 5

13. **hostname** — Show system hostname.

hostname

14. **iconv** — Character encoding conversion.

echo "test" | iconv -f UTF-8 -t ASCII

15. **id** — Display user and group IDs.

id

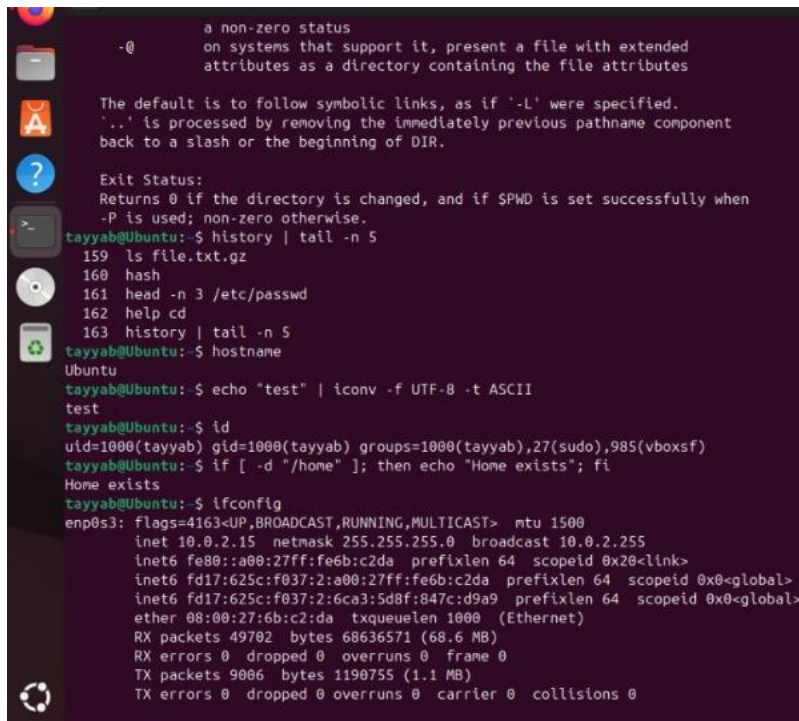
16. **if** — Conditional execution in shell scripts.

if [ -d /home ]; then echo "Home exists"; fi

17. **ifconfig** — Display network interface configuration.

ifconfig

**Screenshot :**

A screenshot of a terminal window with a dark background and light-colored text. The terminal shows a series of commands and their outputs. At the top, there's a man page for 'ls' showing options like '-l', '-R', and '-d'. Below that, the user runs 'history | tail -n 5', which shows the last five commands: 'ls file.txt.gz', 'hash', 'head -n 3 /etc/passwd', 'help cd', and 'history | tail -n 5'. Then, the user runs 'hostname', which outputs 'Ubuntu'. Next, 'echo "test" | iconv -f UTF-8 -t ASCII' outputs 'test'. Then, 'id' outputs 'uid=1000(tayyab) gid=1000(tayyab) groups=1000(tayyab),27(sudo),985(vboxsf)'. Then, 'if [ -d "/home" ]; then echo "Home exists"; fi' outputs 'Home exists'. Finally, 'ifconfig' outputs detailed network information for the 'enp0s3' interface, including IP address, netmask, broadcast, and various statistics.

```
-@ a non-zero status
 on systems that support it, present a file with extended
 attributes as a directory containing the file attributes

The default is to follow symbolic links, as if '-L' were specified.
'..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when
-P is used; non-zero otherwise.
tayyab@Ubuntu:~$ history | tail -n 5
159 ls file.txt.gz
160 hash
161 head -n 3 /etc/passwd
162 help cd
163 history | tail -n 5
tayyab@Ubuntu:~$ hostname
Ubuntu
tayyab@Ubuntu:~$ echo "test" | iconv -f UTF-8 -t ASCII
test
tayyab@Ubuntu:~$ id
uid=1000(tayyab) gid=1000(tayyab) groups=1000(tayyab),27(sudo),985(vboxsf)
tayyab@Ubuntu:~$ if [-d "/home"]; then echo "Home exists"; fi
Home exists
tayyab@Ubuntu:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
 inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
 inet6 fe80::a00:27ff:fe6b:c2da prefixlen 64 scopeid 0x20<link>
 inet6 fd17:625c:f037:2:a00:27ff:fe6b:c2da prefixlen 64 scopeid 0x0<global>
 inet6 fd17:625c:f037:2:6ca3:5d8f:847c:d9a9 prefixlen 64 scopeid 0x0<global>
 ether 08:00:27:6b:c2:da txqueuelen 1000 (Ethernet)
 RX packets 49702 bytes 68636571 (68.6 MB)
 RX errors 0 dropped 0 overruns 0 frame 0
 TX packets 9006 bytes 1190755 (1.1 MB)
 TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

**Installation / Update Space:**

sudo apt install net-tools

---

18. **install** — Copy files and set ownership and permissions.

touch source.txt

install -m 755 source.txt dest.txt

ls -l dest.txt

19. **ip** — Show and manage network interfaces.

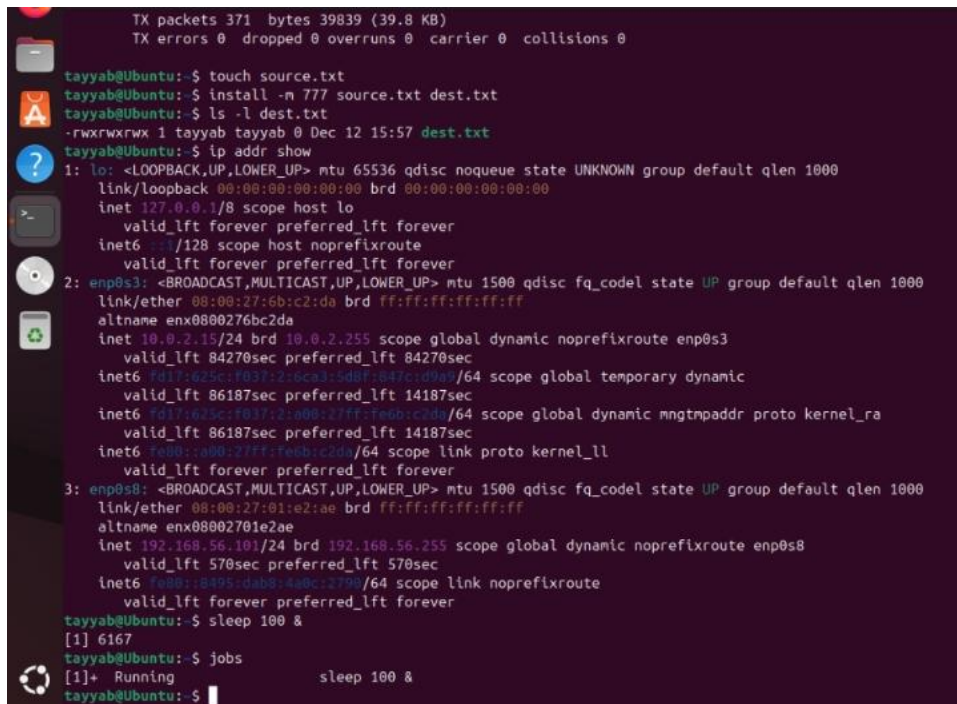
ip addr show

20. **jobs** — Display background jobs.

sleep 60 &

jobs

Screenshot :

A screenshot of a terminal window on a Linux system. The terminal shows the output of several commands. At the top, it displays network statistics: 'TX packets 371 bytes 39839 (39.8 KB)' and 'TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0'. Below this, the user runs 'touch source.txt', 'install -m 777 source.txt dest.txt', and 'ls -l dest.txt', which shows the file permissions as '-rwxrwxrwx 1 tayyab tayyab 0 Dec 12 15:57 dest.txt'. Then, the user runs 'ip addr show', which displays detailed information for three network interfaces: 'lo' (loopback), 'enp0s3' (ethernet), and 'enp0s8' (ethernet). The output for each interface includes its state, MTU, queue discipline, and various IP addresses and MAC addresses. Finally, the user runs 'sleep 100 &' and 'jobs', which shows the job as '[1] 6167 sleep 100 &' in a running state. The terminal prompt is 'tayyab@Ubuntu: \$'.

21. **join** — Join lines of two files on a common field.

echo "1 A" > f1.txt

echo "1 B" > f2.txt

join f1.txt f2.txt

22. **kill** — Terminate a process using PID.

sleep 100 &

kill \$!

23. **killall** — Terminate processes by name.

sleep 100 &

killall sleep

24. **let** — Perform arithmetic operations.

let sum=5+5

echo \$sum

25. **link** — Create a hard link to a file.

touch original.txt

link original.txt hardlink.txt

ls -l hardlink.txt

26. **ln** — Create a symbolic link.

ln -s original.txt softlink.txt

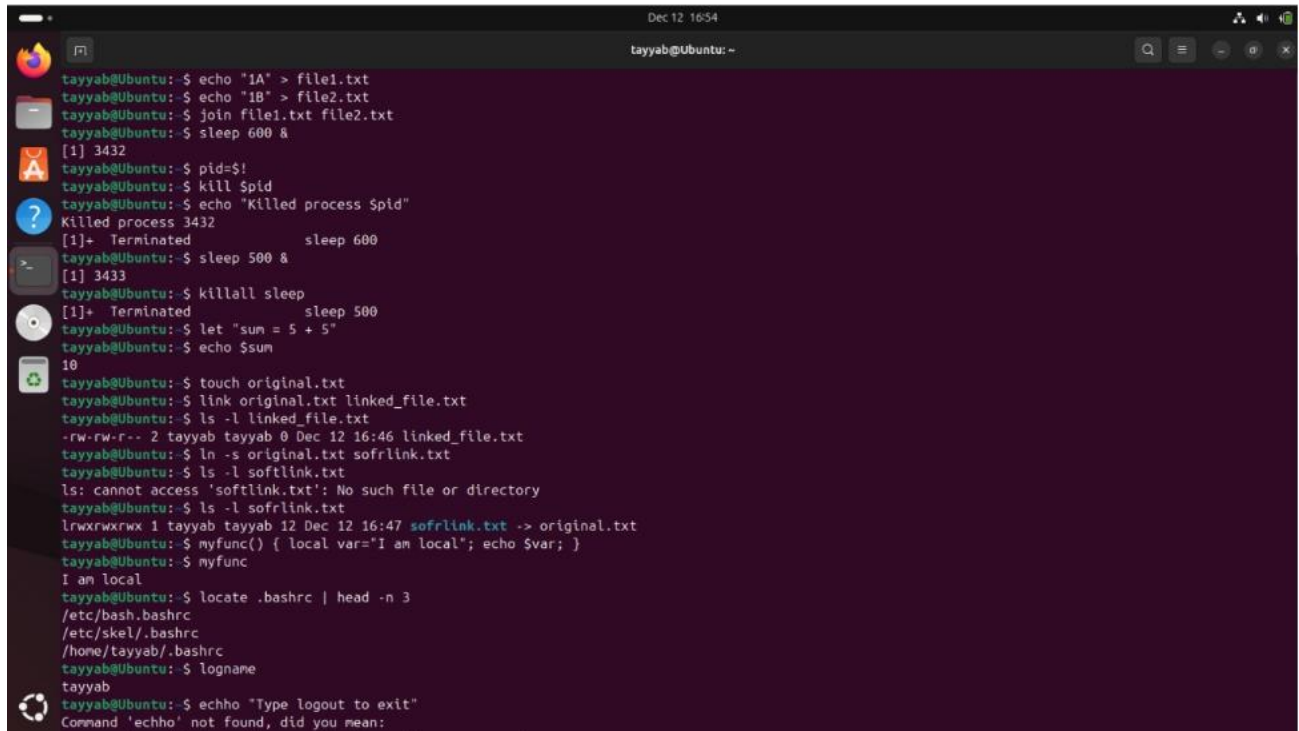
ls -l softlink.txt

27. **local** — Declare local variables in functions.

```
myfunc(){ local x=10; echo $x; }
```

```
myfunc
```

**Screenshot :**



```
tayyab@Ubuntu: ~$ echo "1A" > file1.txt
tayyab@Ubuntu: ~$ echo "1B" > file2.txt
tayyab@Ubuntu: ~$ join file1.txt file2.txt
tayyab@Ubuntu: ~$ sleep 600 &
[1] 3432
tayyab@Ubuntu: ~$ pid=$!
tayyab@Ubuntu: ~$ kill $pid
tayyab@Ubuntu: ~$ echo "Killed process $pid"
Killed process 3432
[1]+ Terminated sleep 600
tayyab@Ubuntu: ~$ sleep 500 &
[1] 3433
tayyab@Ubuntu: ~$ killall sleep
[1]+ Terminated sleep 500
tayyab@Ubuntu: ~$ let "sum = 5 + 5"
tayyab@Ubuntu: ~$ echo $sum
10
tayyab@Ubuntu: ~$ touch original.txt
tayyab@Ubuntu: ~$ link original.txt linked_file.txt
tayyab@Ubuntu: ~$ ls -l linked_file.txt
-rw-rw-r-- 2 tayyab tayyab 0 Dec 12 16:46 linked_file.txt
tayyab@Ubuntu: ~$ ln -s original.txt softlink.txt
tayyab@Ubuntu: ~$ ls -l softlink.txt
ls: cannot access 'softlink.txt': No such file or directory
tayyab@Ubuntu: ~$ ls -l softlink.txt
lrwxrwxrwx 1 tayyab tayyab 12 Dec 12 16:47 softlink.txt -> original.txt
tayyab@Ubuntu: ~$ myfunc() { local var="I am local"; echo $var; }
tayyab@Ubuntu: ~$ myfunc
I am local
tayyab@Ubuntu: ~$ locate .bashrc | head -n 3
/etc/bash.bashrc
/etc/skel/.bashrc
/home/tayyab/.bashrc
tayyab@Ubuntu: ~$ logname
tayyab
tayyab@Ubuntu: ~$ echho "Type logout to exit"
Command 'echho' not found, did you mean:
```

28. **locate** — Search for files using database.

```
locate .bashrc | head
```

29. **logname** — Display login name of the user.

```
logname
```

30. **logout** — Exit login shell.

```
echo "Type logout to exit shell"
```

31. **look** — Display dictionary words starting with pattern.

```
look bas /usr/share/dict/words | head
```

32. **lpc** — Printer control program.

```
/usr/sbin/lpc status
```

33. **lpr** — Send file to printer.

```
echo "Test print" | lpr
```

34. **lprint** — Print files (lp alternative).

```
lp --help
```

**Screenshot :**

```

I am local
tayyab@Ubuntu:~$ locate .bashrc | head -n 3
/etc/bash.bashrc
/etc/skel/.bashrc
/home/tayyab/.bashrc
tayyab@Ubuntu:~$ logname
tayyab
tayyab@Ubuntu:~$ echo "Type logout to exit"
Command 'echo' not found, did you mean:
 command 'echo' from deb coreutils (9.5-1ubuntu1.25.04.2)
Try: sudo apt install <deb name>
tayyab@Ubuntu:~$ echo "Type logout to exit"
Type logout to exit
tayyab@Ubuntu:~$ look "bas" /usr/share/dict/words | head -n 5
tayyab@Ubuntu:~$ /usr/sbin/lpc status
tayyab@Ubuntu:~$ echo "test print" | lpr
lpr: Error - No default destination.
tayyab@Ubuntu:~$ echo "test print" | lpr
Command 'lpr' not found, did you mean:
 command 'lpr' from deb cups-bsd (2.4.12-0ubuntu1.6)
 command 'pr' from deb coreutils (9.5-1ubuntu1.25.04.2)
Try: sudo apt install <deb name>
tayyab@Ubuntu:~$ lp --help
Usage: lp [options] [...] [file(s)]
 lp [options] -i id
Options:
 -c Make a copy of the print file(s)
 -d destination Specify the destination
 -E Encrypt the connection to the server
 -h server[:port] Connect to the named server and port
 -H HH:MM Hold the job until the specified UTC time
 -H hold Hold the job until released/resumed
 -H immediate Print the job as soon as possible
 -H restart Reprint the job
 -H resume Resume a held job
 -i id Specify an existing job ID to modify
 -m Send an email notification when the job completes

```

### 35. **lpq** — Display printer queue status.

lpq

### 36. **ls** — List directory contents.

ls -l

**Screenshot :**

```

tayyab@Ubuntu:~$ ls -l
total 120
-rwxrwxr-x 1 tayyab tayyab 233 Dec 12 05:12 apt_example.sh
-rw-rw-r-- 1 tayyab tayyab 78 Oct 21 02:46 both.log
-rw-rw-r-- 1 tayyab tayyab 0 Oct 21 02:54 clear_test.txt
-rw-rw-r-- 1 tayyab tayyab 57 Oct 31 09:31 decrypted.txt
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 Desktop
-rwxrwxrwx 1 tayyab tayyab 0 Dec 12 15:57 dest.txt
drwxr-xr-x 3 tayyab tayyab 4096 Oct 31 09:38 Documents
drwxr-xr-x 2 tayyab tayyab 4096 Dec 12 10:01 Downloads
-rw-rw-r-- 1 tayyab tayyab 256 Oct 31 09:29 encrypted.bin
-rw-rw-r-- 1 tayyab tayyab 66 Oct 21 02:43 error.log
-rw-rw-r-- 1 tayyab tayyab 5 Oct 21 02:50 existing_file.txt
-rw-rw-r-- 1 tayyab tayyab 3 Dec 12 16:44 file1.txt
-rw-rw-r-- 1 tayyab tayyab 3 Dec 12 16:44 file2.txt
-rw-rw-r-- 1 tayyab tayyab 46 Dec 12 15:51 file.txt.gz
-rw-rw-r-- 1 tayyab tayyab 18 Oct 21 02:52 input.txt
-rw-rw-r-- 1 tayyab tayyab 6 Oct 21 02:51 input.txy
-rw-rw-r-- 2 tayyab tayyab 0 Dec 12 16:46 linked_file.txt
-rw-rw-r-- 1 tayyab tayyab 80 Oct 17 13:18 message.enc
-rw-rw-r-- 1 tayyab tayyab 57 Oct 31 09:25 message.txt
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 Music
-rwxrwxr-x 1 tayyab tayyab 206 Dec 12 05:20 opt_example.sh
-rw-rw-r-- 2 tayyab tayyab 0 Dec 12 16:46 original.txt
-rw-rw-r-- 1 tayyab tayyab 12 Oct 21 02:43 output.log
drwxrwxr-x 2 tayyab tayyab 4096 Oct 31 09:38 Pictures
drwxr-xr-x 1 tayyab tayyab 1708 Oct 31 09:21 private_key.pem
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 Public
-rw-rw-r-- 1 tayyab tayyab 451 Oct 31 09:22 public_key.pem
-rw-rw-r-- 1 tayyab tayyab 37 Dec 12 05:04 sample.txt
drwx----- 7 tayyab tayyab 4096 Nov 28 06:53 snap

```



### 37. Isuf — List open files and associated processes.

Isuf | head -n 5

```
Dec 12 16:56
tayyab@Ubuntu: ~
drwxr-xr-x 3 tayyab tayyab 4096 Oct 31 09:38 Documents
drwxr-xr-x 2 tayyab tayyab 4096 Dec 12 10:01 Downloads
-rw-rw-r-- 1 tayyab tayyab 256 Oct 31 09:29 encrypted.bin
-rw-rw-r-- 1 tayyab tayyab 66 Oct 21 02:43 error.log
-rw-rw-r-- 1 tayyab tayyab 5 Oct 21 02:50 existing_file.txt
-rw-rw-r-- 1 tayyab tayyab 3 Dec 12 16:44 file1.txt
-rw-rw-r-- 1 tayyab tayyab 3 Dec 12 16:44 file2.txt
-rw-rw-r-- 1 tayyab tayyab 46 Dec 12 15:51 file.txt.gz
-rw-rw-r-- 1 tayyab tayyab 18 Oct 21 02:52 input.txt
-rw-rw-r-- 1 tayyab tayyab 6 Oct 21 02:51 input.txy
-rw-rw-r-- 2 tayyab tayyab 0 Dec 12 16:46 linked_file.txt
-rw-rw-r-- 1 tayyab tayyab 80 Oct 17 13:18 message.enc
-rw-rw-r-- 1 tayyab tayyab 57 Oct 31 09:25 message.txt
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 music
-rw-rw-r-- 1 tayyab tayyab 206 Dec 12 05:20 opt_example.sh
-rw-rw-r-- 2 tayyab tayyab 0 Dec 12 16:46 original.txt
-rw-rw-r-- 1 tayyab tayyab 12 Oct 21 02:43 output.log
drwxr-xr-x 2 tayyab tayyab 4096 Oct 31 09:38 "p"
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 Pictures
-rw-rw-r-- 1 tayyab tayyab 1708 Oct 31 09:21 private_key.pem
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 Public
-rw-rw-r-- 1 tayyab tayyab 451 Oct 31 09:22 public_key.pem
-rw-rw-r-- 1 tayyab tayyab 37 Dec 12 05:04 sample.txt
drwx----- 7 tayyab tayyab 4096 Nov 28 06:53 snap
lrwxrwxrwx 1 tayyab tayyab 12 Dec 12 16:47 sofflink.txt -> original.txt
-rw-rw-r-- 1 tayyab tayyab 0 Dec 12 15:57 source.txt
-rwxrwxr-x 1 tayyab tayyab 902 Nov 3 16:12 system_monitor.sh
-rw-rw-r-- 1 tayyab tayyab 759 Nov 3 15:44 systems_monitor.sh
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 templates
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41 Videos
tayyab@Ubuntu: ~$ lsuf | head -n 5
COMMAND PID TID TASKCMD USER FD TYPE DEVICE SIZE/OFF NODE NAME
systemd 1 root cwd unknown /proc/1/cwd (readlink: Permission denied)
systemd 1 root rtd unknown /proc/1/root (readlink: Permission denied)
systemd 1 root txt unknown /proc/1/exe (readlink: Permission denied)
systemd 1 root NOFD 0000 /proc/1/fd (opendir: Permission denied)
tayyab@Ubuntu: ~$
```

### MMV:

```
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
vboxuser@satti:~$ mvv
Usage: mvv [-m|-x|-r|-c|-o|-a|-l|-s] [-h] [-d|-p] [-g|-t] [-v|-n] FROM TO
move/copy/link multiple files by wildcard patterns

The FROM pattern is a shell glob pattern, in which '*' stands for any number
of characters and '?' stands for a single character.

Use #[l|u]N in the TO pattern to get the string matched by the Nth
FROM pattern wildcard [lowercased|uppercased].

Patterns should be quoted on the command line.

--help Print help and exit
-V, --version Print version and exit
-h, --hidden treat dot files normally (default=off)
-D, --makedirs create non-existent directories (default=off)

Group: mode
Mode of operation
-m, --move move source file to target name
-x, --copydel copy source to target, then delete source
-r, --rename rename source to target in same directory
-c, --copy copy source to target, preserving source permissions
-o, --overwrite overwrite target with source, preserving target permissions
```

```

Group: mode
Mode of operation
-m, --move move source file to target name
-x, --copydel copy source to target, then delete source
-r, --rename rename source to target in same directory
-c, --copy copy source to target, preserving source permissions
-o, --overwrite overwrite target with source, preserving target permissions
-l, --hardlink link target name to source file
-s, --symlink symlink target name to source file

Group: delete
How to handle file deletions and overwrites
-d, --force perform file deletes and overwrites without confirmation
-p, --protect treat file deletes and overwrites as errors

Group: erroneous
How to handle erroneous actions
-g, --go skip any erroneous actions
-t, --terminate erroneous actions are treated as errors

Group: report
Reporting actions
-v, --verbose report all actions performed
-n, --dryrun only report which actions would be performed
boxuser@satti:~$

```

Mtools:

```

vboxuser@satti:~$ mtools
Supported commands:
mattrib, mbadbblocks, mcat, mcd, mcopy, mdel, mdeltree, mdir
mdoctorfat, mdu, mformat, minfo, mlabel, mmd, mmount, mpartition
mrd, mread, mmove, mren, mshowfat, mshortname, mtoolstest, mtype
mwrite, mzip

```

MMM

```

My traceroute [v0.95]
satti (::1) -> localhost (::1) 2025-12-20T13:27:59+0000
Keys: Help Display mode Restart statistics Order of fields quit

 Host Packets Pings
 Loss% Snt Last Avg Best Wrst StDev
1. localhost6.localdomain6 0.0% 13 0.1 0.1 0.0 0.2 0.0

```

MV:

```

vboxuser@satti:~$ mv --help
Usage: mv [OPTION]... [-t] SOURCE DEST
or: mv [OPTION]... SOURCE... DIRECTORY
or: mv [OPTION]... -t DIRECTORY SOURCE...
Rename SOURCE to DEST, or move SOURCE(s) to DIRECTORY.

Mandatory arguments to long options are mandatory for short options too.
 --backup[=CONTROL] make a backup of each existing destination file
 -b like --backup but does not accept an argument
 --debug explain how a file is copied. Implies -v
 -f, --force do not prompt before overwriting
 -i, --interactive prompt before overwrite
 -n, --no-clobber do not overwrite an existing file
If you specify more than one of -i, -f, -n, only the final one takes effect.
 --no-copy do not copy if renaming fails
 --strip-trailing-slashes remove any trailing slashes from each SOURCE argument
 -S, --suffix=SUFFIX override the usual backup suffix
 -t, --target-directory=DIRECTORY move all SOURCE arguments into DIRECTORY
 -T, --no-target-directory treat DEST as a normal file
 --update[=UPDATE] control which existing files are updated;
 UPDATE={all,none,older(default)}. See below
 equivalent to --update[=older]
 -u equivalent to --update[=older]
 -v, --verbose explain what is being done
 -Z, --context set SELinux security context of destination

```

## N COMMANDS:

Nc:

```
vboxuser@satti:~$ nc
usage: nc [-46CDdFhklNnrStUuvZz] [-I length] [-i interval] [-M ttl]
 [-m minttl] [-O length] [-P proxy_username] [-p source_port]
 [-q seconds] [-s sourceaddr] [-T keyword] [-V rtable] [-W recvlimit]
 [-w timeout] [-X proxy_protocol] [-x proxy_address[:port]]
 [destination] [port]
```

## \NETSTAT:

```
vboxuser@satti:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 satti:60406 ubuntu-archive-mir:http TIME_WAIT
udp 0 0 satti:bootpc _gateway:bootps ESTABLISHED

Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags Type State I-Node Path
unix 2 [] DGRAM 16579
unix 3 [] STREAM CONNECTED 16523
unix 3 [] STREAM CONNECTED 8578
unix 3 [] STREAM CONNECTED 8514
unix 3 [] STREAM CONNECTED 15597
unix 3 [] STREAM CONNECTED 15421
unix 3 [] STREAM CONNECTED 16760 /run/systemd/journal/
stdout
unix 3 [] STREAM CONNECTED 16716
unix 3 [] STREAM CONNECTED 16194
unix 3 [] STREAM CONNECTED 16605 /run/dbus/system_bus_
socket
unix 3 [] STREAM CONNECTED 8711
unix 3 [] STREAM CONNECTED 15412 /run/user/1000/bus
unix 3 [] STREAM CONNECTED 8782
unix 2 [] DGRAM 8732
unix 3 [] STREAM CONNECTED 16525 /run/user/1000/bus
unix 3 [] STREAM CONNECTED 15600
unix 3 [] STREAM CONNECTED 8515 /run/dbus/system_bus_
```

```
vboxuser@satti:~$ netstat
unix 3 [] STREAM CONNECTED 8782
unix 2 [] DGRAM 8732
unix 3 [] STREAM CONNECTED 16525 /run/user/1000/bus
unix 3 [] STREAM CONNECTED 15600
unix 3 [] STREAM CONNECTED 8515 /run/dbus/system_bus_
socket
unix 3 [] STREAM CONNECTED 16604
unix 3 [] STREAM CONNECTED 16589 /run/snapd-snap.socke
t
unix 3 [] STREAM CONNECTED 17151 /run/systemd/journal/
stdout
unix 3 [] STREAM CONNECTED 9997
unix 3 [] STREAM CONNECTED 36028
unix 3 [] STREAM CONNECTED 16526 /run/user/1000/bus
unix 3 [] STREAM CONNECTED 15405
unix 3 [] STREAM CONNECTED 13378 /run/dbus/system_bus_
socket
unix 3 [] STREAM CONNECTED 16608
unix 3 [] STREAM CONNECTED 8712 /run/systemd/journal/
stdout
unix 3 [] STREAM CONNECTED 16759
unix 3 [] STREAM CONNECTED 16588
unix 3 [] STREAM CONNECTED 16179 /run/user/1000/gvfsd/
socket-MSOMfI3e
unix 3 [] STREAM CONNECTED 13908 /run/dbus/system_bus_
```

unix	3	[ ]	STREAM	CONNECTED	13908	/run/dbus/system_bus_
socket						
unix	3	[ ]	STREAM	CONNECTED	16909	
unix	3	[ ]	STREAM	CONNECTED	16297	/run/user/1000/waylan
d-0						
unix	2	[ ]	DGRAM	CONNECTED	8781	
unix	3	[ ]	STREAM	CONNECTED	15408	
unix	3	[ ]	STREAM	CONNECTED	13907	
unix	3	[ ]	STREAM	CONNECTED	16786	/run/dbus/system_bus_
socket						
unix	3	[ ]	STREAM	CONNECTED	16761	
unix	3	[ ]	STREAM	CONNECTED	16592	
unix	3	[ ]	STREAM	CONNECTED	16195	/run/systemd/journal/
stdout						
unix	2	[ ]	DGRAM		8518	
unix	3	[ ]	STREAM	CONNECTED	16528	
unix	3	[ ]	STREAM	CONNECTED	16914	/run/user/1000/at-spi
/bus						
unix	3	[ ]	STREAM	CONNECTED	8668	/run/systemd/journal/
stdout						
unix	3	[ ]	STREAM	CONNECTED	17152	
unix	3	[ ]	STREAM	CONNECTED	16557	
unix	2	[ ]	DGRAM		15385	
unix	3	[ ]	STREAM	CONNECTED	13726	/run/systemd/journal/
stdout						

unix	3	[ ]	STREAM	CONNECTED	16557	
unix	2	[ ]	DGRAM		15385	
unix	3	[ ]	STREAM	CONNECTED	13726	/run/systemd/journal/
stdout						
unix	3	[ ]	STREAM	CONNECTED	12449	/run/dbus/system_bus_
socket						
unix	3	[ ]	STREAM	CONNECTED	16609	/run/user/1000/pulse/
native						
unix	3	[ ]	STREAM	CONNECTED	16522	
unix	3	[ ]	STREAM	CONNECTED	15601	/run/dbus/system_bus_
socket						
unix	3	[ ]	STREAM	CONNECTED	15387	/run/systemd/journal/
stdout						
unix	3	[ ]	STREAM	CONNECTED	8949	
unix	3	[ ]	STREAM	CONNECTED	8519	
unix	3	[ ]	DGRAM	CONNECTED	13569	
unix	3	[ ]	STREAM	CONNECTED	17149	
unix	3	[ ]	STREAM	CONNECTED	8757	/run/dbus/system_bus_
socket						
unix	3	[ ]	STREAM	CONNECTED	16558	/run/user/1000/at-spi
/bus						
unix	3	[ ]	STREAM	CONNECTED	13504	
unix	3	[ ]	STREAM	CONNECTED	8579	
unix	3	[ ]	STREAM	CONNECTED	36029	
unix	3	[ ]	STREAM	CONNECTED	16530	/run/user/1000/bus

unix	3	[ ]	STREAM	CONNECTED	16530	/run/user/1000/bus
unix	3	[ ]	STREAM	CONNECTED	15386	
unix	3	[ ]	STREAM	CONNECTED	9998	/run/systemd/journal/
stdout						
unix	3	[ ]	STREAM	CONNECTED	16881	/run/user/1000/bus
unix	3	[ ]	STREAM	CONNECTED	12352	
unix	3	[ ]	STREAM	CONNECTED	17153	/run/user/1000/waylan
d-0						
re-0						
unix	3	[ ]	STREAM	CONNECTED	16854	/run/user/1000/pipewl
unix	3	[ ]	STREAM	CONNECTED	12447	
unix	3	[ ]	STREAM	CONNECTED	15573	/run/user/1000/bus
unix	3	[ ]	STREAM	CONNECTED	15406	/run/systemd/journal/
stdout						
unix	3	[ ]	STREAM	CONNECTED	16907	/run/user/1000/bus
unix	3	[ ]	STREAM	CONNECTED	8950	/run/dbus/system_bus_
socket						
unix	3	[ ]	STREAM	CONNECTED	8520	/run/dbus/system_bus_
socket						
unix	3	[ ]	STREAM	CONNECTED	17154	
unix	3	[ ]	STREAM	CONNECTED	13505	/run/systemd/journal/
stdout						
unix	3	[ ]	STREAM	CONNECTED	15768	/home/vboxuser/.cache
/ibus/dbus-bkrvxOpK						
unix	3	[ ]	STREAM	CONNECTED	16355	



NICE:

```
vboxuser@satti:~$ nice
0
```

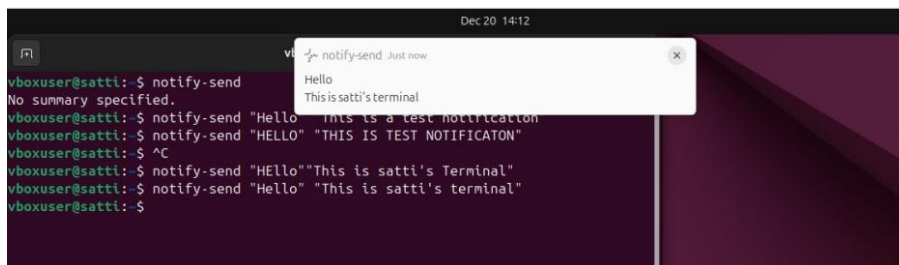
NL:

```
vboxuser@satti:~$ nl
nl
1 nl
```

NOHUP:

```
vboxuser@satti:~$ nohup
nohup: missing operand
Try 'nohup --help' for more information.
vboxuser@satti:~$ nohup.out
nohup.out: command not found
vboxuser@satti:~$ nohup sleep 100 &
[1] 4027
```

NOTIFY:



The screenshot shows a terminal window with the following commands and output:

```
vboxuser@satti:~$ notify-send
No summary specified.
vboxuser@satti:~$ notify-send "Hello" "This is a test notification"
vboxuser@satti:~$ notify-send "HELLO" "THIS IS TEST NOTIFICATON"
vboxuser@satti:~$ ^C
vboxuser@satti:~$ notify-send "Hello""This is satti's Terminal"
vboxuser@satti:~$ notify-send "Hello" "This is satti's terminal"
vboxuser@satti:~$
```

A notification popup titled "notify-send Just now" is displayed over the terminal, showing the text "Hello" and "This is satti's terminal".

NS-Look UP:

```
> nslookup google.com
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: google.com
Address: 142.250.202.142
Name: google.com
Address: 2a00:1450:4019:815::200e
```

O commands:

OP:

```
vboxuser@satti:~$ op
op: command not found
vboxuser@satti:~$ alias op='xdg-open'
vboxuser@satti:~$
vboxuser@satti:~$
```

OPEN:

```
vboxuser@satti:~$ open
xdg-open - opens a file or URL in the user's preferred application

Synopsis

xdg-open { file | URL }

xdg-open { --help | --manual | --version }

Use 'man xdg-open' or 'xdg-open --manual' for additional info.
vboxuser@satti:~$ open youtube.com
gio: file:///home/vboxuser/youtube.com: Error when getting information for file
"/home/vboxuser/youtube.com": No such file or directory
vboxuser@satti:~$ open google.com
gio: file:///home/vboxuser/google.com: Error when getting information for file
"/home/vboxuser/google.com": No such file or directory
vboxuser@satti:~$
```

P COMMANDS:

PASSWD:

```
vboxuser@satti:~$ passwd
Changing password for vboxuser.
Current password:
New password:
BAD PASSWORD: The password is shorter than 8 characters
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/sys
tematic
New password:
Retype new password:
passwd: password updated successfully
```

PASTE:

```
vboxuser@satti:~$ paste
vboxuser@satti:~$ type paste
paste is hashed (/usr/bin/paste)
vboxuser@satti:~$ alias paste
bash: alias: paste: not found
vboxuser@satti:~$ unalias paste
bash: unalias: paste: not found
vboxuser@satti:~$
```

```
vboxuser@satti:~$ pgrep bash
2686
vboxuser@satti:~$ pgrep firefox
vboxuser@satti:~$ pgrep python
vboxuser@satti:~$
```

```
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=66 ttl=255
time=51.0 ms
d
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=67 ttl=255
time=50.0 ms
dd
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=68 ttl=255
time=50.4 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=69 ttl=255
time=49.9 ms
^X64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=70 ttl=25
5 time=53.2 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=71 ttl=255
time=50.4 ms
^C
--- youtube.com ping statistics ---
71 packets transmitted, 70 received, 1.40845% packet loss, time 70603ms
rtt min/avg/max/mdev = 45.331/410.036/4170.404/770.318 ms, pipe 5
```

PING:

```
vboxuser@satti:~$ ping youtube.com
PING youtube.com (142.250.201.238) 56(84) bytes of data.
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=1 ttl=255 t
ime=1218 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=2 ttl=255 t
ime=1206 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=3 ttl=255 t
ime=209 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=5 ttl=255 t
ime=1302 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=6 ttl=255 t
ime=749 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=7 ttl=255 t
```

PKILL:

```
vboxuser@satti:~$ pkill 5MB testfile
pkill: only one pattern can be provided
Try `pkill --help' for more information.
vboxuser@satti:~$ pkill
pkill: no matching criteria specified
Try `pkill --help' for more information.
vboxuser@satti:~$ pkill chrome
vboxuser@satti:~$ pkill firefox
vboxuser@satti:~$
```

POPD:

```
vboxuser@satti:~$ popd
bash: popd: directory stack empty
vboxuser@satti:~$ popd /tmp
bash: popd: /tmp: invalid argument
popd: usage: popd [-n] [+N | -N]
vboxuser@satti:~$ popd +1
bash: popd: directory stack empty
vboxuser@satti:~$
```

PR:

```
vboxuser@satti:~$ pr SMB testfile
pr: SMB: No such file or directory

2025-12-19 13:39 testfile Page 1
```

PRINTCAP:

```
vboxuser@satti:~$ printcap
Command 'printcap' not found, did you mean:
 command 'printcal' from deb argyll (3.1.0+repack-1)
Try: sudo apt install <deb name>
vboxuser@satti:~$ ls -la /etc/printcap
lrwxrwxrwx 1 root root 18 Dec 21 13:42 /etc/printcap -> /run/cups/printcap
```

PRINTENV:

```
vboxuser@satti:~$ printenv
SHELL=/bin/bash
SESSION_MANAGER=local/satti:@/tmp/.ICE-unix/1890,unix/satti:/tmp/.ICE-unix/1890
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/vboxuser
LOGNAME=vboxuser
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
SYSTEMD_EXEC_PID=1934
XAUTHORITY=/run/user/1000/.mutter-Xwaylandauth.9K1UH3
HOME=/home/vboxuser
USERNAME=vboxuser
```



```

USERNAME=vboxuser
IM_CONFIG_PHASE=1
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd
=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:e
x=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:
.lz4=01;31:.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=0
1;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01
;31:*.xz=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz
2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.
sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;
31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif
=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;
35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif
=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35
:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.web
p=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
.nuv=01;35:.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01
;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv
=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.f
lac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xsp

```

```

SHLVL=1
GSM_SKIP_SSH_AGENT_WORKAROUND=true
QT_IM_MODULE=ibus
XDG_RUNTIME_DIR=/run/user/1000
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/
var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/us
r/local/games:/snap/bin:/snap/bin
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
_=usr/bin/printenv

```

```

f=00;36:*=00;90:*=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpk
g-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00
;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp
=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
XDG_CURRENT_DESKTOP=ubuntu:GNOME
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/user.slice/user-1000.slice/user@1000.servic
e/session.slice/org.gnome.Shell@wayland.service/memory.pressure
VTE_VERSION=7600
WAYLAND_DISPLAY=wayland-0
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/077e92ab_9bf0_4641_85cf_fdb42b5
a3661
GNOME_SETUP_DISPLAY=:1
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=vboxuser
GNOME_TERMINAL_SERVICE=:1.97
CLUTTER_DISABLE_MIPMAPPED_TEXT=1
DISPLAY=:0
SHLVL=1

```

PRINTF:

```

vboxuser@satti:~$ printf
printf: usage: printf [-v var] format [arguments]
vboxuser@satti:~$

```

PS:

```
vboxuser@satti:~$ ps
 PID TTY TIME CMD
 2686 pts/0 00:00:00 bash
 4185 pts/0 00:00:00 ps
```

PUSHD:

```
vboxuser@satti:~$ pushd
bash: pushd: no other directory
vboxuser@satti:~$ pushd 5MB testfile
bash: pushd: too many arguments
```

PV:

```
vboxuser@satti:~$ pv
^D0.00 B 0:00:30 [0.00 B/s] [<=>
1.00 B 0:00:33 [30.3mB/s] [<=>
```

```
vboxuser@satti:~$ pv
0.00 B 0:00:11 [0.00 B/s] [<=>
```

PWD:

```
vboxuser@satti:~$ pwd
/home/vboxuser
```

Q COMMANDS:

QUOTA:

```
vboxuser@satti:~$ quota
vboxuser@satti:~$ quota
vboxuser@satti:~$
```

```
-R, --exclude-root exclude root when checking all filesystems
-F, --format=formatname check quota files of specific format
-a, --all check all filesystems
-h, --help display this message and exit
-V, --version display version information and exit

Bugs to jack@suse.cz
vboxuser@satti:~$
```

```

vboxuser@satti:~$ quota
vboxuser@satti:~$ quota
vboxuser@satti:~$ quotacheck
Bad number of arguments.
Utility for checking and repairing quota files.
quotacheck [-gucbfinvdmMR] [-F <quota-format>] filesystem|-a

-u, --user check user files
-g, --group check group files
-c, --create-files create new quota files
-b, --backup create backups of old quota files
-f, --force force check even if quotas are enabled
-i, --interactive interactive mode
-n, --use-first-dquot use the first copy of duplicated structure
-v, --verbose print more information
-d, --debug print even more messages
-m, --no-remount do not remount filesystem read-only
-M, --try-remount try remounting filesystem read-only,
 continue even if it fails
-R, --exclude-root exclude root when checking all filesystems
-F, --format=formatname check quota files of specific format
-a, --all check all filesystems
-h, --help display this message and exit
-V, --version display version information and exit

```

R COMMANDS:

RAM:

```

vboxuser@satti:~$ ram
Command 'ram' not found, did you mean:
 command 'rem' from snap rem (0.17.0)
 command 'rar' from deb rar (2:6.23-1)
 command 'ra6' from deb ipv6toolkit (2.0+ds.1-2)
 command 'ra' from deb argus-client (1:3.0.8.2-6.2ubuntu1)
 command 'jam' from deb jam (2.6.1-2build2)
 command 'vam' from deb vim-addon-manager (0.5.10)
 command 'nam' from deb nam (1.15-6)
 command 'rmm' from deb mailutils-mh (1:3.16-1build1)
 command 'rmm' from deb mmh (0.4-6)
 command 'rmm' from deb nmh (1.8-1)
 command 'bam' from deb bam (0.5.1-3)
 command 'cam' from deb libcamera-tools (0.1.0-3ubuntu1)
 command 'rpm' from deb rpm (4.18.0+dfsg-1build2)
 command 'rm' from deb coreutils (9.4-3ubuntu6.1)
 command 'rem' from deb remind (04.02.08-1)
See 'snap info <snapname>' for additional versions.
vboxuser@satti:~$ ram free -h
Command 'ram' not found, did you mean:
 command 'rem' from snap rem (0.17.0)
 command 'rem' from deb remind (04.02.08-1)

```



```
vboxuser@satti:~$ rar

RAR 7.00 Copyright (c) 1993-2024 Alexander Roshal 26 Feb 2024
Trial version Type 'rar -?' for help

Usage: rar <command> [-<switch 1> -<switch N> <archive> <files...>
<@listfiles...> <path_to_extract/>

<Commands>
a Add files to archive
c Add archive comment
ch Change archive parameters
cw Write archive comment to file
d Delete files from archive
e Extract files without archived paths
f Freshen files in archive
i[par]=<str> Find string in archives
k Lock archive
l[t[a],b] List archive contents [technical[all], bare]
m[f] Move to archive [files only]
p Print file to stdout
r Repair archive
rc Reconstruct missing volumes
```

```
ds Disable name sort for solid archive
dw Wipe files after archiving
e[+]<attr> Set file exclude and include attributes
ed Do not add empty directories
ep Exclude paths from names
ep1 Exclude base directory from names
ep3 Expand paths to full including the drive letter
ep4<path> Exclude the path prefix from names
f Freshen files
hp[password] Encrypt both file data and headers
ht[b|c] Select hash type [BLAKE2,CRC32] for file checksum
id[c,d,n,p,q] Display or disable messages
ierr Send all messages to stderr
ilog[name] Log errors to file
inul Disable all messages
isnd[-] Control notification sounds
iver Display the version number
k Lock archive
kb Keep broken extracted files
log[f][=name] Write names to log file
m<0..5> Set compression level (0-store...3-default...5-maximal)
mc<par> Set advanced compression parameters
md[x]<n>[kmg] Dictionary size in KB, MB or GB
me[par] Set encryption parameters
```

```
mc<par> Set advanced compression parameters
md[x]<n>[kmg] Dictionary size in KB, MB or GB
me[par] Set encryption parameters
ms[ext;ext] Specify file types to store
mt<threads> Set the number of threads
n<file> Additionally filter included files
n@ Read additional filter masks from stdin
n@<list> Read additional filter masks from list file
o[+|-] Set the overwrite mode
oh Save hard links as the link instead of the file
oi[0-4][:min] Save identical files as references
ol[a,-] Process symbolic links as the link [absolute paths, skip]
op<path> Set the output path for extracted files
or Rename files automatically
ow Save or restore file owner and group
p[password] Set password
qo[-|+] Add quick open information [none|force]
r Recurse subdirectories
r- Disable recursion
r0 Recurse subdirectories for wildcard names only
rr[N] Add data recovery record
rv[N] Create recovery volumes
s[<N>,v[-],e] Create solid archive
s- Disable solid archiving
```

```

sc<chr>[obj] Specify the character set
sfx[name] Create SFX archive
si[name] Read data from standard input (stdin)
sl<size>[u] Process files with size less than specified
sm<size>[u] Process files with size more than specified
t Test files after archiving
ta[mcao]<d> Process files modified after <d> YYYYMMDDHHMMSS date
tb[mcao]<d> Process files modified before <d> YYYYMMDDHHMMSS date
tk Keep original archive time
tl Set archive time to latest file
tn[mcao]<t> Process files newer than <t> time
to[mcao]<t> Process files older than <t> time
ts[m,c,a,p] Save or restore time (modification, creation, access, preserve)
u Update files
v<size>[u] Create volumes with size in [bBkKmMgGtT] units
ver[n] File version control
vp Pause before each volume
w<path> Assign work directory
x<file> Exclude specified file
x@ Read file names to exclude from stdin
x@<list> Exclude files listed in specified list file
y Assume Yes on all queries
z[file] Read archive comment from file

```

```

rn Rename archived files
rr[N] Add data recovery record
rv[N] Create recovery volumes
s[name|.] Convert archive to or from SFX
t Test archive files
u Update files in archive
v[t[a],b] Verbosely list archive contents [technical[all],bare]
x Extract files with full path

Switches>
- Stop switches scanning
@[+] Disable [enable] file lists
ad[1,2] Alternate destination path
ag[format] Generate archive name using the current date
ai Ignore file attributes
am[s,r] Archive name and time [save, restore]
ap<path> Set path inside archive
as Synchronize archive contents
c- Disable comments show
cfg- Disable read configuration
cl Convert names to lower case
cu Convert names to upper case
df Delete files after archiving
dh Open shared files

```

RCP:

```

vboxuser@satti:~$ rcp
Command 'rcp' not found, but can be installed with:
sudo apt install rsh-client
vboxuser@satti:~$ sudo apt install rsh-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package rsh-client is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'rsh-client' has no installation candidate

```

READ:

```

vboxuser@satti:~$ read
vboxuser@satti:~$ read 5MB testfile
bash: read: `5MB': not a valid identifier
vboxuser@satti:~$

```

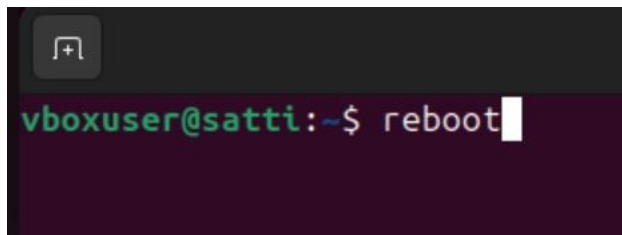
READ ARRAY:

```
vboxuser@satti:~$ readarray
vboxuser@satti:~$ readarray 12345
bash: readarray: `12345': not a valid identifier
vboxuser@satti:~$
```

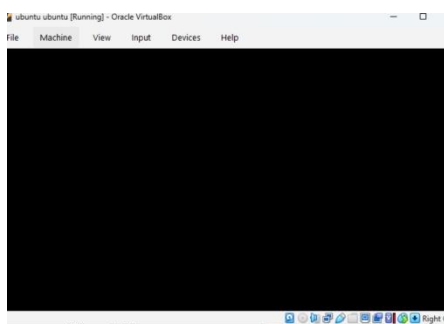
READONLY:

```
vboxuser@satti:~$ readonly
declare -r BASHOPTS="checkwinsize:cmdhist:complete_fullquote:expand_aliases:extg
lob:extquote:force_ignores:globasciiranges:globskipdots:histappend:interactive_c
omments:patsub_replacement:progcomp:promptvars:sourcepath"
declare -ar BASH_VERSION=([0]="5" [1]="2" [2]="21" [3]="1" [4]="release" [5]="x
86_64-pc-linux-gnu")
declare -ir EUID="1000"
declare -ir PPID="2679"
declare -r SHELLOPTS="braceexpand:emacs:hashall:histexpand:history:interactive-c
omments:monitor"
declare -ir UID="1000"
```

REBOOT:



```
vboxuser@satti:~$ reboot
```



REMSYNC:

```
vboxuser@satti:~$ remsync
Command 'remsync' not found, did you mean:
 command 'regsync' from snap regclient (0.11.1)
See 'snap info <snapname>' for additional versions.
```

RENAME:

```
vboxuser@satti:~$ rename
Usage:
 file-rename [-h|-m|-V] [-v] [-0] [-n] [-f] [-d] [-u [enc]]
 [-e|-E perlexpr]*|perlexpr [files]
```

RENICE:

```
vboxuser@satti:~$ renice
renice: not enough arguments
Try 'renice --help' for more information.
vboxuser@satti:~$ renice -p
renice: not enough arguments
Try 'renice --help' for more information.
vboxuser@satti:~$ renice --help

Usage:
 renice [-n|--priority|--relative] <priority> [-p|--pid] <pid>...
 renice [-n|--priority|--relative] <priority> -g|--pgrp <pgid>...
 renice [-n|--priority|--relative] <priority> -u|--user <user>...

Alter the priority of running processes.

Options:
 -n <num> specify the nice value
 If POSIXLY_CORRECT flag is set in environment
 then the priority is 'relative' to current
```

RETURN:

```
vboxuser@satti:~$ return
bash: return: can only 'return' from a function or sourced script
vboxuser@satti:~$ return {n}
bash: return: {n}: numeric argument required
bash: return: can only 'return' from a function or sourced script
vboxuser@satti:~$ return 300
bash: return: can only 'return' from a function or sourced script
vboxuser@satti:~$
```

REV:

```
vboxuser@satti:~$ rev
vboxuser@satti:~$ rev 5MB testfile
rev: cannot open 5MB: No such file or directory
vboxuser@satti:~$
```

RM:

```
vboxuser@satti:~$ rm
rm: missing operand
Try 'rm --help' for more information.
vboxuser@satti:~$ rm -f
vboxuser@satti:~$ rm -i
rm: missing operand
Try 'rm --help' for more information.
vboxuser@satti:~$ rm "$file"
rm: cannot remove ':': No such file or directory
vboxuser@satti:~$
```



**RMDIR:**

```
vboxuser@satti:~$ rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.
vboxuser@satti:~$ rmdir vboxuser@satti
rmdir: failed to remove 'vboxuser@satti': No such file or directory
vboxuser@satti:~$ rmdir SMB testfile
rmdir: failed to remove 'SMB': No such file or directory
rmdir: failed to remove 'testfile': Not a directory
```

**Dependencies/Installation:** sudo apt install screen

**Screenshot:**

```
sohaib@sohaib:~$ sudo apt install screen
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
screen is already the newest version (4.9.1-1build1).
The following package was automatically installed and is no longer required:
 libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sohaib@sohaib:~$
```

**Command Name:** scp

**Command Syntax:** scp [options] source\_file

user@host:destination\_path

**Description:** Securely copies files between hosts over a network using SSH.

**Dependencies/Installation:** sudo apt install openssh-client

**Screenshot:**

```
sohaib@sohaib: ~
sohaib@sohaib:~$ sudo apt install openssh-clint openssh-server
[sudo] password for sohaib:
Sorry, try again.
[sudo] password for sohaib:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package openssh-clint
sohaib@sohaib:~$
```

**Command Name:** sdiff

**Command Syntax:** sdiff [options] file1 file2

**Description:** Compares two files side-by-side and shows differences.

**Dependencies/Installation:** Part of diffutils, typically pre-installed.

**Screenshot:**

```
sohaib@sohaib: ~
sohaib@sohaib:~$ scp -r myfolder user@192.168.1.10:/home/user/
scp: stat local "myfolder": No such file or directory
sohaib@sohaib:~$ touch myfolder user@192.168.1.10:/home/user/
touch: cannot touch 'user@192.168.1.10:/home/user/': No such file or directory
sohaib@sohaib:~$ touch file1.txt file2
sohaib@sohaib:~$ sdiff file1.txt file2.txt
diff: file2.txt: No such file or directory
sohaib@sohaib:~$ touch file1.txt
sohaib@sohaib:~$ touch file2.txt
sohaib@sohaib:~$ sdiff file1.txt file2.txt
sohaib@sohaib:~$
```

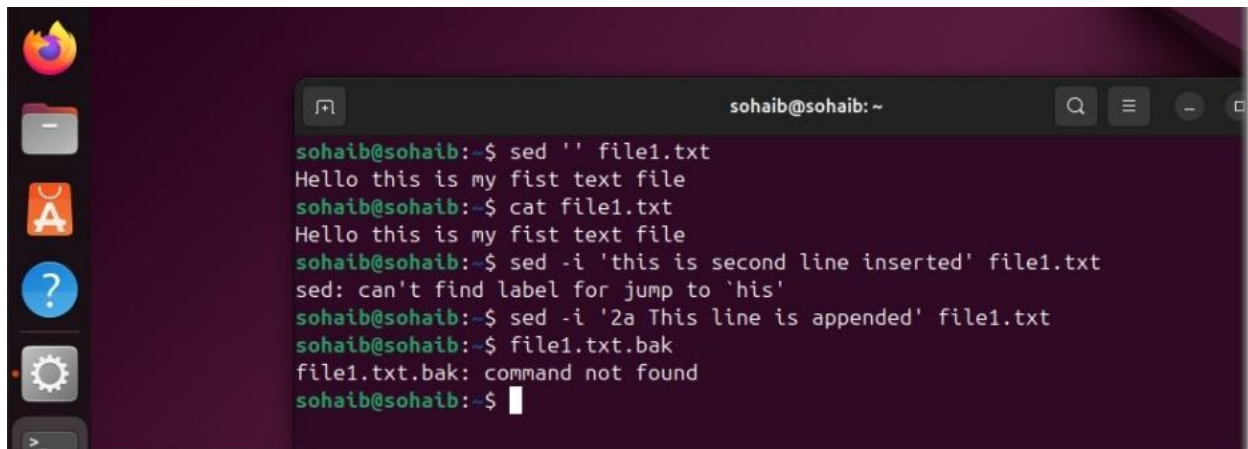
**Command Name:** sed

**Command Syntax:** sed [options] 'script' input\_file

**Description:** Parses and transforms text by applying scripts to an input stream.

**Dependencies/Installation:** Part of sed package, typically pre-installed.

**Screenshot:**

A terminal window with a dark purple background. The prompt is 'sohaib@sohaib: ~'. The user enters 'sed '' file1.txt', which outputs 'Hello this is my fist text file'. Then they enter 'cat file1.txt', which also outputs 'Hello this is my fist text file'. Next, they enter 'sed -i 'this is second line inserted' file1.txt', which outputs an error: 'sed: can't find label for jump to 'his''. Then they enter 'sed -i '2a This line is appended' file1.txt', which outputs nothing. Finally, they enter 'file1.txt.bak', which outputs 'file1.txt.bak: command not found'.

```
sohaib@sohaib:~$ sed '' file1.txt
Hello this is my fist text file
sohaib@sohaib:~$ cat file1.txt
Hello this is my fist text file
sohaib@sohaib:~$ sed -i 'this is second line inserted' file1.txt
sed: can't find label for jump to 'his'
sohaib@sohaib:~$ sed -i '2a This line is appended' file1.txt
sohaib@sohaib:~$ file1.txt.bak
file1.txt.bak: command not found
sohaib@sohaib:~$
```

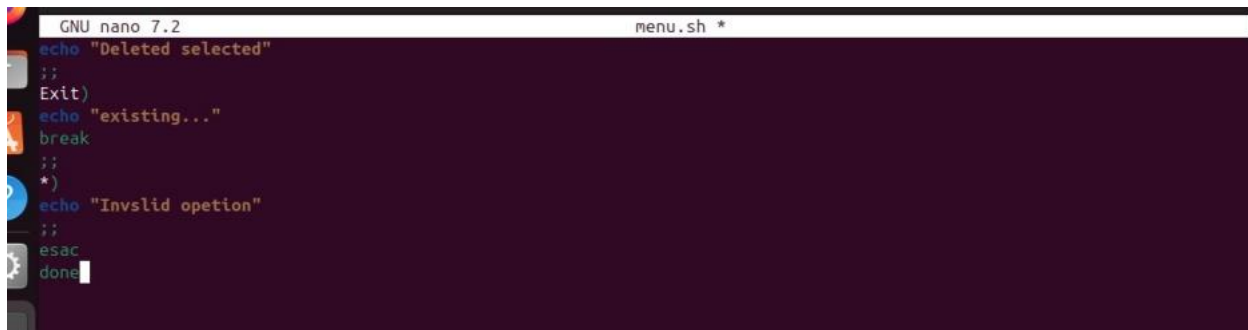
**Command Name:** select

**Command Syntax:** select name [in words ...]; do commands; done

**Description:** A shell built-in that generates simple text menus from a list.

**Dependencies/Installation:** Built into Bash, no installation needed.

**Screenshot:**

A terminal window showing a 'select' menu. The prompt is 'GNU nano 7.2' and the file is 'menu.sh \*'. The menu has three options: 'Deleted selected', 'existing...', and 'Invslid opetion'. The user has selected 'Invslid opetion', and the terminal shows 'esac' and 'done' at the bottom.

```
GNU nano 7.2 menu.sh *
echo "Deleted selected"
;;
Exit)
echo "existing..."
break
;;
*)
echo "Invslid opetion"
;;
esac
done
```

**Command Name:** seq

**Command Syntax:** seq [options] first last

**Description:** Generates a sequence of numbers.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**



```
sohaib@sohaib: ~
sohaib@sohaib:~$ seq 5
1
2
3
4
5
sohaib@sohaib:~$ seq 5 6
5
6
sohaib@sohaib:~$ seq 5 5 6
5
sohaib@sohaib:~$ seq 6
1
2
3
4
5
6
sohaib@sohaib:~$
```

**Command Name:** set

**Command Syntax:** set [options] [arguments]

**Description:** Displays or sets shell environment variables and positional parameters.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

```
command_not_found_handle ()
{
 if [-x /usr/lib/command-not-found]; then
 /usr/lib/command-not-found -- "$1";
 return $?;
 else
 if [-x /usr/share/command-not-found/command-not-found]; then
 /usr/share/command-not-found/command-not-found -- "$1";
 return $?;
 else
 printf "ks: command not found\n" "$1" 1>&2;
 return 127;
 fi;
 fi;
}
dequote ()
{
 eval printf %s "$1" 2> /dev/null
}
quote ()
{
 local quoted=${1//\'/\'\'\'\'};
 printf "%s" "$quoted"
}
quote_readline ()
{
 local ret;
 _quote_readline_by_ref "$1" ret;
 printf %s "$ret"
}
sohaib@sohaib:~$
```

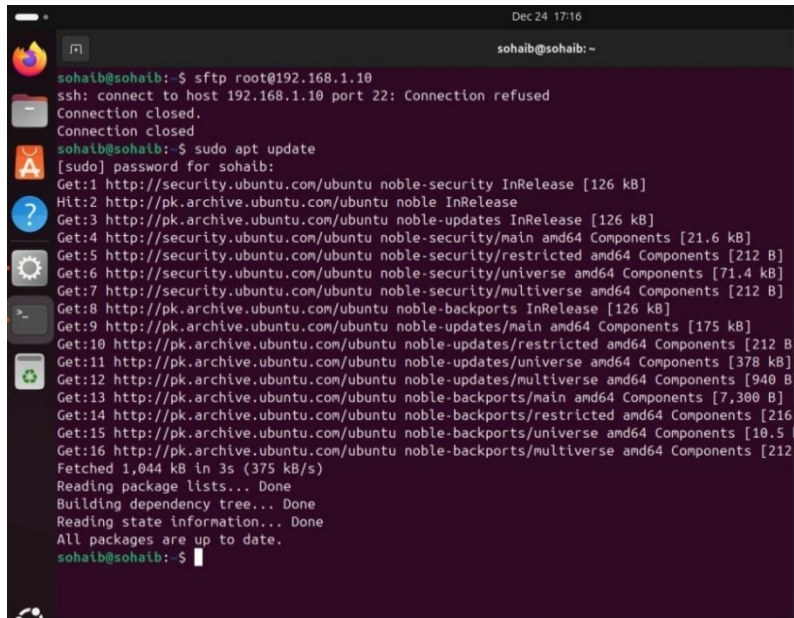
**Command Name:** sftp

**Command Syntax:** sftp [options] user@host

**Description:** Provides an interactive, encrypted file transfer session over SSH.

**Dependencies/Installation:** sudo apt install openssh-client

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' showing the execution of 'sudo apt update'. The terminal output displays the progress of updating package lists from various Ubuntu repositories, including security, updates, and backports. The process concludes with the message 'All packages are up to date.'

```
sohaib@sohaib:~$ sftp root@192.168.1.10
ssh: connect to host 192.168.1.10 port 22: Connection refused
Connection closed.
Connection closed.
sohaib@sohaib:~$ sudo apt update
[sudo] password for sohaib:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.4 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:8 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:10 http://pk.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:11 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:12 http://pk.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:13 http://pk.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,300 B]
Get:14 http://pk.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:15 http://pk.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:16 http://pk.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 1,044 kB in 3s (375 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
sohaib@sohaib:~$
```

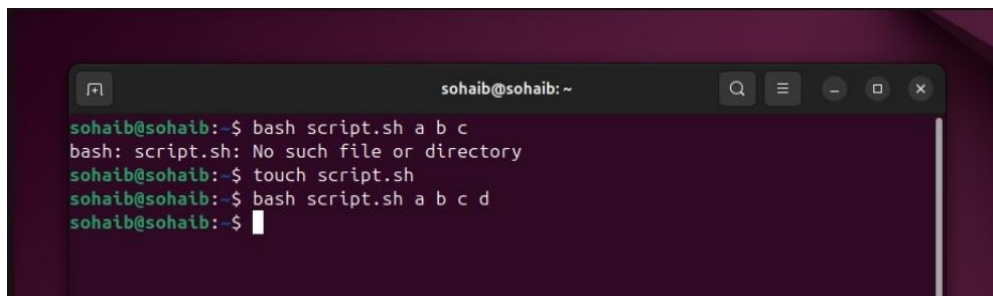
**Command Name:** shift

**Command Syntax:** shift [n]

**Description:** Shifts positional parameters to the left, discarding the first argument.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' showing the execution of 'bash script.sh a b c' and 'touch script.sh'. The first command results in an error 'bash: script.sh: No such file or directory'. The second command 'touch script.sh' is executed successfully, creating the file. The third command 'bash script.sh a b c d' is also shown.

```
sohaib@sohaib:~$ bash script.sh a b c
bash: script.sh: No such file or directory
sohaib@sohaib:~$ touch script.sh
sohaib@sohaib:~$ bash script.sh a b c d
sohaib@sohaib:~$
```

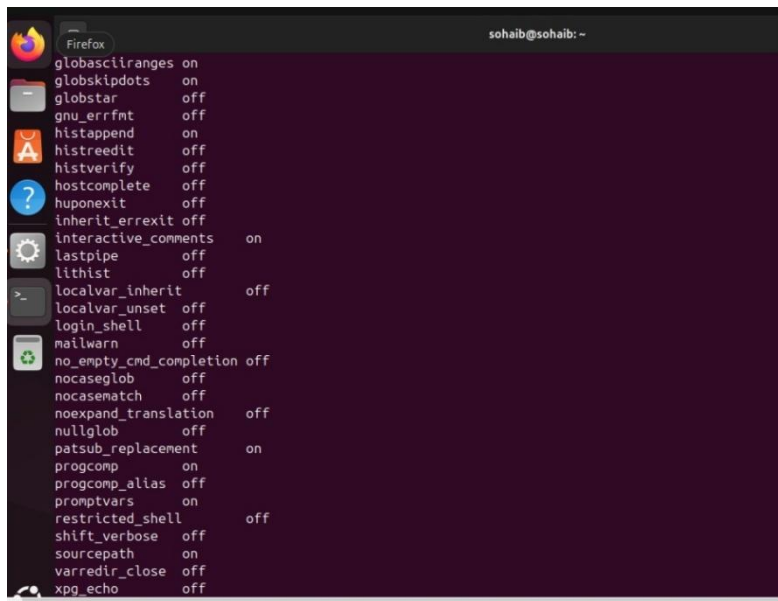
**Command Name:** shopt

**Command Syntax:** shopt [options] [optname ...]

**Description:** Manages optional shell behavior settings in Bash.

**Dependencies/Installation:** Built into Bash, no installation needed.

**Screenshot:**



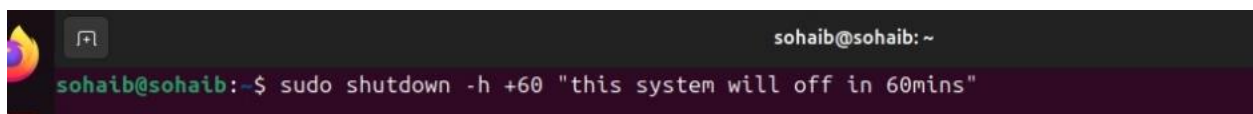
**Command Name:** shutdown

**Command Syntax:** sudo shutdown [options] time [message]

**Description:** Powers off, halts, or reboots the system at a specified time.

**Dependencies/Installation:** Typically pre-installed via system utilities.

**Screenshot:**



**Command Name:** sleep

**Command Syntax:** sleep number[suffix]

**Description:** Pauses execution for a specified number of seconds.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

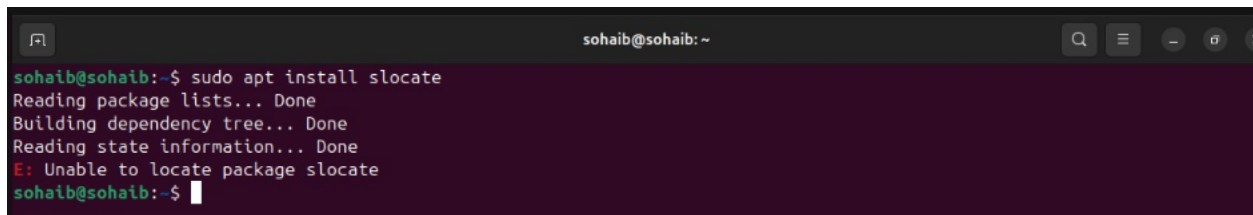
**Command Name:** slocate

**Command Syntax:** slocate [options] pattern

**Description:** Finds files by name using a pre-built, secure database.

**Dependencies/Installation:** sudo apt install mlocate

**Screenshot:**

A terminal window with a dark purple background. The prompt is 'sohaib@sohaib: ~'. The user enters 'sudo apt install slocate'. The output shows progress: 'Reading package lists... Done', 'Building dependency tree... Done', 'Reading state information... Done'. It then shows an error: 'E: Unable to locate package slocate'. The prompt returns to 'sohaib@sohaib:~\$'.

**Command Name:** sort

**Command Syntax:** sort [options] [file]

**Description:** Sorts lines of text files alphabetically, numerically, or by other criteria.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

A terminal window with a dark purple background. The prompt is 'sohaib@sohaib:~\$'. The user enters 'sort --version'. The output shows version information for GNU coreutils 9.4, including copyright and license details. The user then enters 'sort file1.txt'. The output shows the contents of file1.txt, which are 'Hello this is my first text file'. The prompt returns to 'sohaib@sohaib:~\$'.

**Command Name:** source

**Command Syntax:** source filename or . filename

**Description:** Executes commands from a file in the current shell environment.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

```
sohaib@sohaib:~$ source file1.txt
Command 'Hello' not found, did you mean:
 command 'hello' from snap hello (2.10)
 command 'jello' from deb jello (1.6.0-1)
 command 'hello' from deb hello (2.10-3)
 command 'hello' from deb hello-traditional (2.10-6)
See 'snap info <snapname>' for additional versions.
sohaib@sohaib:~$
```

**Command Name:** split

**Command Syntax:** split [options] input\_file [prefix]

**Description:** Splits a large file into smaller pieces by lines or bytes.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

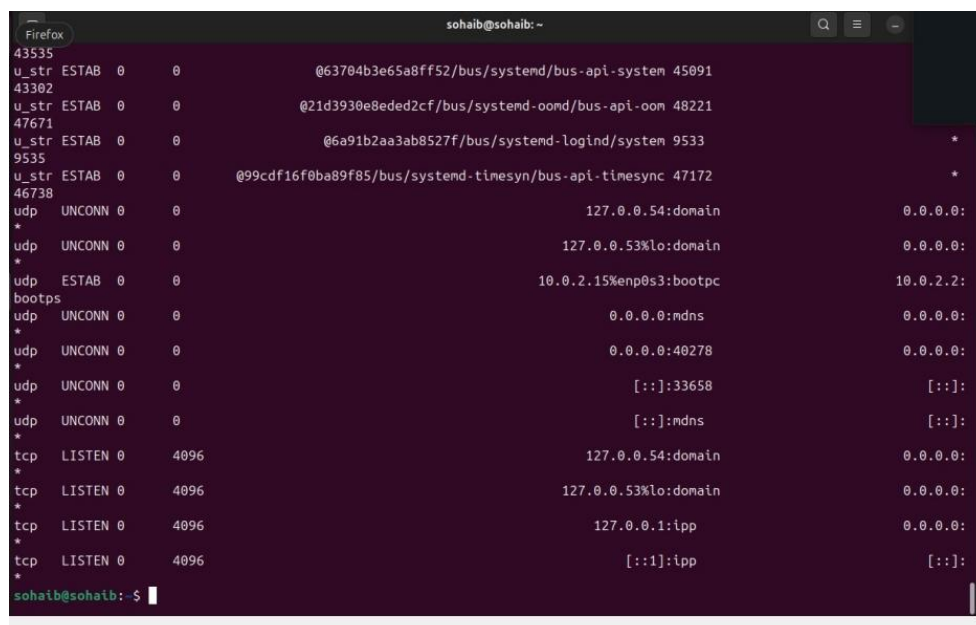
**Command Name:** ss

**Command Syntax:** ss [options] [filter]

**Description:** Displays detailed information about network sockets.

**Dependencies/Installation:** Part of iproute2, typically pre-installed.

**Screenshot:**



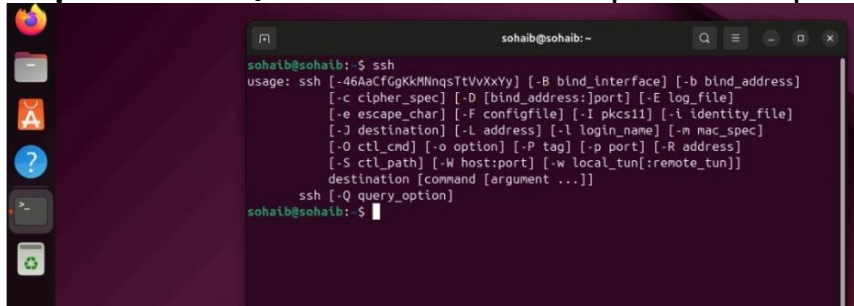
```
sohaib@sohaib: ~
43535 u_str ESTAB 0 0 @63704b3e65a8ff52/bus/systemd/bus-api-system 45091
43302 u_str ESTAB 0 0 @21d3930e8eded2cf/bus/systemd-oomd/bus-api-oom 48221
47671 u_str ESTAB 0 0 @6a91b2aa3ab8527f/bus/systemd-logind/system 9533
9535 u_str ESTAB 0 0 @99cdf16f0ba89f85/bus/systemd-timesyn/bus-api-timesync 47172
46738 udp UNCONN 0 0 127.0.0.54:domain 0.0.0.0:
*
udp UNCONN 0 0 127.0.0.53%lo:domain 0.0.0.0:
*
udp ESTAB 0 0 10.0.2.15%enp0s3:bootpc 10.0.2.2:
bootpc
udp UNCONN 0 0 0.0.0.0:mdns 0.0.0.0:
*
udp UNCONN 0 0 0.0.0.0:40278 0.0.0.0:
*
udp UNCONN 0 0 [::]:33658 [::]:
*
udp UNCONN 0 0 [::]:mdns [::]:
*
tcp LISTEN 0 4096 127.0.0.54:domain 0.0.0.0:
*
tcp LISTEN 0 4096 127.0.0.53%lo:domain 0.0.0.0:
*
tcp LISTEN 0 4096 127.0.0.1:ipp 0.0.0.0:
*
tcp LISTEN 0 4096 [::]:ipp [::]:
*
sohaib@sohaib:~$
```

**Command Name:** ssh

**Command Syntax:** ssh [options] user@host [command]

**Description:** Establishes an encrypted, secure remote login session.

**Dependencies/Installation:** sudo apt install openssh-client

A screenshot of a Linux terminal window. The window title is 'sohaib@sohaib: ~'. The user has entered the command 'ssh'. The terminal displays the usage for the ssh command, listing various options like -B, -b, -c, -D, -E, -e, -F, -I, -i, -j, -l, -m, -O, -o, -P, -p, -R, -S, -w, and -Q, along with their descriptions. The prompt 'sohaib@sohaib:~\$' is visible at the bottom.

```
sohaib@sohaib:~$ ssh
usage: ssh [-46AaCfGgKkMmNnQqStTVvXxYy] [-B bind_interface] [-b bind_address]
[-c cipher_spec] [-D [bind_address:]port] [-E log_file]
[-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
[-j destination] [-L address] [-l login_name] [-m mac_spec]
[-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]
[-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
destination [command [argument ...]]
ssh [-Q query_option]
```

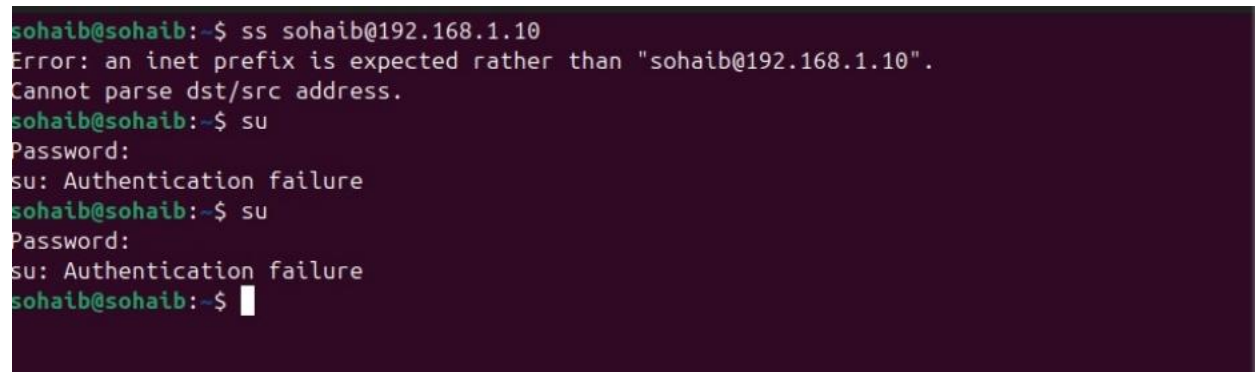
**Command Name:** su

**Command Syntax:** su [options] [username]

**Description:** Switches the current user to another user within a login shell.

**Dependencies/Installation:** Part of util-linux, typically pre-installed.

**Screenshot:**

A screenshot of a Linux terminal window. The user has entered the command 'su sohaib@192.168.1.10'. The terminal displays an error message: 'Error: an inet prefix is expected rather than "sohaib@192.168.1.10". Cannot parse dst/src address.' The user then enters 'su'. The terminal prompts for a password, and the user enters a password. The terminal displays 'su: Authentication failure'. The user then enters 'su' again. The terminal prompts for a password, and the user enters a password. The terminal displays 'su: Authentication failure'. The prompt 'sohaib@sohaib:~\$' is visible at the bottom.

```
sohaib@sohaib:~$ ss sohaib@192.168.1.10
Error: an inet prefix is expected rather than "sohaib@192.168.1.10".
Cannot parse dst/src address.
sohaib@sohaib:~$ su
Password:
su: Authentication failure
sohaib@sohaib:~$ su
Password:
su: Authentication failure
sohaib@sohaib:~$
```

**Command Name:** sudo

**Command Syntax:** sudo [options] command

**Description:** Executes a command with superuser or another user's privileges.



**Dependencies/Installation:** sudo apt install sudo

**Screenshot:**

A screenshot of a Linux terminal window. The window title is 'sohaib@sohaib: ~'. The user has entered the command 'sudo', which has triggered a display of usage information. The usage text shows various options for the sudo command, including flags for help, keys, version, verbosity, group, host, prompt, user, role, type, count, directory, timeout, and file paths. The prompt returns to the shell after the command is executed.

```
sohaib@sohaib:~$ sudo
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
 [-u user] [command [arg ...]]
usage: sudo [-AbbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
 [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
 [-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
 [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
 [-u user] file ...
sohaib@sohaib:~$
```

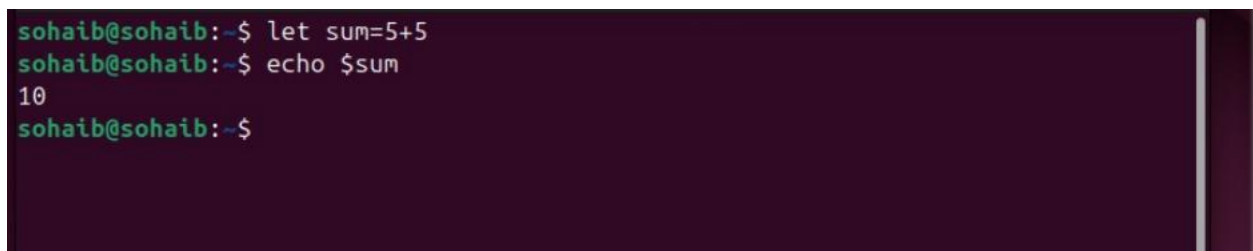
**Command Name:** sum

**Command Syntax:** sum [options] file

**Description:** Calculates a checksum and block count for a file (legacy command).

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

A screenshot of a Linux terminal window. The user enters a command to calculate the sum of 5 and 5 using the 'let' command, followed by an 'echo' command to display the result. The output shows the value 10.

```
sohaib@sohaib:~$ let sum=5+5
sohaib@sohaib:~$ echo $sum
10
sohaib@sohaib:~$
```

**Command Name:** suspend

**Command Syntax:** suspend [-f]

**Description:** Suspends the execution of the current shell.

**Dependencies/Installation:** Built into Bash, no installation needed.

**Screenshot:**

```
sohaib@sohaib: ~
sohaib@sohaib:~$ let sum=5+5
sohaib@sohaib:~$ echo $sum
10
sohaib@sohaib:~$ Ctrl + Z
Ctrl: command not found
sohaib@sohaib:~$ jobs
sohaib@sohaib:~$ fg %1
bash: fg: %1: no such job
sohaib@sohaib:~$ bg %1
bash: bg: %1: no such job
sohaib@sohaib:~$
```

**Command Name:** tail

**Command Syntax:** tail [options] [file]

**Description:** Outputs the last part (default 10 lines) of files.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

```
sohaib@sohaib: ~
sohaib@sohaib:~$ tail /etc/passwd
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-deskto
p:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/no
login
gnome-initial-setup:x:119:65534::/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin
/nologin
sohaib:x:1000:1000:sohaib:/home/sohaib:/bin/bash
sohaib@sohaib:~$
```

**Command Name:** tar

**Command Syntax:** tar [options] [archive\_file] [files\_to\_archive]

**Description:** Creates, extracts, and manipulates archive files.

**Dependencies/Installation:** Part of tar package, typically pre-installed.

**Screenshot:**

```
sohaib@sohaib:~$ tar -cvf archive.tar file1.txt file2.txt
file1.txt
file2.txt
```

**Command Name:** tee

**Command Syntax:** command | tee [options] file

**Description:** Reads from stdin and writes to both stdout and one or more files.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

```
sohaib@sohaib:~$ echo "Hello" | sudo tee /root/file1.txt
Hello
sohaib@sohaib:~$
```

**Command Name:** test

**Command Syntax:** test expression or [ expression ]

**Description:** Evaluates conditional expressions and returns a status.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

```
>_ sohaib@sohaib:~$ test -f file1.txt
sohaib@sohaib:~$ echo $?
0
sohaib@sohaib:~$ test -r file1.txt
sohaib@sohaib:~$ test 10 -gt 5
sohaib@sohaib:~$ echo $?
0
sohaib@sohaib:~$
```

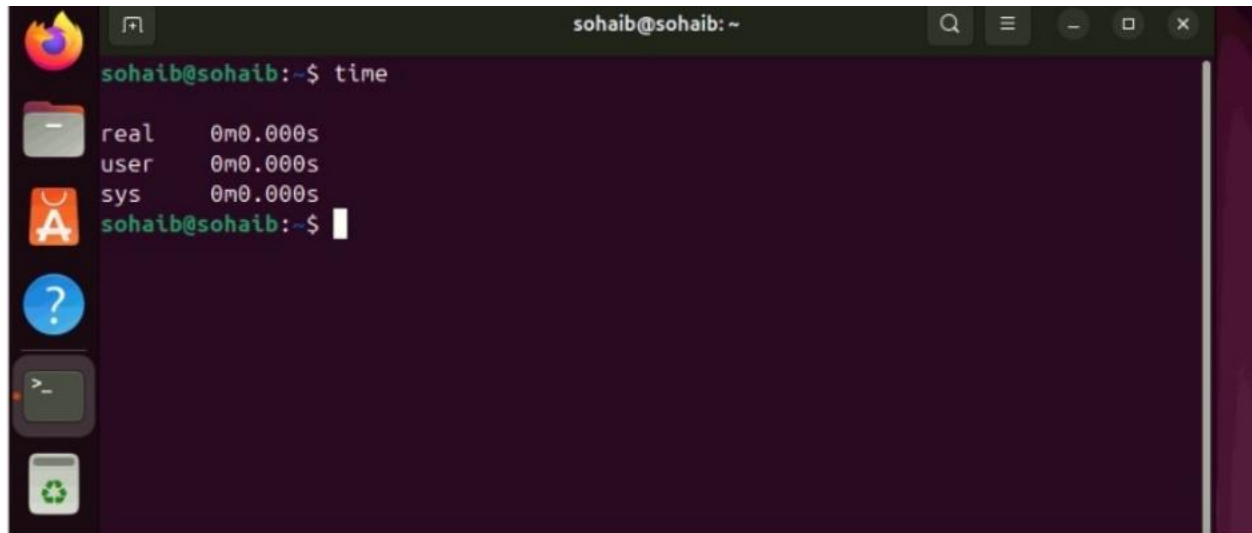
**Command Name:** time

**Command Syntax:** time [options] command [arguments]

**Description:** Measures the execution time of a command.

**Dependencies/Installation:** Part of time package or shell built-in.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' showing the output of the 'time' command. The output displays three categories: 'real' (0m0.000s), 'user' (0m0.000s), and 'sys' (0m0.000s). The terminal has a dark purple background and a sidebar on the left with various application icons.

```
sohaib@sohaib:~$ time
real 0m0.000s
user 0m0.000s
sys 0m0.000s
sohaib@sohaib:~$
```

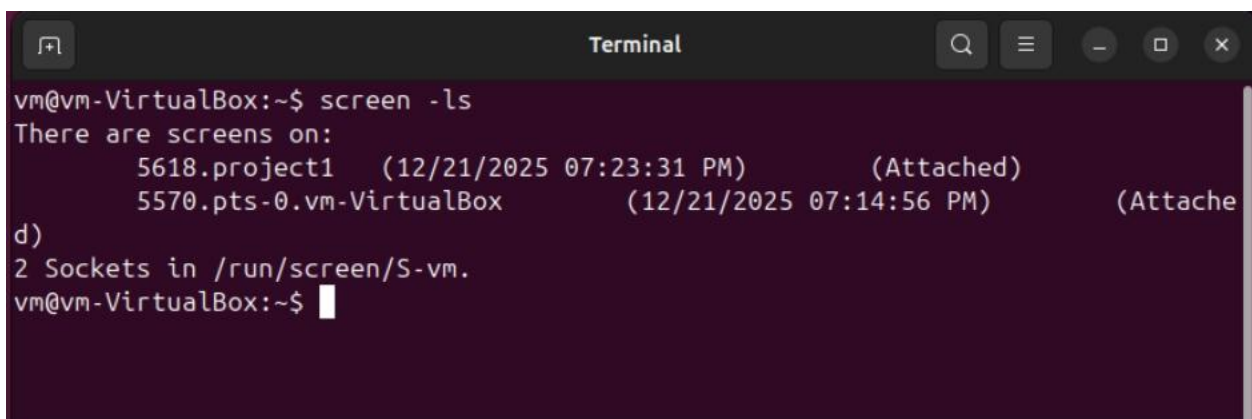
**Command Name:** timeout

**Command Syntax:** timeout [options] duration command [arguments]

**Description:** Runs a command with a time limit, terminating it if exceeded.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

A terminal window titled 'Terminal' showing the output of the 'screen -ls' command. The output lists two active screens: '5618.project1' and '5570.pts-0.vm-VirtualBox', both attached. It also shows '2 Sockets in /run/screen/S-vm.' The terminal has a dark purple background and a sidebar on the left with various application icons.

```
vm@vm-VirtualBox:~$ screen -ls
There are screens on:
 5618.project1 (12/21/2025 07:23:31 PM) (Attached)
 5570.pts-0.vm-VirtualBox (12/21/2025 07:14:56 PM) (Attache
d)
2 Sockets in /run/screen/S-vm.
vm@vm-VirtualBox:~$
```

**Command Name:** times

**Command Syntax:** times

**Description:** Displays accumulated user and system CPU times for the shell.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

```
sohaib@sohaib:~$ times
0m0.051s 0m0.055s
0m0.377s 0m0.163s
sohaib@sohaib:~$
```

**Command Name:** touch

**Command Syntax:** touch [options] file

**Description:** Creates empty files or updates timestamps of existing files.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

```
sohaib@sohaib:~$ touch file.txt
sohaib@sohaib:~$ touch sohaib.txt
sohaib@sohaib:~$
```

**Command Name:** top

**Command Syntax:** top [options]

**Description:** Displays a dynamic, real-time view of system processes.

**Dependencies/Installation:** Part of procps, typically pre-installed.

```
top - 11:36:57 up 1:02, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 192 total, 1 running, 191 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.0 us, 0.5 sy, 0.0 ni, 97.8 id, 0.2 wa, 0.0 hi, 0.5 si, 0.0 st
MiB Mem : 3916.6 total, 379.1 free, 1028.3 used, 2773.7 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 2888.3 avail Mem

 PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
 2986 sohaib 20 0 3946644 388188 147976 S 2.6 9.7 0:50.56 gnome-s+
 47 root 20 0 0 0 0 I 0.3 0.0 0:02.10 kworker+
 1279 root 20 0 316880 9328 7536 S 0.3 0.2 0:01.11 upowerd
 4395 sohaib 20 0 553396 52836 42220 S 0.3 1.3 0:01.02 gnome-t+
 1 root 20 0 23260 14444 9572 S 0.0 0.4 0:06.64 systemd
 2 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kthreadd
 3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_wor
 47 root -20 0 0 0 0 I 0.0 0.0 0:00.00 kworker+
 5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
 6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
 7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
 8 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
 10 root 20 0 0 0 0 I 0.0 0.0 0:00.14 kworker+
 12 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker+
 13 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
 14 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tas+
 15 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tas+
```

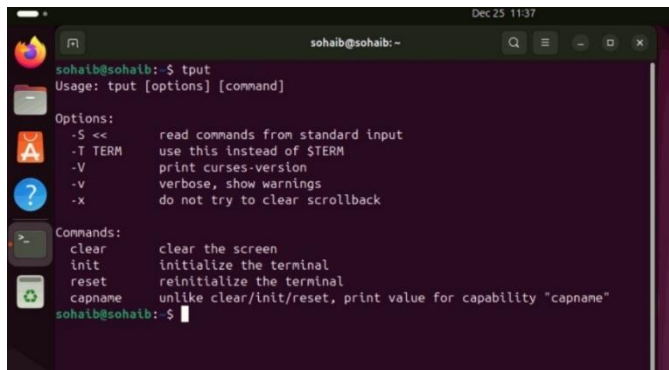
**Command Name:** tput

**Command Syntax:** tput [option] [capability\_name]

**Description:** Queries and manipulates terminal capabilities.

**Dependencies/Installation:** Part of ncurses-bin, typically pre-installed.

**Screenshot:**



```
sohaib@sohaib:~$ tput
Usage: tput [options] [command]

Options:
-S << read commands from standard input
-T TERM use this instead of $TERM
-V print curses-version
-v verbose, show warnings
-x do not try to clear scrollbar

Commands:
clear clear the screen
init initialize the terminal
reset reinitialize the terminal
capname unlike clear/init/reset, print value for capability "capname"

sohaib@sohaib:~$
```

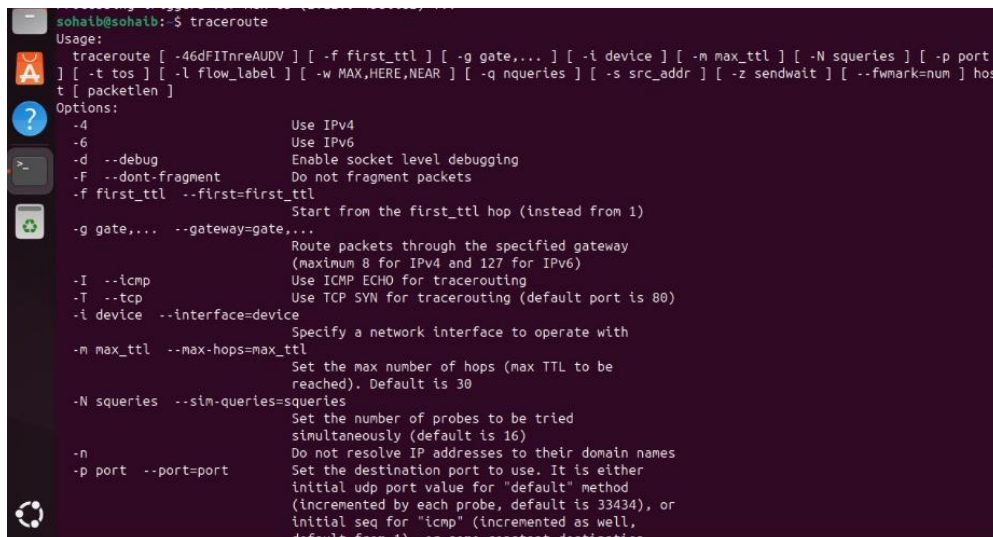
**Command Name:** traceroute

**Command Syntax:** traceroute [options] host

**Description:** Traces the network path packets take to a destination host.

**Dependencies/Installation:** sudo apt install traceroute

**Screenshot:**



```
sohaib@sohaib:~$ traceroute
Usage:
traceroute [-4dFIInreAUDV] [-f first_ttl] [-g gate,...] [-i device] [-m max_ttl] [-N squeries] [-p port] [-t tos] [-l flow_label] [-w MAX,HERE,NEAR] [-q nqueries] [-s src_addr] [-z sendwait] [--fwmark=num] host [packetlen]

Options:
-4 Use IPv4
-6 Use IPv6
-d --debug Enable socket level debugging
-F --dont-fragment Do not fragment packets
-f first_ttl --first=first_ttl Start from the first_ttl hop (instead from 1)
-g gate,... --gateway=gate,... Route packets through the specified gateway
 (maximum 8 for IPv4 and 127 for IPv6)
-I --icmp Use ICMP ECHO for tracerouting
-T --tcp Use TCP SYN for tracerouting (default port is 80)
-i device --interface=device Specify a network interface to operate with
-m max_ttl --max-hops=max_ttl Set the max number of hops (max TTL to be
 reached). Default is 30
-N squeries --sim-queries=squeries Set the number of probes to be tried
 simultaneously (default is 16)
-n Do not resolve IP addresses to their domain names
-p port --port=port Set the destination port to use. It is either
 initial udp port value for "default" method
 (incremented by each probe, default is 33434), or
 initial seq for "icmp" (incremented as well,
 default from 1), or some constant destination
```



**Command Name:** trap

**Command Syntax:** trap 'commands' signals

**Description:** Specifies commands to execute when the shell receives signals.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

The trap command may not work if it is used in the wrong shell, written with incorrect signal syntax, or the script exits before the signal is generated.

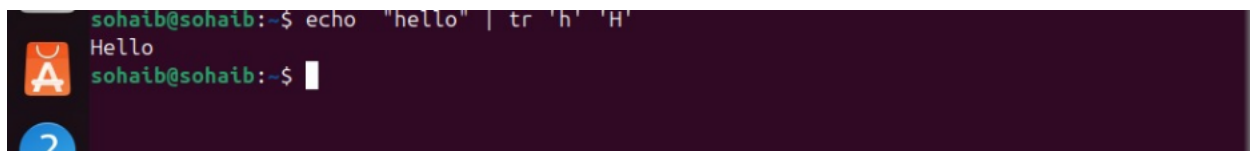
**Command Name:** tr

**Command Syntax:** tr [options] set1 set2

**Description:** Translates, squeezes, or deletes characters from input.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**



```
sohaib@sohaib:~$ echo "hello" | tr 'h' 'H'
Hello
sohaib@sohaib:~$
```

**Command Name:** true

**Command Syntax:** true

**Description:** Returns a successful (zero) exit status.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**



```
sohaib@sohaib:~$ true
sohaib@sohaib:~$ echo $?
0
sohaib@sohaib:~$
```



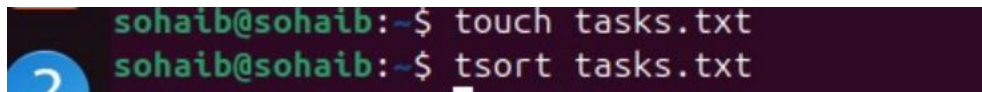
**Command Name:** tsort

**Command Syntax:** tsort [options] [file]

**Description:** Performs a topological sort on partially ordered items.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**



```
sohaib@sohaib:~$ touch tasks.txt
sohaib@sohaib:~$ tsort tasks.txt
```

**Command Name:** type

**Command Syntax:** type [options] command\_name

**Description:** Indicates how a command name is interpreted by the shell.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**



```
sohaib@sohaib:~$ type ls
ls is aliased to `ls --color=auto'
sohaib@sohaib:~$
```

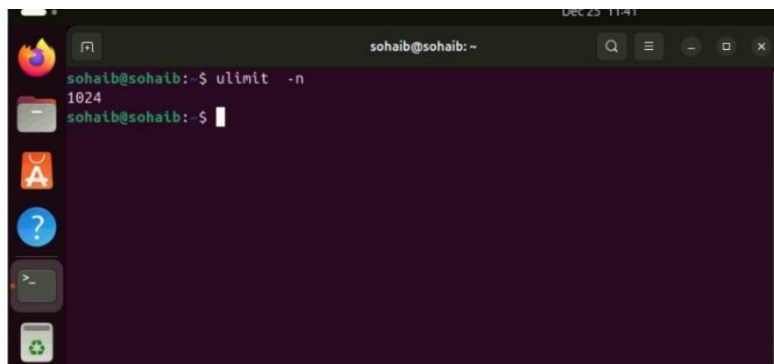
**Command Name:** ulimit

**Command Syntax:** ulimit [options] [limit]

**Description:** Gets or sets user-level limits on system resources.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**



```
sohaib@sohaib:~$ ulimit -n
1024
sohaib@sohaib:~$
```

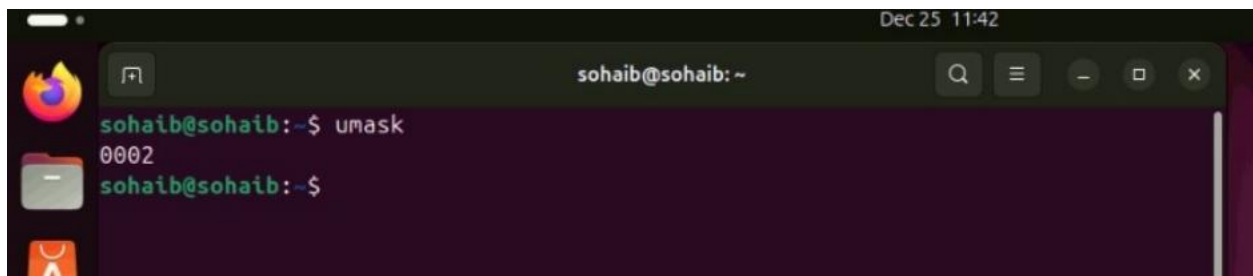
**Command Name:** umask

**Command Syntax:** umask [options] [mask]

**Description:** Sets or displays the file mode creation mask.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' with a search bar and window controls. The terminal shows the command 'umask' being executed, which returns '0002'. The prompt then returns to 'sohaib@sohaib:~\$'.

```
sohaib@sohaib:~$ umask
0002
sohaib@sohaib:~$
```

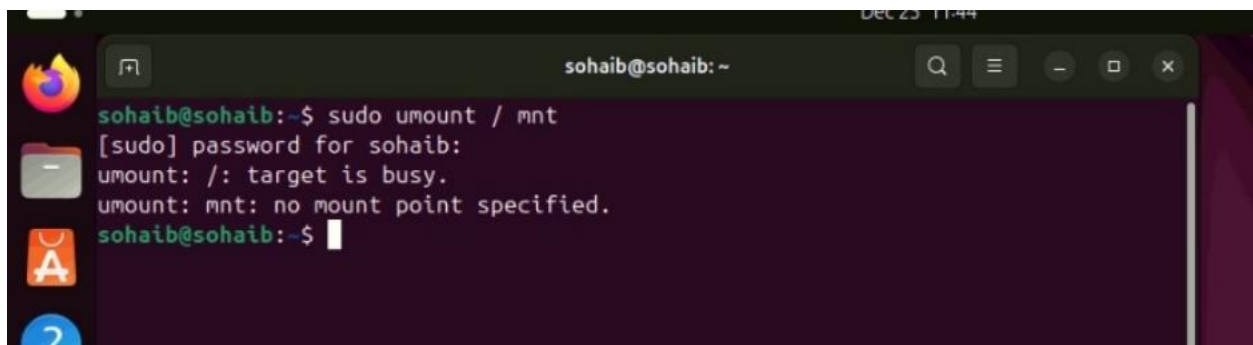
**Command Name:** umount

**Command Syntax:** sudo umount [options] mount\_point

**Description:** Detaches a previously mounted filesystem.

**Dependencies/Installation:** Part of util-linux, typically pre-installed.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' with a search bar and window controls. The terminal shows the command 'sudo umount /mnt' being executed. It prompts for a password, then shows error messages: 'umount: /: target is busy.' and 'umount: mnt: no mount point specified.' The prompt then returns to 'sohaib@sohaib:~\$'.

```
sohaib@sohaib:~$ sudo umount /mnt
[sudo] password for sohaib:
umount: /: target is busy.
umount: mnt: no mount point specified.
sohaib@sohaib:~$
```

**Command Name:** unalias

**Command Syntax:** unalias [options] alias\_name

**Description:** Removes alias definitions from the current shell.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

```
sohaib@sohaib:~$ unalias 55
bash: unalias: 55: not found
sohaib@sohaib:~$ unalias -a
sohaib@sohaib:~$ unalias 11
bash: unalias: 11: not found
sohaib@sohaib:~$ alias 11='ls -l'
sohaib@sohaib:~$
```

**Command Name:** uniq

**Command Syntax:** uniq [options] [input\_file]

**Description:** Filters out or reports adjacent duplicate lines from sorted input.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

```
sohaib@sohaib:~$ uniq tasks.txt
sohaib@sohaib:~$ uniq -c file1.txt
sohaib@sohaib:~$
```

**Command Name:** units

**Command Syntax:** units [options] [from\_unit to\_unit]

**Description:** Interactive program for converting between units of measurement.

**Dependencies/Installation:** sudo apt install units

**Screenshot:**

```
sohaib@sohaib:~$ units
Currency exchange rates from exchangerate-api.com (USD base) on 2024-02-18
Consumer price index data from US BLS, 2024-02-18
7235 units, 125 prefixes, 134 nonlinear units

You have:
```

**Command Name:** until

**Command Syntax:** until test-commands; do consequent-commands; done

**Description:** Loop that executes commands until a condition becomes

true.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

The until loop does not run because its condition is already true at the start, so the loop terminates immediately.

**Command Name:** uptime

**Command Syntax:** uptime [options]

**Description:** Shows system uptime, current time, user count, and load averages.

**Dependencies/Installation:** Part of procs, typically pre-installed.

**Screenshot:**

A screenshot of a Linux terminal window. The window title is "sohaib@sohaib: ~". The terminal shows the command "uptime" being executed, which returns the output: "11:48:09 up 1:13, 1 user, load average: 0.10, 0.07, 0.03". The prompt "sohaib@sohaib:~\$" is visible on the line below the output. The terminal has a dark background with light-colored text. On the left side of the terminal window, there is a vertical sidebar with several icons: a Firefox logo, a folder icon, an application icon, and a question mark icon. The top right corner of the window shows the date and time "Dec 25 11:48".

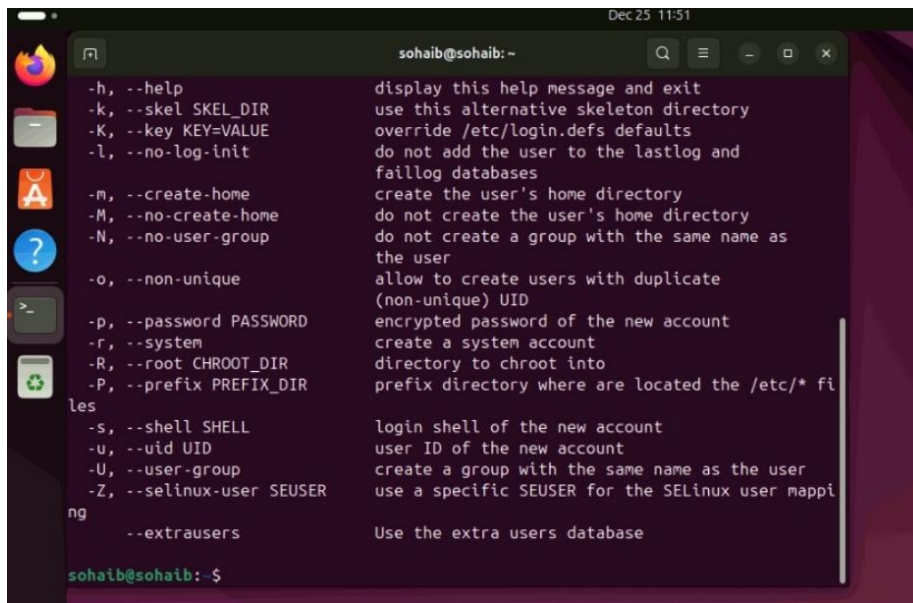
**Command Name:** useradd

**Command Syntax:** sudo useradd [options] username

**Description:** Creates a new user account or updates default new user info.

**Dependencies/Installation:** Part of passwd package, typically pre-installed.

**Screenshot:**



```
sohaib@sohaib:~
-h, --help display this help message and exit
-k, --skel SKEL_DIR use this alternative skeleton directory
-K, --key KEY=VALUE override /etc/login.defs defaults
-l, --no-log-init do not add the user to the lastlog and
 faillog databases
-m, --create-home create the user's home directory
-M, --no-create-home do not create the user's home directory
-N, --no-user-group do not create a group with the same name as
 the user
-o, --non-unique allow to create users with duplicate
 (non-unique) UID
-p, --password PASSWORD encrypted password of the new account
-r, --system create a system account
-R, --root CHROOT_DIR directory to chroot into
-P, --prefix PREFIX_DIR prefix directory where are located the /etc/* fi
les
-s, --shell SHELL login shell of the new account
-u, --uid UID user ID of the new account
-U, --user-group create a group with the same name as the user
-Z, --selinux-user SEUSER use a specific SEUSER for the SELinux user mappi
ng
--extrausers Use the extra users database
sohaib@sohaib:~$
```

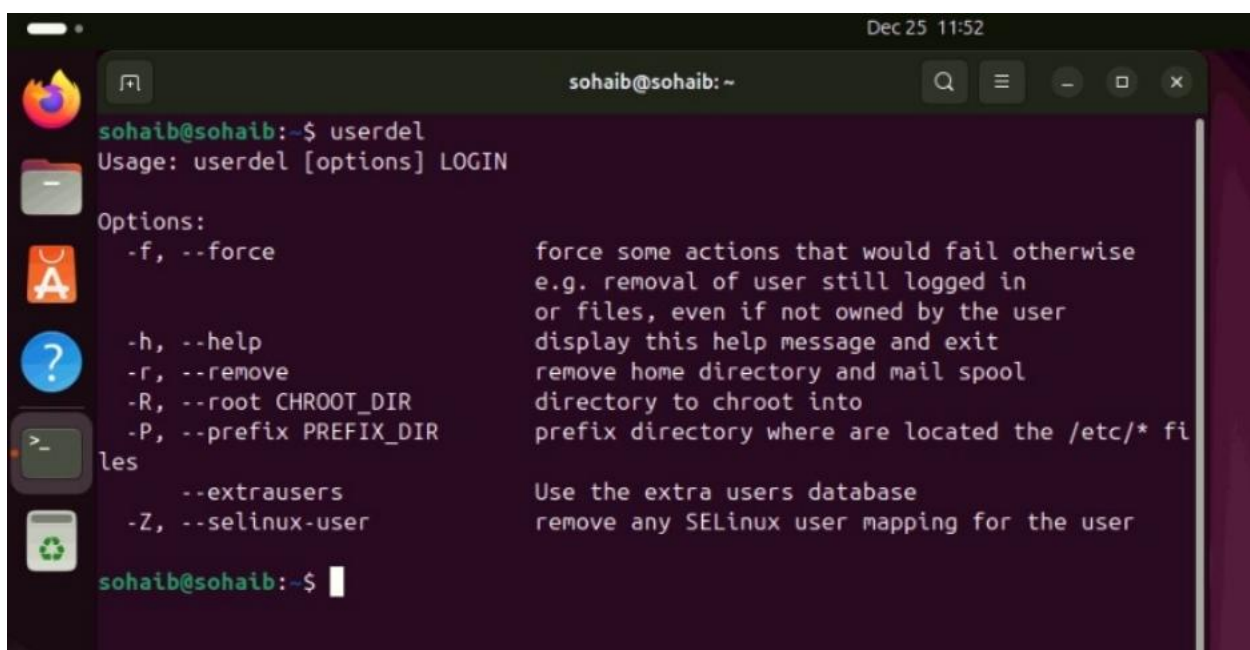
**Command Name:** userdel

**Command Syntax:** sudo userdel [options] username

**Description:** Deletes a user account and related files.

**Dependencies/Installation:** Part of passwd package, typically pre-installed.

**Screenshot:**



```
sohaib@sohaib:~$ userdel
Usage: userdel [options] LOGIN

Options:
-f, --force force some actions that would fail otherwise
 e.g. removal of user still logged in
 or files, even if not owned by the user
-h, --help display this help message and exit
-r, --remove remove home directory and mail spool
-R, --root CHROOT_DIR directory to chroot into
-P, --prefix PREFIX_DIR prefix directory where are located the /etc/* fi
les
--extrausers Use the extra users database
-Z, --selinux-user remove any SELinux user mapping for the user
sohaib@sohaib:~$
```

**Command Name:** usermod

**Command Syntax:** sudo usermod [options] username

**Description:** Modifies an existing user account's attributes.

**Dependencies/Installation:** Part of passwd package, typically pre-installed.

**Screenshot:**

User mode has limited privileges, so certain commands or operations cannot run due to lack of administrative (root) permissions.

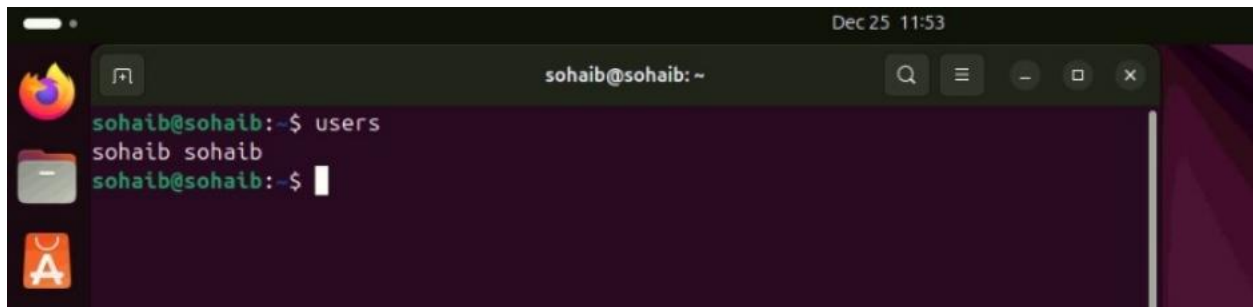
**Command Name:** users

**Command Syntax:** users [options] [file]

**Description:** Lists usernames of users currently logged in.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

A screenshot of a Linux terminal window. The window title is "sohaib@sohaib: ~". The terminal shows the command "users" being executed, which outputs "sohaib sohaib". The prompt "sohaib@sohaib: ~\$" is visible at the bottom. The terminal has a dark background with light-colored text. On the left side of the terminal window, there are icons for Firefox, a file manager, and the Ubuntu logo. The top right corner of the window shows the date and time "Dec 25 11:53".

**Command Name:** vdir

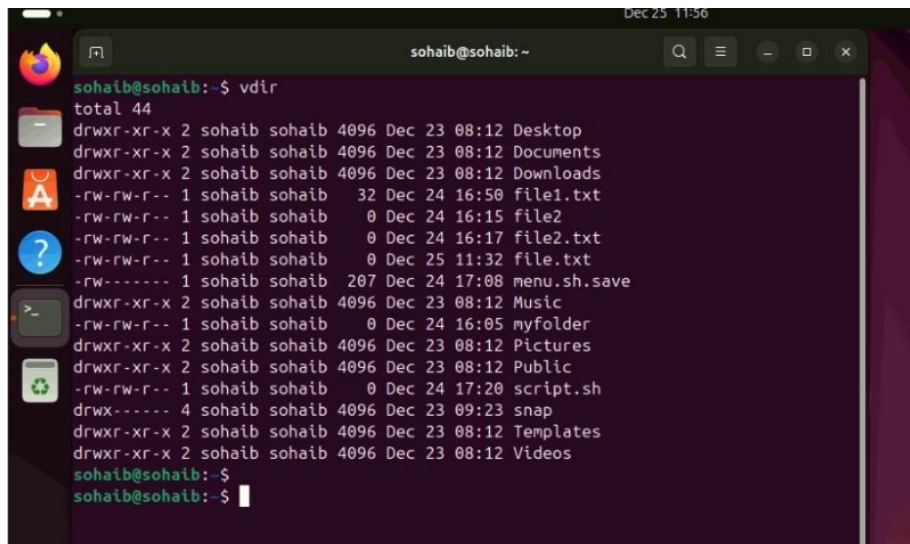
**Command Syntax:** vdir [options] [file]

**Description:** Lists directory contents in long format (equivalent to ls -l -b).

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**



A terminal window titled 'sohaib@sohaib: ~' showing the output of the 'vdir' command. The output lists the contents of the current directory with permissions, owner, size, date, and name. The files include Desktop, Documents, Downloads, file1.txt, file2, file2.txt, file.txt, menu.sh.save, Music, myfolder, Pictures, Public, script.sh, snap, Templates, and Videos.

```
sohaib@sohaib:~$ vdir
total 44
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Desktop
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Documents
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Downloads
-rw-rw-r-- 1 sohaib sohaib 32 Dec 24 16:50 file1.txt
-rw-rw-r-- 1 sohaib sohaib 0 Dec 24 16:15 file2
-rw-rw-r-- 1 sohaib sohaib 0 Dec 24 16:17 file2.txt
-rw-rw-r-- 1 sohaib sohaib 0 Dec 25 11:32 file.txt
-rw----- 1 sohaib sohaib 207 Dec 24 17:08 menu.sh.save
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Music
-rw-rw-r-- 1 sohaib sohaib 0 Dec 24 16:05 myfolder
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Pictures
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Public
-rw-rw-r-- 1 sohaib sohaib 0 Dec 24 17:20 script.sh
drwx----- 4 sohaib sohaib 4096 Dec 23 09:23 snap
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Templates
drwxr-xr-x 2 sohaib sohaib 4096 Dec 23 08:12 Videos
sohaib@sohaib:~$
sohaib@sohaib:~$
```

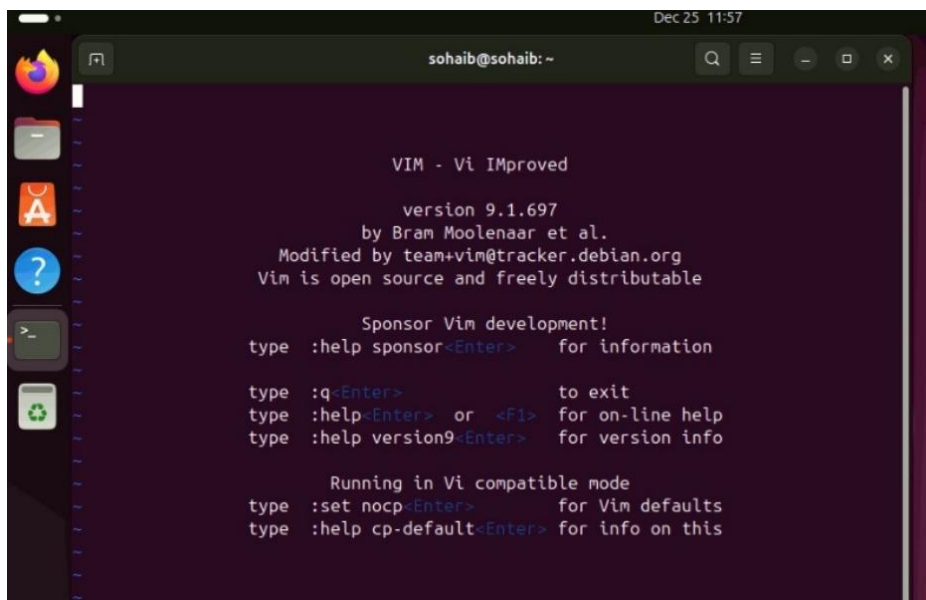
**Command Name:** vi

**Command Syntax:** vi [options] [file]

**Description:** A classic, screen-oriented text editor.

**Dependencies/Installation:** sudo apt install vim

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' showing the Vim startup screen. It displays the Vim logo, version 9.1.697, and information about the developers and the license. It also lists various commands and their functions, such as ':q' to exit, ':help' for on-line help, and ':set nocp' for Vim defaults.

```
VIM - Vi IMproved

version 9.1.697
by Bram Moolenaar et al.
Modified by team+vim@tracker.debian.org
Vim is open source and freely distributable

Sponsor Vim development!
type :help sponsor<Enter> for information

type :q<Enter> to exit
type :help<Enter> or <F1> for on-line help
type :help version9<Enter> for version info

Running in Vi compatible mode
type :set nocp<Enter> for Vim defaults
type :help cp-default<Enter> for info on this
```

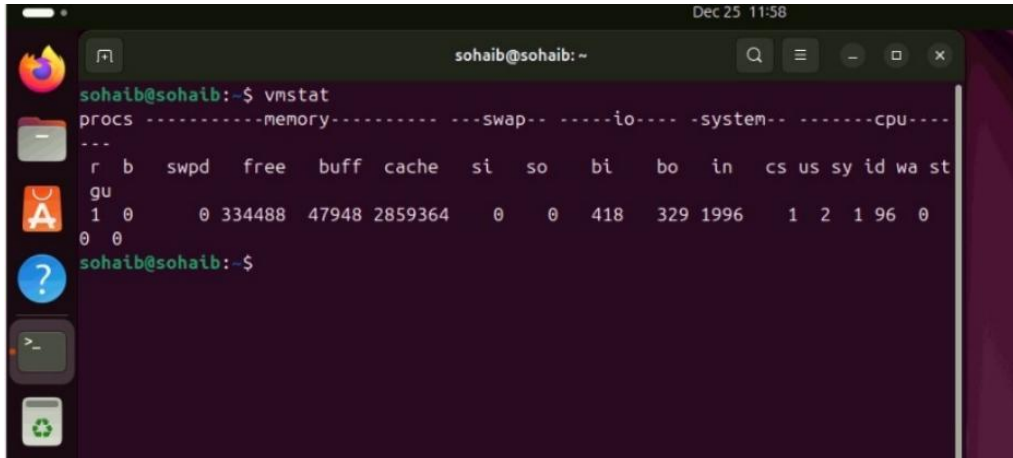
**Command Name:** vmstat

**Command Syntax:** vmstat [options] [delay [count]]

**Description:** Reports virtual memory statistics and system activity.

**Dependencies/Installation:** Part of procps, typically pre-installed.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' showing the output of the 'vmstat' command. The output is a table with columns forprocs, memory, swap, io, system, and cpu. The data row shows values for various statistics including r, b, swpd, free, buff, cache, si, so, bi, bo, in, cs, us, sy, id, wa, and st.

```
sohaib@sohaib:~$ vmstat
procs-----memory----- --swap-- ----io---- -system-- -----cpu----

r b swpd free buff cache si so bi bo in cs us sy id wa st
gu
1 0 0 334488 47948 2859364 0 0 418 329 1996 1 2 1 96 0
0 0
sohaib@sohaib:~$
```

**Command Name:** w

**Command Syntax:** w [options] [user]

**Description:** Shows logged-in users and their processes, plus system load.

**Dependencies/Installation:** Part of procps, typically pre-installed.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' showing the output of the 'w' command. The output displays system load averages and a table of logged-in users with columns for USER, TTY, FROM, LOGIN@, IDLE, JCPU, PCPU, and WHAT.

```
sohaib@sohaib:~$ w
11:58:44 up 1:24, 1 user, load average: 0.18, 0.05, 0.01
USER TTY FROM LOGIN@ IDLE JCPU PCPU WHAT
sohaib tty2 - 10:45 1:24m 0.05s 0.05s /usr/libexec/gn
sohaib@sohaib:~$
```

**Command Name:** wait

**Command Syntax:** wait [options] [pid ...]

**Description:** Waits for child processes to complete and returns their status.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

The wait command does not produce output unless it is waiting for a background process, so it appears not to run.

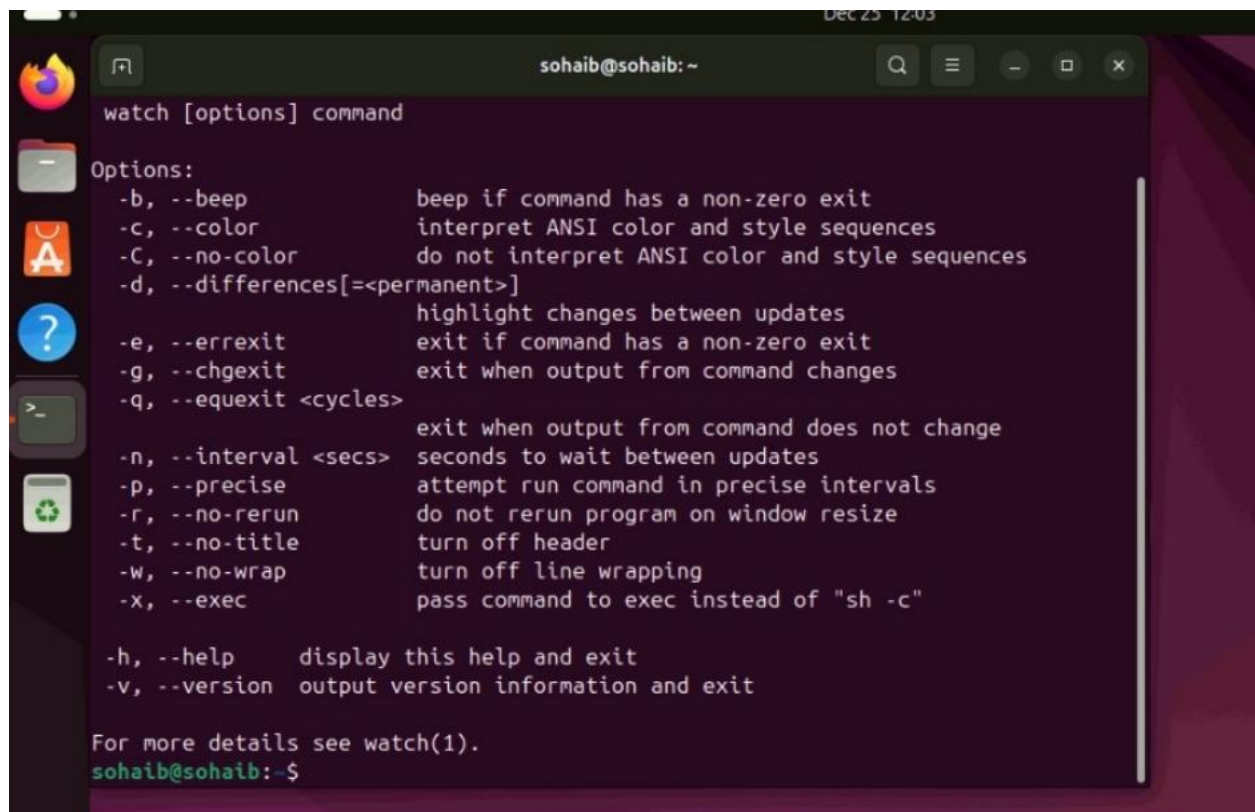
**Command Name:** watch

**Command Syntax:** watch [options] command

**Description:** Executes a command repeatedly at regular intervals.

**Dependencies/Installation:** sudo apt install watch

**Screenshot:**

A terminal window with a dark purple background and light green text. The window title is 'sohaib@sohaib: ~'. The prompt is 'sohaib@sohaib:~\$'. The user has entered 'watch [options] command'. The terminal displays the help text for the 'watch' command, listing various options and their descriptions. The options are: -b, --beep; -c, --color; -C, --no-color; -d, --differences[=<permanent>]; -e, --errexit; -g, --chgexit; -q, --equexit <cycles>; -n, --interval <secs>; -p, --precise; -r, --no-rerun; -t, --no-title; -W, --no-wrap; -X, --exec; -h, --help; -v, --version. The text is formatted with some options in bold and some descriptions in a lighter green color. At the bottom, it says 'For more details see watch(1).' and the prompt 'sohaib@sohaib:~\$' is visible.

```
sohaib@sohaib:~$ watch [options] command

Options:
 -b, --beep beep if command has a non-zero exit
 -c, --color interpret ANSI color and style sequences
 -C, --no-color do not interpret ANSI color and style sequences
 -d, --differences[=<permanent>] highlight changes between updates
 -e, --errexit exit if command has a non-zero exit
 -g, --chgexit exit when output from command changes
 -q, --equexit <cycles> exit when output from command does not change
 -n, --interval <secs> seconds to wait between updates
 -p, --precise attempt run command in precise intervals
 -r, --no-rerun do not rerun program on window resize
 -t, --no-title turn off header
 -W, --no-wrap turn off line wrapping
 -X, --exec pass command to exec instead of "sh -c"

 -h, --help display this help and exit
 -v, --version output version information and exit

For more details see watch(1).
sohaib@sohaib:~$
```

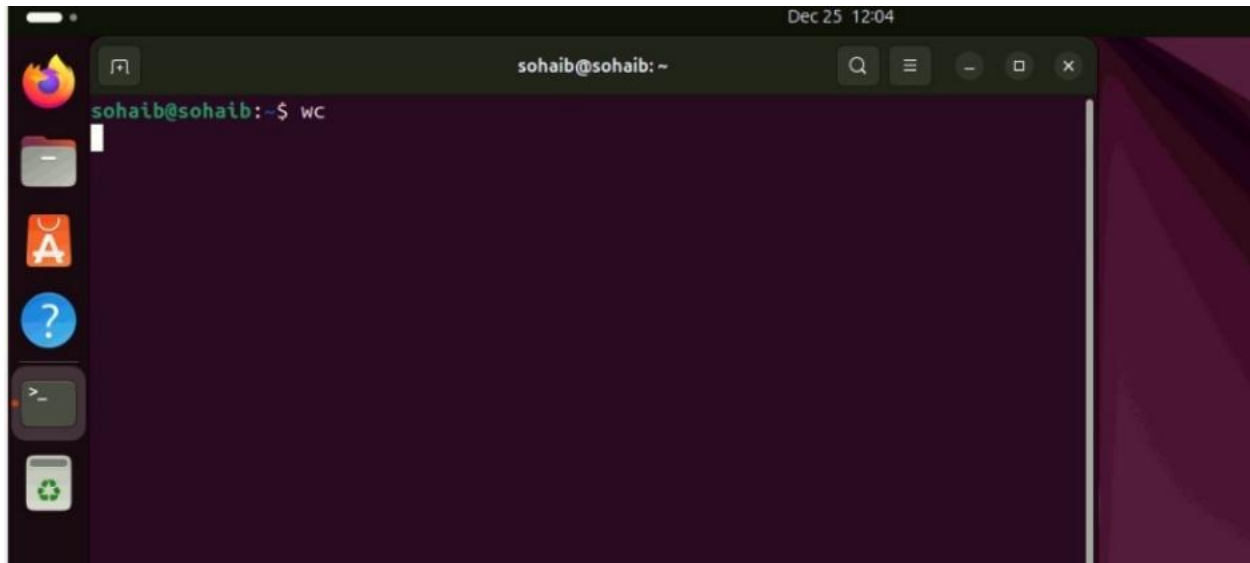
**Command Name:** wc

**Command Syntax:** wc [options] [file]

**Description:** Counts lines, words, and bytes/characters in files.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**



**Command Name:** whereis

**Command Syntax:** whereis [options] command\_name

**Description:** Locates binary, source, and manual page files for a command.

**Dependencies/Installation:** Part of util-linux, typically pre-installed.

**Screenshot:**

The whereis command may show no output if the program is not installed or not present in standard binary directories.

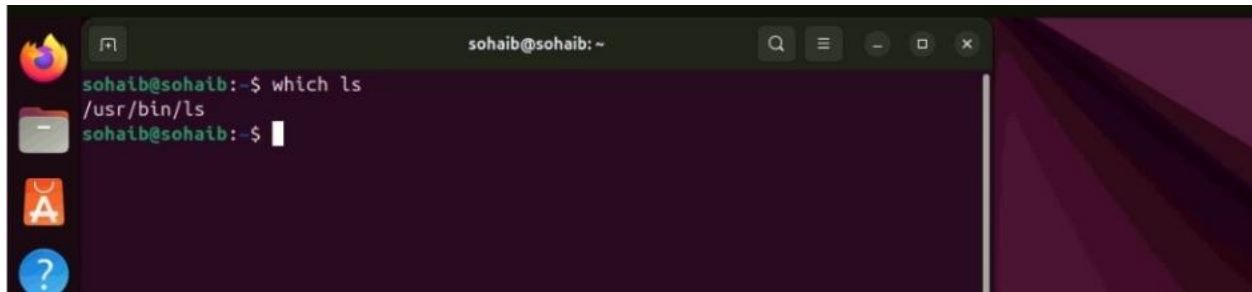
**Command Name:** which

**Command Syntax:** which [options] [--] command\_name

**Description:** Displays the full path of executable commands.

**Dependencies/Installation:** Part of debianutils, typically pre-installed.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' with search, menu, and window control icons. The prompt is 'sohaib@sohaib:~\$'. The command 'which ls' has been entered, and the output '/usr/bin/ls' is displayed on the next line. The prompt 'sohaib@sohaib:~\$' is shown again on the third line.

```
sohaib@sohaib:~$ which ls
/usr/bin/ls
sohaib@sohaib:~$
```

**Command Name:** while

**Command Syntax:** while test-commands; do consequent-commands;  
done

**Description:** Loop that executes commands as long as a condition is true.

**Dependencies/Installation:** Built into the shell, no installation needed.

**Screenshot:**

The while loop does not execute because its condition is false at the beginning.

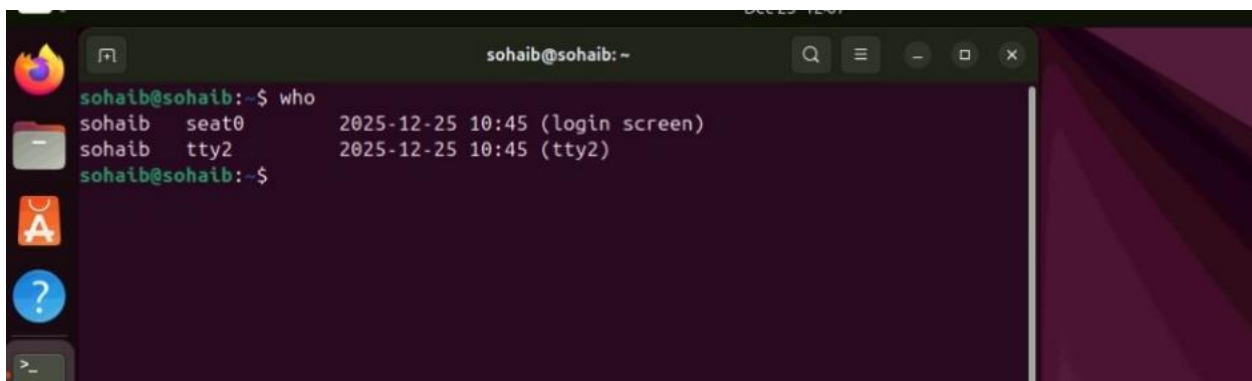
**Command Name:** who

**Command Syntax:** who [options] [file]

**Description:** Lists users currently logged into the system.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

A terminal window titled 'sohaib@sohaib: ~' with search, menu, and window control icons. The prompt is 'sohaib@sohaib:~\$'. The command 'who' has been entered, and the output shows two users: 'sohaib' on 'seat0' and 'sohaib' on 'tty2', both logged in at '2025-12-25 10:45'. The prompt 'sohaib@sohaib:~\$' is shown on the next line.

```
sohaib@sohaib:~$ who
sohaib seat0 2025-12-25 10:45 (login screen)
sohaib tty2 2025-12-25 10:45 (tty2)
sohaib@sohaib:~$
```

**Command Name:** whoami

**Command Syntax:** whoami

**Description:** Prints the effective username of the current user.

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

A screenshot of a Linux terminal window. The window title is 'sohaib@sohaib: ~'. The terminal shows the command 'whoami' being entered at the prompt 'sohaib@sohaib:~\$'. The output of the command is 'sohaib', displayed on the next line. The terminal has a dark background with light-colored text. On the left side of the terminal window, there is a vertical dock with several application icons: a Firefox browser icon, a file manager icon, an application icon, and a help icon.

**Command Name:** wget

**Command Syntax:** wget [options] [URL]

**Description:** A non-interactive network downloader for HTTP, HTTPS, and FTP.

**Dependencies/Installation:** sudo apt install wget

**Screenshot:**

wget may fail if it is **not installed**, the **URL is incorrect**, there is **no internet connection**, or the system **blocks network access**.

**Command Name:** write

**Command Syntax:** write user [terminal]

**Description:** Sends a message to another logged-in user on the same system.

**Dependencies/Installation:** Part of util-linux, typically pre-installed.

**Screenshot:**

The target user is not logged in or has disabled message receiving (mesg n).

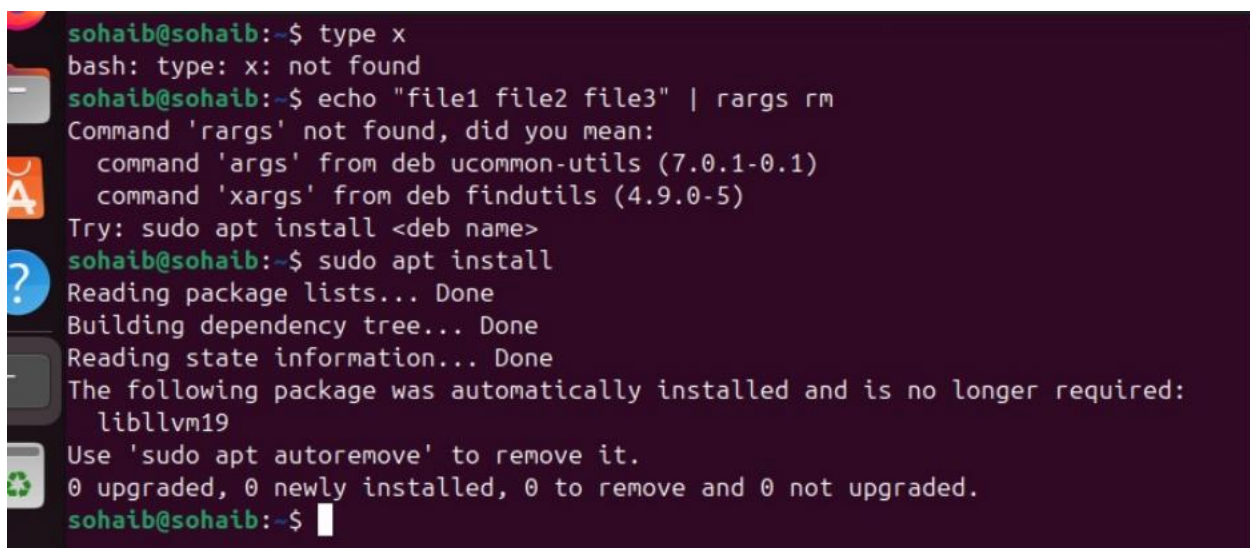
**Command Name:** xargs

**Command Syntax:** command | xargs [options] [command]

**Description:** Builds and executes command lines from standard input.

**Dependencies/Installation:** Part of findutils, typically pre-installed.

**Screenshot:**

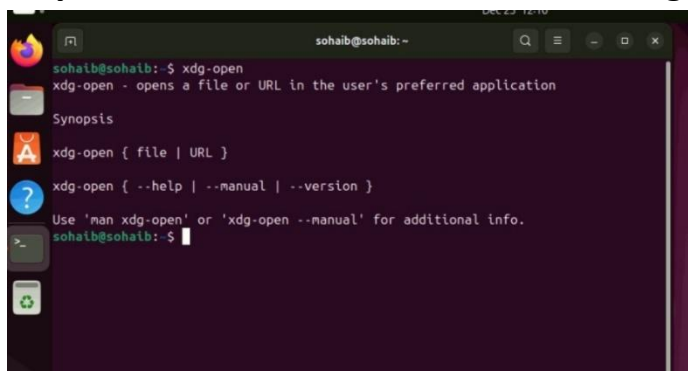
A terminal window with a dark purple background. The user 'sohaib' is at the prompt. They run 'type x' and get 'bash: type: x: not found'. Then they run 'echo "file1 file2 file3" | rargs rm', which gives an error: 'Command 'rargs' not found, did you mean: command 'args' from deb ucommon-utils (7.0.1-0.1) command 'xargs' from deb findutils (4.9.0-5)'. They are prompted to try 'sudo apt install <deb name>'. They then run 'sudo apt install', which shows package lists, dependency tree, and state information. It notes that 'libllvm19' was automatically installed and is no longer required, and suggests using 'sudo apt autoremove' to remove it. The summary shows 0 upgraded, 0 newly installed, 0 to remove, and 0 not upgraded. The prompt returns to 'sohaib@sohaib:~\$'.

**Command Name:** xdg-open

**Command Syntax:** xdg-open {file|URL}

**Description:** Opens a file or URL using the user's preferred application.

**Dependencies/Installation:** Part of xdg-utils, typically pre-installed.

A terminal window showing the help output for the 'xdg-open' command. The user 'sohaib' runs 'xdg-open' and gets a description: 'xdg-open - opens a file or URL in the user's preferred application'. Below this is the synopsis: 'xdg-open { file | URL }'. Then it shows the options: 'xdg-open { --help | --manual | --version }'. At the bottom, it says 'Use 'man xdg-open' or 'xdg-open --manual' for additional info.' The prompt returns to 'sohaib@sohaib:~\$'.



**Command Name:** xz

**Command Syntax:** xz [options] [file]

**Description:** Compresses or decompresses files using LZMA/LZMA2 algorithm.

**Dependencies/Installation:** sudo apt install xz-utils

**Screenshot:**

A terminal window with a dark purple background. The prompt is 'sohaib@sohaib:~\$'. The first command is 'xz --version', which outputs 'xz (XZ Utils) 5.4.5' and 'liblzma 5.4.5'. The second command is 'xz file.txt', which is partially visible.

```
sohaib@sohaib:~$ xz --version
xz (XZ Utils) 5.4.5
liblzma 5.4.5
sohaib@sohaib:~$ xz file.txt
```

**Command Name:** yes

**Command Syntax:** yes [string]

**Description:** Outputs a string repeatedly until killed (default "y").

**Dependencies/Installation:** Part of coreutils, typically pre-installed.

**Screenshot:**

A terminal window with a dark purple background. The prompt is 'sohaib@sohaib:~\$'. The command 'yes' is entered, followed by a cursor.

```
sohaib@sohaib:~$ yes
```

**Command Name:** zip

**Command Syntax:** zip [options] zipfile list\_of\_files

**Description:** Packages and compresses files into a .zip archive.

**Dependencies/Installation:** sudo apt install zip

**Screenshot:**

The zip command may not produce output if the **file/directory does not exist**, the **syntax is wrong**, or the **zip package is not installed**.